Coercive Growth:
Forced Resettlement and Ethnicity-Based Agglomeration

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Abstract

I study how social divisions shape the gains from agglomeration, leveraging a 1950s resettlement program in British Malaya that forcibly relocated half a million Chinese. Decades later, areas with higher resettlement had greater population density and Chinese shares, and were more industrialized, with higher incomes and greater labor market specialization. However, the agglomeration benefits were concentrated among the Chinese. Evidence suggests that segregation and cultural and linguistic barriers hindered cross-ethnic spillovers. Estimating a quantitative spatial model where agglomeration externalities vary by sector and ethnic composition, I find that output gains from forced relocation were insufficient to offset its welfare losses. (*JEL* J15, N15, R11, R23, O15)

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I Introduction

The clustering of people and economic activity can raise productivity by reducing transaction and search costs and fostering knowledge spillovers (Marshall, 1890; Duranton and Puga, 2004). Yet, these gains from proximity hinge on the density and nature of local social interactions. Social divisions such as ethnic segregation and tension, common in many urban settings, can hinder these productive interactions. This raises a fundamental question: How do ethnic divisions shape agglomeration economies and, more broadly, the level and distribution of economic development?

To answer this question, I study an ethnic-based resettlement program implemented during the 1950s Malayan Emergency in British Malaya, which forcibly relocated roughly half a million rural Chinese into villages across the country. Leveraging this variation, I examine how the concentration of one ethnic group (Chinese) affected local economic outcomes across regions and groups over the following 50 years. To quantify the aggregate effects, I develop a quantitative spatial model that allows agglomeration externalities to vary by sector and ethnic composition, which I identify using the resettlement as a population shifter. I then use the model to evaluate counterfactual policies for economic development in the presence of heterogeneous agglomeration forces.

This setting provides a useful context for studying how ethnic divisions shape agglomeration economies for three reasons. First, the resettlement program shifted both population size and ethnic composition across Malaysia, where deep cultural, religious, and linguistic divisions existed between ethnic groups. Second, the relocation was driven by a clear military objective, which limits endogenous sorting and enables me to model counterfactual resettlements while conditioning on key observables. Finally, because of the siting strategy, receiving areas were more urban and industrial, where agglomeration externalities typically emerge. Yet, given the underlying ethnic divisions, different groups may have benefited differently from the influx of Chinese.

To estimate the local effects of resettlement, I leverage the program's wartime objectives to construct counterfactual resettlement and isolate quasi-random variation. During the Malayan Emergency (1948–1960), a guerrilla conflict between the British and Malayan communists, the British forcibly relocated nearly one-tenth of the population (or one-third of the Chinese population) into roughly 500 "New Villages." The goal was to sever ties to communist insurgents, so the British relocated ethnic Chinese from remote, jungle-adjacent areas, where insurgents were based, to more accessible locations. The program was implemented in two stages. First, suitable sites were selected based on security access, with most located along major roads (Figure 1). Second, rural Chinese populations were relocated to these sites in a way that minimized dislocation from their original settlements.

Following this procedure, I construct counterfactual resettlement in two steps. First, I randomly permute New Village locations, conditional on distance to transportation networks and other key covariates the British considered, such as land use and the local Chinese population. Second, I use a gravity model to predict the number of people resettled to each site, assuming that resettlement costs increased with distance. I repeat this procedure 1,000 times and take the average predicted resettlement density across permutations to obtain each county's *expected* resettlement density (Borusyak and Hull, 2023).[1]

The estimation compares areas with similar expected resettlement density and prewar characteristics but different realized resettlement. The identifying variation comes from the precise placement of New Villages along major transportation routes and deviations from the dislocation-minimizing plan. Historical accounts suggest that the British lacked the capacity or intent to fine-tune resettlement based on unobserved economic fundamentals. Consistent with this, I show that geographic features and prewar economic activity are balanced across areas with varying residual resettlement densities.

Digitizing population censuses from 1931 to 2000, I document that the program persistently reshaped the population distribution in Peninsular Malaysia. Counties with higher resettlement density saw a sharp increase in Chinese population between 1947 and 1957, while the non-Chinese population remained stable. After mobility restrictions were lifted in 1960, these areas attracted internal migrants from all ethnic groups. By 2000, these areas still had higher population densities and a larger Chinese share. On average, every 1% of the 1947 population resettled led to a 1.3% increase in population by 2000.

The influx of Chinese substantially transformed local economic structure and raised productivity in areas with higher resettlement. Census data from 1980 and 1991 show that these counties had greater employment and a larger share in non-agricultural sectors such as manufacturing, trade, and services. This shift reflected both a higher concentration of Chinese—who were more likely to work outside agriculture—and a smaller agricultural share among non-Chinese. Chinese households earned higher incomes in these counties, especially those employed outside agriculture. In contrast, non-Chinese saw only marginal gains in the non-agricultural sector, with effects less than half as large as those for Chinese.

These comparisons are made within ethnic groups and sectors across counties with varying resettlement densities, so the results are not driven by comparative advantage or broader cultural differences across ethnicities. The ethnic gap in income gains also appears within agriculture and across education levels, suggesting that the productivity benefits of co-ethnic concentration extended beyond specific sectors or skill groups.

[1] Using a similar approach, Dell and Olken (2020) identify the effects of proximity to sugar plants established in colonial Java by specifying counterfactual plant locations.

I find that labor matching and greater specialization contributed to ethnicity-based agglomeration economies. By the 1980s, Chinese individuals in more resettled counties had higher labor force participation, greater occupational and industrial concentration, and were more likely to hold managerial roles. Educational attainment was also higher, especially among younger cohorts who had not yet completed or begun schooling at the time of resettlement. These patterns are consistent with higher co-ethnic density improving worker-firm matching, thereby raising the returns to formal employment, specialization, and education (Kim, 1989; Dauth et al., 2022). In contrast, non-Chinese individuals showed no significant differences in these labor market outcomes, and their educational gains were much smaller. While education accounts for part of the income gains among Chinese, the ethnic gap persists even conditional on education.

The evidence suggests that segregation and social divisions limited cross-ethnic productivity spillovers. Income gains for non-Chinese were larger in regions with lower pre-existing segregation—measured by residential patterns or the distance between religion sites such as mosques and non-Muslim temples—and were statistically similar to those of the Chinese in areas where ethnic groups were fully integrated. Gains were also larger in counties where a higher share of Chinese could speak Malay by 1980—roughly a decade into nation-building efforts. These patterns point to physical, cultural, or linguistic divisions as key obstacles to cross-ethnic spillovers.

These cross-sectional analyses capture differences across receiving counties, but do not speak to the program's aggregate impact, which also depends on outcomes in the origin counties from which the Chinese were removed, as well as in other counties that were neither destinations nor origins but may have been indirectly affected through migration and trade.

To evaluate aggregate effects, I develop a quantitative spatial model with two ethnic groups and two sectors. The model incorporates a Roy (1951)-type framework for migration and occupation choices—the key margins of adjustment following resettlement. I build on the work of Allen and Donaldson (2022) and Peters (2022) by allowing productivity spillovers to vary by sector and the ethnic composition of local employment, so that within-ethnic spillovers may differ from cross-ethnic spillovers. Local amenities also depend on ethnic composition, as in Diamond (2016). In this two-period model, individuals begin with an initial location, choose where to migrate based on heterogeneous preferences, and then select a sector based on heterogeneous productivity.

To bring the model to the data, I use the post-resettlement, 1957 population distribution as the initial population, and treat 1980 as the equilibrium outcome. Due to migration frictions, individuals tended to move near their resettled locations, so the 1957 distribution continued to shape the 1980 equilibrium. The forced resettlement thus provides an exogenous shift in population that helps identify the parameters of agglomeration externalities. Specifically, the relationship between the local Chinese share and the Chinese wage premium (relative to non-Chinese) within a location-sector—adjusted for occupational composition—identifies the strength of within- versus cross-ethnic productivity spillovers. Similarly, the relationship between local employment size and wages—adjusting for composition effects and general equilibrium forces—identifies sector-specific scale economies.

The estimates reveal sizable variation in agglomeration externalities across sectors and regions, shaped by ethnic composition. In a neoclassical model with downward-sloping labor demand, an increase in non-agricultural labor supply would lower relative wages in the sector, pushing workers into agriculture. Instead, I find higher non-agricultural wages and lower agricultural employment shares in more densely resettled counties, consistent with strong external economies of scale in non-agriculture. I estimate that a 1% increase in non-agricultural employment raises labor productivity by 0.22%. By contrast, agriculture exhibits local diminishing returns to scale due to the fixed land supply: a 1% increase in agricultural employment reduces labor productivity by 0.12%. I also estimate stronger within-ethnic productivity spillovers than cross-ethnic spillovers, with the latter being positive only in non-agriculture. Finally, the estimated amenity spillovers suggest homophily—a preference for living near others of the same ethnic background.

I use the estimated model to evaluate the aggregate economic and welfare impacts of the resettlement program. To do so, I simulate a "no resettlement" 1980 equilibrium using the pre-resettlement, 1947 population distribution as the initial condition, holding all parameters and location fundamentals fixed. Assuming the 1947 distribution would have persisted to 1957 absent resettlement, I measure the program's impact by comparing this counterfactual to the observed 1980 equilibrium. I find that resettlement increases aggregate output by 2%, with two-thirds of the gain driven by labor reallocation to more productive regions and sectors. Moving rural Chinese from remote areas to locations with better market access and higher industrial productivity raises their output, while freeing up rural lands for Malays, who benefit from improved agricultural productivity.

While output increases, the welfare effects are more nuanced. The average welfare gain is 4.8%, but this calculation abstracts from the costs of forced movement by treating the shift from the 1947 to the 1957 distribution as costless. To benchmark these costs, I invert the model to calculate the minimum place- and ethnicity-specific wage subsidies needed to induce voluntary relocation replicating the observed 1947–57 population changes. I find that the total required subsidy for the resettled population exceeds the program's aggregate economic gains, suggesting a net welfare loss from forced relocation. These subsidies serve as a lower bound on utility loss (in monetary terms), as resettled individuals likely valued their assigned destinations less than the average resident, biasing the required compensation downward. While younger cohorts benefit from improved economic opportunities, the original resettled generation bears the welfare loss.

Since Marshall (1890), a large literature has documented the productivity benefits of density and geographic concentration (e.g., Greenstone, Hornbeck and Moretti (2010); Kline and Moretti (2014); Ahlfeldt et al. (2015); Peters (2022); Smith and Kulka (2024); Ciccone and Nimczik (2024).[2] While this literature emphasizes the role of density, less is known about how social composition shapes productivity gains from agglomeration. I study an ethnic-based resettlement program that exogenously shifted local ethnic composition alongside population density to examine its effects on group-specific outcomes, complementing research on the distributional consequences of agglomeration across skill groups (Baum-Snow and Pavan, 2013; Baum-Snow, Freedman and Pavan, 2018).

By treating ethnic composition as an endogenous amenity, I build on work showing how local demographics influence residential sorting (Diamond, 2016; Weiwu, 2024; Almagro and Domínguez-Iino, 2024). I extend this line of research by showing that ethnic composition also directly affects local productivity, contributing to a broader literature on how neighborhood exposures shape long-run economic outcomes (Chetty, Hendren and Katz, 2016; Ioannides, 2012; Chyn and Katz, 2021; Chetty et al., 2025; Chyn, Collinson and Sandler, 2025).

This paper also contributes to the longstanding debate on the relationship between diversity and economic performance. In urban settings, Marshall (1890) emphasizes the gains from industrial specialization—supported empirically by Henderson, Kuncoro and Turner (1995)—while Jacobs (1969) and Glaeser et al. (1992) highlight the role of diverse urban environments in fostering innovation. A separate political economy literature studies how ethnic and, more broadly, social divisions affect economic outcomes through firm productivity (Hjort, 2014), social cohesion (Esteban, Mayoral and Ray, 2012; Guarnieri, 2025), trust (Alesina and La Ferrara, 2002), and public goods provision (Easterly and Levine, 1997; Alesina, Baqir and Easterly, 1999; Miguel and Gugerty, 2005; Dippel, 2014).[3] I bridge these two lines of work by showing that in mid-20th century Malaysia, deep ethnic divisions limited cross-ethnic spillovers and reduced the productivity gains from agglomeration, highlighting a new channel through which diversity may constrain development.[4]

Most studies on the economic impacts of forced migration focus on outcomes for the resettled (Becker et al., 2020; Nakamura, Sigurdsson and Steinsson, 2022; Sarvimäki, Uusitalo

[2]See Duranton and Puga (2004) and Rosenthal and Strange (2004) for reviews.

[3]See Alesina and La Ferrara (2005) for a review.

[4]Relatedly, Ananat, Fu and Ross (2013); Ananat, Shihe and Ross (2018) argue that segregation and limited interracial interactions reduce agglomeration benefits and widen racial wage gaps in U.S. cities.

and Jäntti, 2022; Carrillo, Charris and Iglesias, 2023; Michalopoulos et al., 2025).[5] I extend this literature by quantifying the aggregate effects of forced resettlement using a spatial general equilibrium model that incorporates impacts on both receiving and sending areas. I find that much of the aggregate output gain from the program reflects the relocation of Chinese to urban areas better aligned with their comparative advantage. This complements Bazzi et al. (2016), who emphasize that productivity gains from migration depend on the match between migrants' location-specific skills and the productivity fundamentals of destinations. I further show that, in this setting, forced relocation imposed welfare losses despite these economic gains, underscoring the importance of aligning migrants' location-specific preferences with the amenity fundamentals of destinations. These findings relate to broader work on the aggregate effects of place-based policies (Glaeser and Gottlieb, 2008; Fajgelbaum and Gaubert, 2020) and spatial misallocation (Bryan and Morten, 2019; Rossi-Hansberg, Sarte and Schwartzman, 2023).

Finally, this paper contributes to the literature on postwar development in East and Southeast Asia (Haggard, 1990; Amsden, 1992; Mundial, 1993; Wade, 2004; Lane, 2025). After World War II, the East Asian Tigers and "Look East" followers like Malaysia adopted industrial policies aimed at structural transformation. The strong non-agricultural externalities I estimate suggest that such policies may generate self-reinforcing productivity gains. The co-ethnic agglomeration effects may also reflect collective action within Chinese communities, which Dell, Lane and Querubin (2018) link to historical exposure to centralized state institutions.

The rest of the paper proceeds as follows. Section II provides the historical context. Section III describes the data. Section IV discusses the empirical strategy. Section V examines the local impacts of resettlement. Section VI lays out the model, which I estimate in Section VII.B. Section VIII evaluates counterfactual policies. Section IX concludes.

II Historical Background

In this section, I first describe the economic roles of the Chinese population in British Malaya and their relationship with the majority Malay population, which help interpret the agglomeration effects of Chinese clustering and the potential for spillovers to other groups. I then discuss the colonial government's objectives in resettling rural Chinese during the Malayan Emergency.

[5] An exception is Peters (2022), who examines the aggregate economic impact of refugee settlement in postwar Germany. Other studies examine political outcomes, including effects on social capital (Abel, 2019) state- and nation-building (Bazzi et al., 2019; Carlitz et al., 2024), and voting behavior (Kok et al., 2025). In the same Malaysian setting, Kok et al. (2025) find that proximity to New Villages reduced support for the pro-Malay nationalist party. For a broader review, see Becker (2022).

II.A Ethnic Chinese and Social Divisions in Peninsular Malaya

By the end of World War II, the population of Peninsular Malaya (British Malaya excluding Singapore) was 49% Malay, 39% Chinese, and 12% Indian and others (Appendix Table A.2). Chinese immigrants began arriving in British Malaya in the 19th century to work in tin mining and rubber plantations—the colony's main export industries (Purcell, 1947). Many later moved into urban and industrial sectors such as manufacturing and commerce, even though the economy remained largely agricultural by the late 1940s. In contrast, most Malays engaged in paddy rice and coconut cultivation (Ginsburg, 1958; Lee and Tan, 2000). By 1947, 81 percent of Malays worked in agriculture, compared to 54 percent of Chinese (Appendix Figure A.1). Industrial jobs were concentrated in towns, contributing to higher urbanization among the Chinese: 43 percent lived in urban areas by 1947, compared to just 11 percent of Malays (Del Tufo, 1947).

Beyond economic roles, Chinese and Malays differed in religion, culture, and language. Most Malays were Muslim, while Chinese were largely non-Muslim and often practiced customs that conflicted with Islamic norms, such as pork consumption. Intermarriage was rare: by 1980, only 1.3 percent of married Malays had Chinese spouses.

Chinese students typically attended vernacular schools before 1950, and fewer than 1 percent of Chinese spoke Malay in 1947. Malaysia's nation-building efforts began in the 1960s, which promoted Malay as the sole official language and introduced policies that suppressed vernacular schools. Malay fluency among Chinese improved over time, though only about a quarter were fluent by 1980. By the late 1980s, the two groups remained largely segregated, with limited interaction in schools or workplaces (Hirschman, 1986).

II.B The Briggs Plan: Emergency Resettlement

By the late 1940s, about one-third of the Chinese population lived in rural areas near the jungle fringe (Sandhu, 1964*b*, p. 150). Many of these rural Chinese—often referred to as "squatters" due to their lack of legal land titles—had been pushed from towns to the countryside during the Japanese occupation (1942–1945), which disrupted industrial employment.[6]

These Chinese squatters became a security concern during the Malayan Emergency (1948–1960), a guerrilla conflict between British forces and communist insurgents, who were predominantly ethnic Chinese. Many squatters supported the insurgents by providing food and information, and some directly participated in the communists' non-military support network. Their proximity to jungle areas, where the insurgents were based, made British

[6] The Squatter Committee Report, The National Archives in the UK (hereafter, "TNA"), CO 717/178. Many urban Chinese fled to avoid conflict or forced labor (Humphrey, 1971, pp. 39, 47; Loh, 1988, pp. 57–60). Early downturns in tin and rubber production during World War I and the Great Depression may have also contributed to this rural shift (Humphrey, 1971, p. 43; Loh, 1988, pp. 23, 27–29).

surveillance and control especially difficult (Humphrey, 1971, p. 49; Loh, 1988, pp. 106–107).

To address this, the British launched a large-scale resettlement program, the "Briggs Plan," that forcibly relocated squatters to secure areas when their original settlements were deemed unsafe. The goal was to deny insurgents access to supplies and intelligence (Briggs, 1951, p. 7). While the plan prioritized military objectives, it also sought to minimize dislocation and economic disruption (Humphrey, 1971, pp. 181–182).

State governments implemented the program rapidly, as speed was critical to prevent the insurgents from adapting (Sunderland, 1964, p. 161; Humphrey, 1971, p. 106). Most resettlements began in mid-1950 and were completed by the end of 1952 (Sandhu, 1964*b*, p. 159). Each state oversaw the process within its own territory, including site selection, land clearing, layout of house plots and roads, and issuance of removal notices. Squatters were typically given fewer than 14 days' notice. In areas where resistance or escape was expected, relocations were conducted at dawn without prior notice (Humphrey, 1971, p. 102). The military provided transport to the new sites, after which the original settlements were burned down (Sandhu, 1964*b*, p. 160).

By the end of the Emergency, approximately 573,000 people had been resettled into around 500 New Villages. Despite their name, only one-third of the New Villages were built from scratch; the rest were constructed around and integrated into existing settlements, contributing to a substantial increase in Chinese urbanization. Between 1947 and 1957, the share of Chinese living in urban areas increased from 43 percent to 73 percent (Sandhu, 1964*a*, p. 177).

Most villages followed a standard layout and included basic amenities such as a school, police station, and community center. Access and mobility were tightly controlled until the late 1950s, with dusk-to-dawn curfews and police checkpoints at village entrances (Humphrey, 1971, pp. 118, 358).

The resettled population was overwhelmingly Chinese (86%), with smaller shares of Malays (9%) and others (5%) (Sandhu, 1964*b*, p. 159). Around 60 percent of resettled Chinese had been farmers prior to resettlement, cultivating food crops and vegetables or tapping rubber (Sandhu, 1964*b*, p. 169). This agricultural share was slightly higher than that of the average Chinese (54%), consistent with squatters originating from more rural areas with greater suitability for agriculture. Still, it was lower than the average for Malays.

Over time, many former farmers shifted from agriculture to wage work in nearby urban sectors (Sandhu, 1964*b*, p. 169), despite having been allocated two acres of agricultural land in addition to a house lot.

Many of these villages continued to grow after the Emergency and still exist today. In 1972, the Ministry of Housing and Local Government of Malaysia reported 465 remaining

New Villages with a combined population of one million (Lee and Tan, 2000, p. 262); by 2005, about 450 remained.

II.C Determinants of Resettlement Density

Understanding the determinants of resettlement density is crucial for evaluating its impacts. The British outlined several criteria for the program (Humphrey, 1971, pp. 95–97), which was implemented in two stages: first, selecting village sites; second, relocating squatters while minimizing dislocation. This section discusses how these criteria, along with other ad hoc factors, shaped variation in resettlement density.

Security and defensibility. Security was the primary objective. Sites were ideally located near major roads or navigable waterways to facilitate police access (Dhu Renick, 1965, p. 9; Humphrey, 1971, p. 96). Most New Villages were located along main transportation routes (Figure 1). For defensibility, sites were preferably on elevated terrain and away from observation points, though elevation does not correlate with resettlement density in the data.

Land availability. Land acquisition costs also shaped siting decisions. Financial constraints led the British to prioritize state-owned land and low-value alienated land (Humphrey, 1971, p. 367). As a result, many villages were established on public rubber estates. In the analysis, I control for land use patterns, as different land types likely correlate with unobserved productivity.

Economic sustainability. Villages were ideally located on well-drained land with water access and agricultural potential (Dhu Renick, 1965, p. 9; Humphrey, 1971, p. 96). In practice, however, a shortage of trained personnel often led to poor siting decisions: many villages were prone to flooding or unsuitable for farming (Humphrey, 1971, p. 107).[7] An official report in 1954 found that 31% of sites were unlikely to remain viable after the Emergency (Corry, 1954), suggesting that economic considerations were not central to siting decisions. In line with this, I show that location characteristics related to productivity or amenities—such as ruggedness, rice suitability, and access to public goods—do not correlate with resettlement density once I control for proximity to transportation.

Proximity to squatter settlements. To reduce disruption, village sites were ideally located near original squatter communities (Humphrey, 1971, pp. 96–97). Most relocations occurred within 20 miles of the original settlements (Sandhu, 1964*b*, p. 160). The spatial distribution of squatters thus played a key role in shaping local resettlement density. Be-

[7]Examples include Batu Rakit/Pulai (Trengganu), sited on sandy wasteland; Jemaluang (Johore), located on tin tailings; and Kampung Abdullah (Johore), which regularly flooded (Sandhu, 1964*b*, p. 161). See also Notes on Planning and Housing Aspects of Resettlement and the Development of New Villages (Arkib Negara Malaysia, hereafter, "ANM", 1953).

cause these locations were self-selected before resettlement, I control for pre-resettlement population distribution in the analysis, as discussed in Section IV.

Idiosyncratic factors. Although the program aimed to minimize dislocation, limited information and shifting security conditions sometimes led to longer relocations. The British lacked reliable surveys of squatter settlements and continued to find new ones as the program progressed. A 1952 newspaper noted, "The Government had only the haziest idea of the number [of squatters]: it was first believed that there were 318,500 [to be resettled], but the total was nearer 500,000."[8] Because village capacity was determined during site selection stage based on initial estimates of nearby squatter populations, newly discovered squatters often had to be relocated to more distant sites. In addition, as insurgent activity shifted, areas once considered secure could become unsafe, causing the number of squatters requiring resettlement in a given area to change over time. These forces led some sites to receive more people than the surrounding Chinese population would predict.

Summary. The British emphasis on speed and security, combined with limited information and planning capacity, generated plausibly exogenous variation in population resettlement. Sites were selected from many similarly suitable locations along major roads, while poor planning and shifting insurgency risks led to idiosyncratic relocations. In Section IV, I discuss how I leverage these variations to construct a population shifter and examine the local effects of resettlement.

III Data

This section describes the data on the resettlement program and key economic outcomes. Additional details on the data sources and construction are provided in Appendix A.

III.A Emergency Resettlement

I measure the resettled population using a 1954 official report to the High Commissioner (Corry, 1954)—henceforth, the "Corry report"—compiled shortly after most resettlement had been completed. The main purpose was to assess the economic conditions of the villages, including their sustainability after the Emergency, and to estimate the number of remaining squatters requiring resettlement. It documents 439 New Villages established by 1954, including their names, populations, forms of local government, and qualitative descriptions. I use village locations from Baillargeon (2021), who geolocated 430 of these villages based on village names and states, covering roughly 540,000 people—94% of the estimated total resettled population by the end of the Emergency. I cross-validated the resettled population

[8] TNA: CO 1022/29, p. 63. Between 1952 and 1953, the estimated number needing resettlement remained at 80,000–90,000, even though over 150,000 had already been resettled during that period (Humphrey, 1971, p. 123).

figures using a 1958 survey conducted by the Malayan Christian Council (see Appendix A.1).

To measure key covariates required to specify counterfactual resettlement, I collected and digitized several historical maps: a 1942 road and railway map, a 1943 land utilization map, 1945 topographical maps, a 1947 population census map, and a 1955 "Black Areas" map showing regions under Emergency regulations due to communist activity. I measure initial squatter settlements by overlaying the land-use, population, and Black Areas maps.[9] Following historical accounts, I define a cluster of Chinese population in 1947 as a squatter settlement if it lies within a Black Area and within five kilometers of forest. Because the 1947 population map does not report ethnic composition at the settlement level, I use the composition from the county tabulation, assuming uniform Chinese shares across settlements within a county. I assess robustness to alternative definitions of squatter settlements in Appendix B.4.

III.B Outcome Data

Population. I digitized Malaysia's Census of Population at the county level for 1931, 1947, 1957, 1970, 1980, 1991, and 2000.[10] These census tabulations provide population counts by ethnic group for each county. To account for changes in county boundaries over time, I construct time-consistent borders based on the 1947 boundaries, grouping counties with overlapping geographies across years. I exclude nine counties with populations in 1947 but no reported populations in 1957 or 1970, as these are likely enumeration errors. The resulting baseline sample includes 777 counties, with a median width of 8–9 kilometers. For regressions using 1931 data, I create a separate set of 614 grouped counties based on the 1931 boundaries.

Employment. I measure county-level employment by ethnicity and occupation/industry from the 1980 and 1991 Population Censuses. Unlike the 1991 tabulations, the 1980 tabulations do not disaggregate employment by ethnicity within occupation or industry, so I impute the 1980 ethnic breakdown using ethnic shares from the 2% census microdata. For prewar aggregate employment data by industry and ethnicity, I use the 1947 tabulated census.

Income and education. I measure household incomes using both the 1980 census microdata and the Second Malaysian Family Life Survey (MFLS-2) from 1988–1989. The census provides data on household asset ownership but lacks direct measures of income or wages, while the MFLS-2 contains household earnings but has limited geographic coverage.[11] To

[9] See Appendix Figure A.2 for the maps. The Black Areas were relatively stable prior to 1955.

[10] I use "county" to refer to the administrative unit *mukim*, which is larger than a village. The 1931 Census was the first to document population by county.

[11] The MFLS-2 was conducted by RAND and Malaysia's National Population and Family Development Board. It provides demographic and socioeconomic data on nearly 3,000 households and is representative of

generate income measures for the broader population, I train a model using the MFLS-2 data to predict earnings based on observable household characteristics and apply it to the census microdata. The model includes household assets (e.g., automobile, motorcycle, refrigerator, TV), pairwise interactions of these assets, household size, and district fixed effects (Appendix Table A.3).

I measure educational attainment from the 1980 census microdata, which includes years of schooling and indicators of completion for primary, secondary, and higher education.

Migration. I measure migration using the 1980 tabulated census, which reports population by place of last residence across 66 districts (the administrative level above counties). I construct a matrix of bilateral migration flows between district pairs to estimate migration costs. I interpret the flows as spanning about 12 years, which is the average years at current residence reported in the 1980 census microdata. The microdata also provides indicators of internal/external migration status, which I use to distinguish incumbents from the initially resettled and subsequent migrants.

Firms. For historical manufacturing outcomes, I digitized the 1970 Directory of Manufacturing, which lists a universe of approximately 12,000 establishments in Peninsular Malaysia.[12] The directory provides each establishment's name, address, main products, industry classification, and employment size. I geocode establishments to counties based on the provided addresses.

For more recent firm outcomes by ethnic ownership, I extracted and cleaned data from the Orbis Historical Disk (accessed in 2024), following the procedure in Kalemli-Özcan et al. (2024). I identify each firm's ultimate owner(s) as those holding more than 50 percent ownership (or at least 25 percent if no single owner holds a majority), using the provided global ownership indicators. The final sample includes active firms with positive revenue, a Malaysian ultimate person owner, a valid NAICS industry code, and a location in Peninsular Malaysia. I restrict the sample to 2011–2015, the years with the most complete coverage. The resulting sample contains roughly 110,000 firms per year (see Appendix A for details).

IV Empirical Strategy

The resettlement of rural Chinese was not entirely random. This section explains how I isolate the quasi-random component of the program to construct a population shifter. I then assess balance in geographic and pre-resettlement characteristics to evaluate the plausibility of the identifying assumptions.

Peninsular Malaysia. However, it covers only 174 counties.

[12] All establishments were required to register under the Registration of Business Ordinance 1957.

IV.A Empirical Specification

I examine how the increase in Chinese population density from Emergency resettlement affected local economic outcomes, and how these effects varied across ethnic groups. I focus on county-level outcomes because counties are small enough to capture the fine variation in resettlement, yet large enough to account for local spillovers from agglomeration. The key challenge is converting site-level resettlement into an exogenous population shifter at the county level.

Consider the following reduced-form model:

$$Y_c = \beta ResettleDensity_c + \lambda \overline{ResettleDensity}_c + \gamma X_c + \alpha \mathbf{1}\left\{ResettleDensity_c > 0\right\} + \varepsilon_c. \quad (1)$$

I define county resettlement density as the inverse hyperbolic sine of resettled population per county area, standardized so that β is interpreted as the effect of a one-standard deviation increase in resettlement density (or "Higher Resettlement").[13] The main outcomes of interest in Y_c include total employment and average household income by sector and ethnic group.

I control for a set of pre-period characteristics in X_c that shaped resettlement decisions and could directly affect post-period outcomes. First, I include state fixed effects, as the program was implemented by state governments with varying economic and land policies. Second, because areas closer to transportation networks received more resettlement and had better market access, I control for pre-period road density and average distances to roads, rail stations, and the coast. Third, because the program aimed to minimize dislocation of resettled Chinese, and areas with larger initial Chinese populations were often more urbanized, I control for 1947 county population and the Chinese population share. Fourth, since resettlement sites were often located on state-owned rubber or tin estates, I control for pre-period land use shares for rubber plantations and mining. Finally, I control for county area to ensure comparisons are made across counties of similar size, as administrative area often correlates with urbanization and local government structure.[14]

Counties without any resettlement, such as dense west coast cities or remote jungle regions, were generally unsuitable for the program and may differ in unobserved economic

[13] In the structural estimation, I instrument for log population density using county resettlement density. I apply the inverse hyperbolic sine transformation to accommodate zeros and improve first-stage power. Appendix B.4 shows that results are robust to using a log transformation when restricting the sample to resettled counties.

[14] County size may also correlate with measurement error in resettlement density. Since a typical New Village is about 0.9 kilometers wide (Sandhu, 1964*a*, p. 173), I draw a half-kilometer radius buffer around each geolocated village center and assign population uniformly across overlapping counties. Most counties are much larger (median width of 8–9 kilometers), so villages typically fall within a single county. In smaller, often more urban counties, however, this procedure may introduce non-classical measurement error correlated with county size. Appendix B.4 shows robustness to excluding the largest and smallest counties.

potential from resettled counties. To ensure comparisons are made only among similar areas, I include an indicator for whether a county received any resettlement.[15]

Despite these controls, a concern remains that they may not fully account for non-random exposure to resettlement. Exposure depends not only on a county's own characteristics but also its position within the broader transportation network and population distribution. For instance, two counties with similar road densities could receive different resettlement flows if their neighbors varied in connectivity. Likewise, counties near more urbanized areas with larger initial Chinese populations may also receive more resettlement. These neighboring characteristics were correlated with market access (Donaldson and Hornbeck, 2016) and reflected pre-period selection of Chinese populations, which may be based on unobserved productivity or amenities. Such exposures, if not fully captured by X_c, could introduce omitted variable bias (Borusyak and Hull, 2023).

To address these concerns, I leverage institutional knowledge to specify counterfactual resettlement. Specifically, I isolate plausibly exogenous variation by controlling for the expected resettlement density, $\overline{ResettleDensity}_c$, defined as the average density across counterfactual resettlements.[16]

I construct $\overline{ResettleDensity}_c$ using a permutation procedure, conducted separately for each state to reflect the program's decentralized implementation by state governments. I approximate $\overline{ResettleDensity}_c$ by averaging the resulting county-level resettlement densities across 1,000 such permutations. Each permutation proceeds in three steps:

(i). Randomly and uniformly permute counterfactual New Village sites (denoted by i) across factually resettled counties, conditional on (i) distance to roads or rivers; (ii) land use; and (iii) the county's squatter population decile.[17]

(ii). Calculate the counterfactual number of people resettled to each site using the gravity

[15] Counties without resettlement are included in the sample to improve the precision of covariate estimates. Appendix B.4 shows that results are similar when restricting the sample to resettled counties only.

[16] Controlling for or re-centering by the expected resettlement helps purge potential omitted variable bias (Borusyak and Hull, 2023). In practice, they yield qualitatively similar results.

[17] Sites are permuted only among counties with any resettlement, consistent with the inclusion of the county resettlement indicator. Following historical accounts discussed in Section II, if no road is accessible within a 5-kilometer buffer but a river is, the permutation is instead conditional on distance to the nearest river. The baseline assumes that village sites were selected independently, as historical accounts emphasize time pressures and a shortage of qualified staff, which likely limited the capacity to coordinate site placement. Appendix B.4 examines robustness to alternative distance thresholds and to a minimum-spacing constraint that rules out arbitrarily close counterfactual sites.

equation:

$$\sum_{j=1}^{J} n_{j \to i} = \sum_{j=1}^{J} n_j \times \frac{d_{ji}^{-\psi}}{\sum_{s=1}^{I} d_{js}^{-\psi}}, \tag{2}$$

where n_j is the initial Chinese squatter population at origin j and d_{ji} is the distance between j and site i. The resettlement cost elasticity with respect to distance, ψ, governs how costly it was to relocate people to more distance sites. I calibrate $\psi = 0.65$ by minimizing the sum of squared differences between actual and predicted resettlement across villages, and assess robustness to alternative values in Appendix B.4.

(iii). Aggregate the counterfactual resettled population across sites in each county and divide by county area to obtain the counterfactual county resettlement density.

Figure 2 illustrates the procedure for the state of Johor using a single covariate: distance to roads. Panel A shows site selection for one actual village. The gray shaded area marks the potential locations with similar distance to the nearest road, from which the village's counterfactual location can be drawn. In practice, the suitable area is a subset of this region, based on additional covariates. Panel B overlays the counterfactual sites on top of the underlying squatter settlements (orange circles) that were to be resettled to these sites.

The identification assumptions are twofold. First, I assume the British did not select sites based on unobserved productivity or amenities, and were equally likely to choose locations with similar proximity to transportation networks and other observable characteristics. Second, although the British sought to minimize dislocation, limited information about the squatter distribution and the rapidly shifting pattern of communist activity caused actual resettlement numbers in a county to deviate from the dislocation-minimizing plan for reasons orthogonal to unobserved location fundamentals. See Appendix B.1 for a formal discussion of the assumptions.

I include county covariates alongside expected resettlement density for two reasons. First, it relaxes the identification assumptions by requiring only that residualized resettlement density be exogenous conditional on the covariates, making the strategy more robust to misspecification in expected resettlement.[18] Second, it improves statistical power.

Figure 3, Panel A, maps the New Villages against expected resettlement density. The spatial clustering and overlap between actual village locations and expected density reflect

[18] For example, the 1947 census map used to measure pre-period population lacks ethnic breakdowns, so I impute Chinese squatter populations n_j using county-level Chinese shares. To mitigate concerns about insufficient control for pre-period population patterns, I include covariates in X_c related to population and transportation, and assess robustness by varying the specification of expected resettlement density and adding further controls more flexibly (see Appendix B.4).

the British strategy of targeting areas with denser road networks and larger squatter populations. Panel B shows the residual variation after partialling out expected resettlement density and the covariates in Equation (1). This identifying variation comes from counties with similar access to transportation networks and nearby squatter populations but different realized resettlement densities.

For individual and household outcomes, I estimate:

$$Y_{iec} = \beta_e ResettleDensity_c + \lambda_e \overline{ResettleDensity}_c + \gamma_e X_{ic} + \alpha_e \mathbf{1}\{ResettleDensity_c > 0\} + \varepsilon_{iec}, \tag{3}$$

where i denotes individuals or households and e denotes ethnic groups. The controls X_{ic} include the county covariates from Equation (1), and, in some specifications, additional individual or household characteristics to improve precision.

I use the Poisson Pseudo Maximum Likelihood (PPML) estimator to estimate elasticities for outcomes such as the number of establishments or employment, which may be zero in some counties or industries (Silva and Tenreyro, 2006). PPML captures both extensive and intensive margins and is invariant to the units of the dependent variable (Chen and Roth, 2023).

I report Conley standard errors that account for spatial correlation within a 30-kilometer radius (Conley, 1999). The standard errors are similar to those clustered at the district level (66 districts) and increase modestly (by 10–15%) when the cutoff is extended to 50 kilometers.

IV.B Pre-Characteristic Balance

This section assesses whether county resettlement density is orthogonal to pre-resettlement characteristics—such as geography, amenities, and economic activity—as expected if the empirical strategy successfully isolates quasi-random variation.

Table 1 reports the relationship between resettlement density and pre-period characteristics. Columns 1–4 examine geographic features related to productivity, including elevation, terrain ruggedness, and suitability for rice and coconut cultivation—the main food crops in Malaysia. Columns 5–8 examine indicators of state capacity and access to public and social amenities: distance to the nearest police station, post/telegraph office, mosque and Chinese temple. In particular, proximity to mosques and Chinese temples reflects access to ethnic community institutions, and their locations may capture underlying patterns of ethnic segregation or tension. Columns 9–12 assess pre-existing economic activity: land use for rubber and mining—the two major export industries—and distance to industrial facilities (e.g., engineering shops, chemical/power plants, and rubber and tin processing plants) and major cities such as Singapore, George Town, Malacca, Ipoh, and Kuala Lumpur.

Panel A shows that raw within-state correlations align with the program's strategy, which prioritized areas along major transportation networks and in more urbanized regions. Counties with higher resettlement density were closer to public services (Columns 5–6), industrial facilities (Column 11), and major cities (Column 12). These areas also had more rubber cultivation (Column 9), consistent with contemporary reports that many resettlement sites were located on state-owned rubber estates. Notably, these counties were not more suitable for agriculture; if anything, they were slightly less so, despite agricultural suitability being a stated criterion in site selection (Columns 3–4). This finding is consistent with qualitative evidence that emphasized security and speed over economic factors in resettlement.

Panel B adds controls for key resettlement determinants, including road networks and initial population distributions. With these controls, location characteristics are largely balanced, suggesting that baseline covariates account for much of the potential bias from broader network effects that expected resettlement density is intended to capture.

Panel C additionally controls for expected resettlement density, which captures urbanization and population patterns extending beyond county boundaries. For example, rubber land share becomes balanced only after including expected resettlement density, perhaps due to squatter settlements on nearby estates that are not captured by county covariates. To mitigate concerns about unobserved land use patterns, I directly control for rubber and mining land use throughout the paper and examine robustness to other land types in Appendix B.4.[19]

The estimated correlations are economically small. For example, a one-standard-deviation increase in resettlement density is associated with a 0.05 metric ton per hectare decline in rice suitability—about 4% of the national average of 1.2 tons per hectare.[20] Overall, counties with higher (residualized) resettlement density had similar geography, amenities, and economic activity before the program.

V Local Impacts of Resettlement

This section examines the local impacts of resettlement in receiving areas over the following decades. I first show how it persistently shaped the population distribution, then examine its economic impacts across ethnic groups and explore potential mechanisms.

[19] A related concern is that the economic composition of the resettled population may differ across areas with varying resettlement density, perhaps due to unobserved economic geography not captured by county covariates or expected resettlement density. Appendix B.2 shows that observable characteristics of the resettled are also balanced.

[20] Some covariates, such as rice suitability, become statistically insignificant due to larger standard errors rather than smaller point estimates. I assess robustness to directly controlling for these covariates in Appendix B.4.

V.A Population Growth and Ethnic Composition

Figure 4 shows changes in population by ethnic group (Panel A) and Chinese population share (Panel B) from a one standard deviation increase in county resettlement density. Prior to resettlement, population trends from 1931 to 1947 were similar across counties with varying resettlement density, indicating that these areas did not initially differ in growth or labor demand. After resettlement, counties with higher resettlement density saw a sharp and persistent rise in both the Chinese population and its share of the total population. In the short run, only the Chinese population increased, consistent with the program targeting ethnic Chinese. Over time, non-Chinese populations also began moving into these counties.

Table 2 documents these changes. From 1947 to 1957, a one standard deviation increase in county resettlement density raised total population by 9.4% and the Chinese share by 4.8 percentage points (Column 1). Since one standard deviation corresponds to 13.6% of the 1947 population, this implies that each 1% resettled mechanically raised local population by 0.69%. This direct effect accounts for nearly 80 percent of the population growth in resettled counties during the decade. In an average county, the Chinese share in 1957 was 22 percent, implying that resettlement increased the Chinese share by about 21.8%.

After mobility restrictions were lifted in 1960, population increased further in higher-resettlement counties, and the higher Chinese share persisted. By 2000, these counties had 18% greater population density and a 4.1 percentage point higher Chinese share (Column 3). These long-run changes were driven by internal migration, not fertility: Chinese residents in counties with higher resettlement were more likely to have voluntarily moved in after 1960, with no significant difference in fertility rates (Appendix Table A.5).

To help interpret the economic impacts examined in the next section, note that by 1980, a one standard deviation increase in resettlement density raised total population by 11% and the Chinese share by 5 percentage points (Column 2). Given a baseline Chinese share of 0.19, this implies the resettlement increased the share from 0.19 to 0.24—about a 26% increase.

V.B Economic Structure and Household Income

The influx of Chinese constituted a skill-biased labor supply shock, given their comparative advantage in industrial and urban sectors. This section examines how resettlement reshaped the local economic structure and household income in subsequent decades.

Table 3, Panel A, shows that in 1980–1991, counties with higher resettlement density had 11% more agricultural employment (Column 1) and 29% more non-agricultural employment (Column 2).[21] Given that about 60 percent of the resettled Chinese were previously farmers,

[21] The regressions pool data from 1980 and 1991 and include year fixed effects and their fully saturated

the larger increase in Chinese non-agricultural employment suggests that some shifted out of agriculture in these counties (Panel B). Panel C shows a similar shift for non-Chinese populations: although their total population increased slightly (by 6%), their agricultural employment did not rise (Column 1), while their non-agricultural employment was 24% higher (Column 2).

Table 4 shows that average household incomes were higher in more resettled counties, particularly for Chinese households. In 1980, Chinese households in counties with higher resettlement density earned 11% more than their counterparts in less resettled counties (Panel A, Column 1).[22] For non-Chinese households, the income gain was smaller and statistically insignificant at 4% (Column 2).

Panels B and C examine income effects by sector of employment, based on the industry of the household head.[23] Chinese households in more resettled counties earned 7% more in agriculture and 13% more in non-agriculture, compared to their counterparts in less resettled counties (Column 1). Non-Chinese households saw modest income gains only in non-agriculture, with a 5% increase that was not statistically significant (Column 2).[24] The ethnic gap in income gains is statistically significant (Column 3).

V.C Labor Matching, Specialization, and Ethnicity-Based Agglomeration

The clustering of Chinese may have generated productivity gains through agglomeration externalities. One channel discussed in the literature is labor matching, where a denser workforce improves match quality between workers and firms (Marshall, 1890). This section examines this mechanism in more detail.

Table 5, Column 1, shows that by 1980, Chinese individuals in counties with higher resettlement were more likely to participate in the labor force (Panel A). Among those employed, their industries and occupations in 1991 were more concentrated, as measured by the Herfindahl-Hirschman Index (HHI) (Panels B and C), consistent with greater specialization.[25] They were also more likely to work in managerial occupations: the share of Chinese

interactions with the controls to estimate average effects across the two years. Throughout the paper, I use "agricultural sector" or simply "agriculture" to refer to agriculture, hunting, forestry, fishing, and mining, though it primarily reflects agriculture. Within the non-agricultural sector, employment effects were similar across secondary and tertiary industries (Appendix Table A.6) and slightly larger in finance and business activities (Appendix Figure A.6).

[22] Appendix Table A.7 shows similar patterns across multiple household assets, which are strong predictors of income.

[23] For households where the head lacked valid employment information or was unemployed, I use the eldest household member with valid employment.

[24] Within non-agriculture, income gains are similar across the secondary and tertiary sectors.

[25] The occupational HHI for ethnic group e in county n is defined as $HHI^e_{n,occ} \equiv \sum_k (L^e_{nk}/L^e_n)^2$, where L^e_{nk} is total employment of group e in occupation k; and $L^e_n = \sum_k L^e_{nk}$ is total employment of group e in county n. Higher HHI values indicate greater concentration. Occupations and industries are measured at the 1-digit level based on the 1991 tabulated census.

employed as legislators, senior officers, or managers was 1.2 percentage points higher in counties with higher resettlement—an increase of 21% relative to the 5.6 percentage points national average (Panel D). In contrast, non-Chinese individuals in more resettled counties showed no significant differences in these labor market outcomes (Column 2).

Table 6 shows that educational attainment was also higher in counties with higher resettlement density. By 1980, Chinese individuals in these counties had 0.4 additional years of schooling (8%), were 3.7 percentage points (7%) more likely to complete primary education, and 4.1 percentage points (14%) more likely to complete secondary education (Column 1). Non-Chinese individuals also had higher education, but to a lesser extent (Column 5). These improvements were driven entirely by younger cohorts under age 50 (Columns 2–3), who had not completed schooling by the time of resettlement.

Table 7 shows that manufacturing firms in industries more reliant on Chinese labor entered and scaled up in counties with higher resettlement density. By 1970, these counties had 24% more manufacturing establishments in industries where ethnic Chinese made up over 80% of pre-resettlement employment (Column 1).[26] Although most manufacturing firms at the time were owner-operated, firms in these county-industries were 2 percentage points (5%) more likely to employ at least one full-time worker (Column 2), consistent with larger scale and higher productivity.

Finally, Table 8 shows that Chinese-owned firms benefited more from higher Chinese density than non-Chinese-owned firms. Using firm-level data from Orbis (2011–2015), I classify ownership by the ethnicity of the ultimate owner (see Appendix A.1 for details). Restricting to domestic firms with a Malaysian individual as the ultimate owner, I find that counties with higher resettlement had 52% more Chinese-owned firms and 31% more non-Chinese-owned firms (Panel A). Chinese-owned firms in these counties were also larger, earning 19% more revenue than their counterparts in the same NAICS 2-digit industries in less resettled counties (Panel B, Column 1). In contrast, revenue gains for non-Chinese-owned firms were smaller and statistically insignificant at 8% (Column 2).

V.D Social Divisions as Barriers to Cross-Ethnic Spillovers

The previous sections showed that Chinese individuals benefited disproportionately from higher Chinese density, suggesting limited agglomeration spillovers across ethnic groups. One possible explanation is that ethnic divisions constrained inter-group contact and productive interaction. This section explores this mechanism by examining how resettlement effects vary with measures of social separation between ethnic groups.

[26] In 1947, only food products, wood products, textiles, and miscellaneous manufacturing had less than 80% Chinese employment; see Appendix Figure A.7. Regressions include industry fixed effects to absorb industry-specific unobservables.

Table 9 examines how residential segregation and linguistic barriers correlated with the income gains from resettlement. Panel A interacts county resettlement density with the 1947 district-level ethnic dissimilarity index (Duncan and Duncan, 1955), a commonly used measure of segregation in the economics literature (e.g., Cutler, Glaeser and Vigdor (1999); Adukia et al. (2024)).[27] In districts with no initial dissimilarity, income gains from resettlement were sizable and similar across groups. Income gains fell sharply in more segregated districts, especially for non-Chinese households. Moving from zero to the median level of dissimilarity, the income gains for non-Chinese households largely disappear (Column 2).

Panel B uses a distance-based measure of segregation in the spirit of White (1983), following the approach in Harari (2024). Leveraging the fact that Malays were predominantly Muslim, while other ethnic groups followed different religions, I construct a distance segregation index based on the relative spatial separation between mosques and non-Muslim temples. The index is defined as the average distance between mosques and temples, normalized by the average distance between any two religious sites in the county (see Appendix A.2 for details). A value above one indicates that religious (and hence ethnic) groups lived disproportionally farther apart.

Interacting this index with resettlement shows a similar pattern: in more religiously segregated areas, income gains for non-Chinese households were smaller. In counties where mosque-to-temple distances were twice the average distance between any two religious sites, income gains for non-Chinese fell by about a quarter (Column 2). In the least segregated areas, Chinese households still experienced slightly larger income gains than non-Chinese households, but the difference was not statistically significant.

Panel C considers linguistic barriers. Despite nation-building efforts that began in the late 1960s, only about 20 percent of Chinese adults were fluent in Malay by 1980. Areas with slower Malay adoption may reflect underlying ethnic divisions and limited cross-ethnic interaction. I find that in counties where fewer than 10 percent of Chinese adults spoke Malay fluently, non-Chinese households saw no significant income gains from resettlement, and possibly even losses (Column 2). In contrast, in counties where more than 10 percent of Chinese were Malay-fluent, non-Chinese households gained 8.6% more.

V.E Alternative Mechanisms and Robustness

Beyond ethnicity-based agglomeration, resettlement may have affected outcomes through other channels that disproportionately benefited the Chinese. This section examines alternative mechanisms and assesses the robustness of the main results.

[27]This index reflects the share of individuals from either group (Chinese or non-Chinese) who would need to relocate to different counties for all counties in a district to have the district's overall ethnic composition.

Alternative mechanisms. Because Chinese population increased by nearly 40 percent in counties with higher resettlement, a natural question is whether the improved outcomes for Chinese in these counties were driven by local incumbents or the resettled population. Although the 1980 census does not identify resettled individuals, it does report whether a person is a migrant—an indicator that should, in principle, capture both those initially resettled and subsequent voluntary movers.[28] In Appendix B.3, I show that the positive effects of resettlement on household incomes remain qualitatively similar when restricting the sample to non-migrant households (those without any migrant household members). This suggests that income gains were not concentrated among incumbents or migrants, but were shared relatively evenly across the Chinese population in counties with higher resettlement.

More broadly, since the empirical strategy compares destinations with varying resettlement densities—rather than outcomes relative to the origin counties—the extent to which the resettled population could mechanically explain the observed gains depends on their characteristics relative to local incumbents. As discussed in Section II, about 60% of the resettled Chinese were previously engaged in food or rubber cultivation—an agricultural share slightly higher than that of the average Chinese. Survey data suggest that they were also generally less educated and less wealthy than the average Chinese (Appendix Table A.8). Thus, if anything, the inflow of resettled individuals would have shifted the local economy toward agriculture and lowered average Chinese income, not the other way around.

I also consider other potential channels through which the program may have affected outcomes, such as land ownership allocated to resettled households and public infrastructure investments, including roads and Chinese schools. While these factors may have played a role, the evidence suggests they are unlikely to be the main drivers of the results. For instance, consistent with historical accounts that the British built roads within New Villages and connected remote ones to major transport networks, I find a slightly higher density of local roads by 1961 in counties with higher resettlement. The magnitude, however, is modest—about 400 meters closer to local roads—and estimated income gains remain similar when directly controlling for road access. See Appendix B.3 for further discussion.

Robustness. Appendix B.4 shows that the main results are robust to a range of alternative specifications and sample restrictions. First, I assess the role of expected resettlement density by excluding it from the baseline specification, which reduces the estimated effect on agricultural employment. This suggests that counties with similar baseline covariates but higher exposure to resettlement may have been slightly more urbanized or industrialized (or closer to these areas) prior to the program. Next, I show that the estimates remain stable

[28] Internal migrants are defined as individuals whose last previous residence was in a different "locality" than where they were enumerated, where locality refers to a village (kampung) or local authority area.

when controlling for alternative measures of expected resettlement density or when using a logarithmic transformation of resettlement density.

Second, while expected resettlement density helps account for omitted variable bias from broader network effects of nearby roads and population, the results are robust to additionally controlling for these neighboring characteristics directly. Third, I further show robustness to flexibly controlling for topographic and land-use patterns, which may correlate with drainage and land prices considered by the British during site selection but which I do not directly observe. Finally, the results are robust to excluding counties with the largest or smallest areas, high-density prewar towns, and counties with extreme resettlement densities. See Appendix B.4 for details.

V.F Discussion

The resettlement of Chinese persistently reshaped the distribution of population and economic activity. Industries where the Chinese had a comparative advantage expanded in areas with higher Chinese density, generating productivity externalities that spurred internal migration and the structural transformation of local economies.

Contrary to the neoclassical prediction that marginal entrants into non-agriculture would be less productive than incumbents, thereby lowering average wages, the opposite occurred: non-agricultural wage rose in counties with higher resettlement, and the agricultural employment share declined. This suggests external economies of scale in the non-agricultural sector. By contrast, agricultural productivity improved little in more resettled counties—consistent with local diminishing returns to labor in agriculture, given fixed land supply.

A key mechanism behind the scale economies appears to be improved labor matching. A denser Chinese workforce may have reduced search frictions and facilitated better worker-firm matches (Dauth et al., 2022), raising the returns to specialization and education (Kim, 1989). Consistent with this, Chinese workers in higher-resettlement counties were more educated, took on more specialized roles, and were more likely to hold managerial positions. Chinese-owned firms in these counties were also larger, which indicates higher productivity. Together, these patterns point to greater division of labor and higher organizational complexity—features typical of urban economies.

However, the benefits of higher Chinese density varied across ethnic groups. Non-Chinese communities benefited less, and the evidence suggests that ethnic segregation and social divisions may have limited the scope for cross-ethnic spillovers. Indeed, non-Chinese households experienced larger productivity gains from resettlement in areas where pre-existing ethnic divisions were milder.

These divisions may have hindered agglomeration spillovers through multiple channels.

One such channel is labor market discrimination: many industrial firms were small and Chinese-owned, and may have favored coethnic hiring to avoid the costs of interethnic tensions (Hjort, 2014).[29] Search frictions may also have been higher between workers and firms of different ethnicities due to limited overlap in social networks. Cultural and linguistic barriers may have constrained learning spillovers and hindered supplier-client linkages across ethnic lines (Boken et al., 2024).

Motivated by prior work documenting a hump-shaped relationship between diversity and economic performance (Ashraf and Galor, 2013), I test for nonlinear effects by adding the square of county resettlement density in the income regression (Appendix Table A.18). The estimated coefficient on the squared term is small and statistically insignificant, suggesting that the relationship is approximately linear over the observed range. This pattern is consistent with Malaysia lying to the right of the optimal diversity level estimated in Ashraf and Galor (2013), where additional local diversity tends to reduce economic performance.

VI A Quantitative Model of Migration, Occupation, and Agglomeration

The cross-sectional analysis shows *relative* differences in economic composition and income across receiving counties with varying resettlement density. But resettlement may have also generated *aggregate* economic gains by relocating Chinese squatters from remote areas to locations with better market access, while freeing up rural lands in origin counties. Assessing the welfare implications, however, requires weighing these gains against the disutility from coercive relocation, which disregarded individuals' location preferences.

In this section, I develop a spatial general equilibrium model that extends Allen and Donaldson (2022) and Peters (2022) to quantify the program's aggregate economic and welfare impacts. Motivated by the heterogeneous effects of resettlement across sectors and ethnic groups, the model features agglomeration externalities that vary by sector and local ethnic composition. A key innovation is that productivity spillovers depend not only on overall population size but also on own-group share, such that an increase in Chinese population can affect the productivity of Chinese and non-Chinese workers differently.

The model incorporates migration and occupational choices—the main adjustment margins following resettlement—in a Roy (1951)-type framework. Individuals from different ethnic groups vary in location preferences and sectoral comparative advantage. Some locations may be more appealing to certain groups due to historical amenities, and groups may differ in productivity across sectors and space.

[29] For example, Lee and Khalid (2016) conducted a resume audit study in Malaysia in 2011 and found that Chinese applicants received more callbacks than equally qualified applicants from other ethnic groups, particularly from Chinese-owned firms. Other studies note that some firms used Mandarin language requirements as a pretext to exclude non-Chinese applicants.

Regions are connected through migration and trade. Resettlement shifts individuals' initial locations prior to migration decisions, independently of fixed location amenities and productivity. This initial population distribution continues to shape equilibrium outcomes due to mobility and trade frictions.

VI.A Environment

The model features N regions (counties in the data) and two sectors $k \in \{A, M\}$: Agriculture (A) and Non-Agriculture (M). Individuals are characterized by two ethnic groups $e \in \{c, m\}$: Chinese (c) and Malays (m), and are initially endowed with a location. They choose where to migrate after drawing a regional taste shock. After moving, they draw productivity for each sector and decide whether to work in agriculture or non-agriculture. Finally, production and consumption take place.

Production. Each region n produces a unique good in each sector, following Armington (1969). In each region-sector, a continuum of perfectly competitive firms produces a homogeneous regional variety. Production exhibits constant returns to scale with labor as the only input. Output is given by $Q_{nk} = H_{nk}$, where H_{nk} denotes the total amount of labor (in efficiency units) employed in sector k of region n, summed across ethnic groups.[30] Agglomeration economies are modeled through their effect on labor efficiency, which I discuss in sectoral labor supply.

Firms in sector k and region n choose labor H_{nk} to maximize profit, taking the local sectoral wage (per efficiency unit) and output prices as given. In equilibrium, no-arbitrage implies that the price of sector-k goods produced in region n and sold in region r is given by $p_{nrk} = (\tau_{nr}/\tau_{nn})p_{nnk}$, where p_{nnk} is the price of sector-k goods sold locally; $\tau_{nr} \geq 1$ is the iceberg trade cost between regions n and r; and τ_{nn} is the within-region trade cost.[31]

Under perfect competition, firms earn zero profit in equilibrium. This implies $w_{nk} = p_{nnk}/\tau_{nn}$, where w_{nk} is the wage per efficiency unit in sector k of region n.

Consumption. Individuals of ethnicity e living in region n derive utility from consuming agricultural and non-agricultural goods and from the local amenity in n:

$$U_n^e(C_A, C_M) = a_n^e \left(\frac{C_A}{\alpha}\right)^{\alpha} \left(\frac{C_M}{1-\alpha}\right)^{1-\alpha}$$

$$C_k = \left(\sum_{r=1}^{N} c_{rk}^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}},$$

[30] One efficiency unit of labor contributes equally to output regardless of ethnicity, although Chinese and Malay individuals may differ in their average efficiency due to, say, differences in education or skills.

[31] Without loss of generality, within-region trade can be costly ($\tau_{nn} > 1$). An increase in τ_{nn} lowers local wages and is isomorphic to worse production fundamentals.

where $\sigma > 1$ is the constant elasticity of substitution across regional varieties, assumed identical across sectors. The term a_n^e captures the (endogenous) amenity value for individuals of ethnicity e in region n.[32]

Utility maximization implies that the indirect utility of a group-e individual in region n with income y_n^e is $a_n^e y_n^e / P_n$, where $P_n \equiv P_{nA}^{\alpha} P_{nM}^{1-\alpha}$ is the ideal price index in region n, and $P_{nk} \equiv (\sum_{l=1}^{N} \tau_{ln}^{1-\sigma} w_{lk}^{1-\sigma})^{1/(1-\sigma)}$ is the local price index for sector-k goods.

Sectoral labor supply. Individuals supply labor inelastically and earn income based on their heterogeneous productivity. After migrating, each individual draws a vector of efficiency units in agriculture and non-agriculture, denoted $\Lambda_i = (\Lambda_{iA}, \Lambda_{iM})$, where Λ_{ik} is the the effective labor individual i provides if employed in sector k.[33] These draws follow a Fréchet distribution:

$$F_{nk}^e(\Lambda) = \exp\left(-\phi_{nk}^e \Lambda^{-\theta}\right),$$

where ϕ_{nk}^e is the scale parameter, capturing the average productivity of ethnicity e in sector k of region n. This parameter reflects both ethnic comparative advantage and location-sector fundamentals, allowing, for example, Chinese to sort into industrial regions and Malays into agricultural regions. The shape parameter θ governs the dispersion of individual productivity, with higher θ indicating less heterogeneity.

Productivity depends on both idiosyncratic ability and local agglomeration externalities. Let λ_{ink}^e denote the effective labor efficiency of individual i of ethnicity e working in sector k and region n. It is given by:

$$\lambda_{ink}^e = \Lambda_{ink}^e f_\lambda(L_{nk}^e, L_{nk}^{e'}),$$

where Λ_{ink}^e is an idiosyncratic skill draw, and $f_\lambda(\cdot)$ captures how agglomeration spillovers vary with local population distribution.

While agglomeration effects could in principle take flexible forms, the data lacks sufficient variation to estimate them non-parametrically. I therefore parameterize $f_\lambda(\cdot)$ as a function of sectoral employment size and ethnic composition:

$$f_\lambda(L_{nk}^e, L_{nk}^{e'}) \equiv (L_{nk})^{\gamma_k} \left(\frac{L_{nk}^e}{L_{nk}}\right)^{\gamma^e},$$

where γ_k and γ^e govern the strength of productivity spillovers with respect to sector size and

[32] Technically, a_n^e varies at the individual level (e.g., a_{in}^e). Individual subscripts are omitted for simplicity.

[33] The assumption that individuals learn their sectoral productivity after migrating is not essential. What matters is that ethnic groups can differ in their comparative advantage across regions—for instance, Chinese individuals may be more likely to work in non-agriculture in some regions than others. Whether this arises from productivity revealed before or after migration is not important for the counterfactual analysis.

own-group share, respectively.

This functional form extends naturally from the urban literature, which typically assumes a constant elasticity with respect to local employment.[34] When $\gamma^e = 0$, the formulation reduces to the standard case in which productivity depends solely on employment size. A positive γ^e implies that, holding total population size fixed, a higher own-group share raises productivity—capturing additional benefits from within-group interaction.[35] A negative γ^e reflects gains from diversity, where a larger share of the other group increases productivity.

In a partial equilibrium with prices fixed, the elasticity of group-e productivity in sector k and region n with respect to own-group population is:

$$\frac{\partial \ln \lambda^e_{nk}}{\partial \ln L^e_n} = \gamma_k \frac{L^e_{nk}}{L_{nk}} + \gamma^e \left(1 - \frac{L^e_{nk}}{L_{nk}}\right). \tag{4}$$

The marginal productivity effect of increased own-group population is a weighted average of γ_k and γ^e, where the weights correspond to the share of group e in sector k. When ethnic group e accounts for most of the local employment in the sector, the elasticity approaches γ_k; when group e' dominates, it approaches γ^e.

Cross-ethnic spillovers are also allowed. The elasticity of group-e' productivity ($e' \neq e$) with respect to the size of group e is:

$$\frac{\partial \ln \lambda^{e'}_{nk}}{\partial \ln L^e_n} = (\gamma_k - \gamma^e) \frac{L^e_{nk}}{L_{nk}}. \tag{5}$$

The marginal effect is positive when sectoral spillovers are stronger than own-group spillovers ($\gamma_k > \gamma^e$). In contrast, when own-group spillovers dominate ($\gamma_k < \gamma^e$), an increase in group-e population reduces the productivity of group e'. The magnitude of the effect scales with group e's local employment share in the sector.

Given the Fréchet distributed efficiency draws, the share of group-e individuals in region n who work in sector k is given by:

$$\pi^e_{nk} = \phi^e_{nk} \left(\frac{w^e_{nk}}{\bar{w}^e_n}\right)^{\theta}, \tag{6}$$

[34] Kline and Moretti (2014) provides empirical support for constant size elasticities in the context of the Tennessee Valley Authority in 1930s U.S. I also assumes a constant elasticity with respect to ethnic share, motivated by the linearity result in the income regression.

[35] Another interpretation of a positive γ^e is that a higher own-group share mitigates productivity losses from inter-group conflict.

where w^e_{nk} is the effective wage per efficiency unit (adjusted for externalities):

$$w^e_{nk} = w_{nk} \left(L_{nk}\right)^{\gamma_k} \left(\frac{L^e_{nk}}{L_{nk}}\right)^{\gamma^e}, \tag{7}$$

and the average wage (up to a scale) for group e in region n is:

$$\bar{w}^e_n = \left(\phi^e_{nA} \left(w^e_{nA}\right)^{\theta} + \phi^e_{nM} \left(w^e_{nM}\right)^{\theta}\right)^{1/\theta}.$$

Migration. Each individual starts with an initial location and chooses where to migrate after drawing a regional taste shock, subject to a moving cost. At the time of migration, individuals know their ethnicity but have not yet observed their sector-specific skills. Each individual i of ethnicity e draws an idiosyncratic taste u^e_{in} for each region n from an ethnicity-specific Fréchet distribution:

$$F^e_n(u) = \exp\left(-\bar{a}^e_n u^{-\nu}\right),$$

where the scale parameter $\bar{a}^e_n$ captures exogenous, ethnicity-specific amenities in region n, and ν is the shape parameter, with a higher ν implying less dispersion in location preferences across individuals.

The realized amenity value for individual i of ethnicity e in region n is:

$$a^e_{in} = u^e_{in} f_a(L^e_n, L^{e'}_n),$$

where, following the specification for productivity spillovers, I parameterize amenity spillovers $f_a(\cdot)$ as a function of local population size and own-group share:

$$f_a(L^e_n, L^{e'}_n) = \left(L_n\right)^{\beta} \left(\frac{L^e_n}{L_n}\right)^{\beta^e}.$$

Here, β captures congestion effects, while β^e governs the strength of homophily—a preference for living near co-ethnics—which may also reflect ethnic tensions.

The indirect utility of individual i from group e, moving from origin r to destination n, is given by:

$$V^e_{irn} = \eta^{-1}_{rn} a^e_{in} \Gamma_\theta \bar{w}^e_n P^{-1}_n,$$

where η_{rn} denotes the iceberg migration cost from r to n, and $\Gamma_\theta \bar{w}^e_n P^{-1}_n$ is the real wage in region n.[36]

Since V^e_{irn} is a Fréchet random variable scaled by a constant, it is itself Fréchet distributed. This implies that the share of individuals from ethnicity e initially in r who choose to move to n is:

$$m^e_{rn} = \frac{\left(\eta^{-1}_{rn} V^e_n\right)^{\nu}}{\sum_{l=1}^{N} \left(\eta^{-1}_{rl} V^e_l\right)^{\nu}},$$

where the mean utility of residing in region n for ethnicity e is:

$$V^e_n = \left(\bar{a}^e_n\right)^{1/\nu} \left(L_n\right)^{\beta} \left(\frac{L^e_n}{L_n}\right)^{\beta^e} \bar{w}^e_n P^{-1}_n. \tag{8}$$

Thus, the bilateral migration flow of ethnic group e from r to region n is:

$$L^e_{rn} = \eta^{-\nu}_{rn} \times \frac{\check{L}^e_r}{(\Pi^e_r)^{\nu}} \times \frac{L^e_n/\bar{L}}{(\mathcal{V}^e_n)^{-\nu}}, \tag{9}$$

where the two migration market access terms are defined as:

$$\Pi^e_r \equiv \left(\sum_{l=1}^{N} (\eta^{-1}_{rl} V^e_l)^{\nu}\right)^{1/\nu}, \tag{10}$$

$$\mathcal{V}^e_n \equiv V^e_n \left(L^e_n/\bar{L}\right)^{-1/\nu}, \tag{11}$$

and $\bar{L}$ denotes total population in the economy, normalized to 1.

The term Π^e_r represents the overall value for individuals from group e to move out of origin region r, while the inverse of $\mathcal{V}^e_n$ captures the value of moving into destination region n. In the trade literature, these terms are referred to as outward and inward migration market access, respectively (Anderson and van Wincoop, 2003).

Trade. Goods flowing from region n to r face exogenous iceberg costs, denoted $\tau_{nr} \geq 1$, where $\tau_{nr} = 1$ represents frictionless trade. Given CES preferences, the trade flow expenditure on sector-k goods from r to n (with goods flowing from n to r), denoted by X_{nrk}, follows the standard gravity equation:

$$X_{nrk} = X_{rk} \frac{\tau^{1-\sigma}_{nr} (w_{nk})^{1-\sigma}}{\sum_{l=1}^{N} \tau^{1-\sigma}_{lr} (w_{lk})^{1-\sigma}},$$

[36] The term $\Gamma_\theta \equiv \Gamma(1 - 1/\theta)$ is a constant, where $\Gamma(\cdot)$ denotes the Gamma function.

where $X_{rk} = \alpha_k Y_r$ is total expenditure of region r on sector-k goods, with $\alpha_A = \alpha$ and $\alpha_M = 1 - \alpha$. The total income of region r is $Y_n = w_{rA} H_{rA} + w_{rM} H_{rM}$.

The gravity equation can be rewritten as:

$$X_{nrk} = \alpha_k \tau_{nr}^{1-\sigma} \times \frac{Y_n/\bar{Y}}{\mathcal{P}_{nk}^{1-\sigma}} \times \frac{Y_r}{P_{rk}^{1-\sigma}}, \tag{12}$$

where I define two trade market access terms:

$$P_{rk} \equiv \left(\sum_{l=1}^{N} \tau_{lr}^{1-\sigma} w_{lk}^{1-\sigma} \right)^{1/(1-\sigma)}, \tag{13}$$

$$\mathcal{P}_{nk} \equiv w_{nk}^{-1} \left(Y_n/\bar{Y} \right)^{1/(1-\sigma)}. \tag{14}$$

Here, $\bar{Y} \equiv \sum_r Y_r$ is the total income of the economy and is normalized to one as the numeraire.

As with migration, (the inverse of) P_{rk} represents the inward trade market access for sector-k goods in region r, while (the inverse of) $\mathcal{P}_{nr}$ represents the outward trade market access for sector-k goods from region n.

VI.B Static Equilibrium

Given any strictly positive initial population vector $\{\check{L}_r^e\}$ and a set of location fundamentals $\{\phi_{nk}^e, \bar{a}_n^e, \tau_{nr}, \eta_{nr}\}$, an equilibrium consists of a set of prices $\{w_{nk}, p_{nk}\}$ and quantities $\{L_{nk}^e, H_{nk}\}$, such that (i) firms and consumers act optimally, and (ii) goods and labor markets clear in every region.

The goods market clearing condition is:

$$w_{nk} H_{nk} = \sum_{r=1}^{N} \alpha_k (w_{rA} H_{rA} + w_{rM} H_{rM}) \frac{\tau_{nr}^{1-\sigma} w_{nk}^{1-\sigma}}{\sum_{l=1}^{N} \tau_{lr}^{1-\sigma} w_{lk}^{1-\sigma}}. \tag{15}$$

This condition states that total income earned in sector k of region n equals total sales of sector-k goods to all destinations. It embeds two underlying conditions: (i) all sectoral revenue is paid out as wages, and (ii) all regional income is fully spent on goods from all regions.

The labor market clearing condition is:

$$H_{nk} = \sum_e H^e_{nk} = \sum_e L^e_n \pi^e_{nk} \left(\Gamma_\theta \bar{w}^e_n w^{-1}_{nk} \right) \tag{16}$$

$$L^e_n = \sum_r \check{L}^e_r \frac{\left(\eta^{-1}_{rn} V^e_n\right)^\nu}{\sum_{l=1}^N \left(\eta^{-1}_{rl} V^e_l\right)^\nu}. \tag{17}$$

Equation (16) states that total labor in sector k of region n equals the sum of efficiency units contributed by both ethnic groups. Each group's contribution is the product of its sectoral employment ($L^e_n \pi^e_{nk}$) and average efficiency units ($\Gamma_\theta \bar{w}^e_n w^{-1}_{nk}$). Equation (17) follows from the migration flow identity: the equilibrium population of group e in region n equals the sum of migration flows from all origins.

Using Equations (6)–(8), we can substitute out V^e_n, π^e_{nk}, and $\bar{w}^e_n$, replacing them with exogenous parameters and endogenous outcomes $\{w_{nk}, L^e_n\}$. The equilibrium is then characterized by a system of $6 \times N$ equations (15)–(17) in $6 \times N$ unknowns $\{w_{nk}, H_{nk}, L^e_n\}$.

Existence and uniqueness. I prove the existence of equilibrium by construction, using the iterative procedure described in Appendix C.1.4. The process features three nested loops: the outer loop updates population distribution; the middle loop solves for sectoral wages and prices given the population; and the inner loop solves for occupational shares given population and wages. Convergence is ensured by congestion forces, such as idiosyncratic migration preferences, sectoral heterogeneity in productivity, imperfect substitution across regional varieties, and external congestion externalities.

When agglomeration forces dominate congestion forces, the model may admit multiple equilibria. In this setting, multiplicity can arise if total population concentrates in different sets of location-sectors across equilibria due to strong size-based agglomeration forces, or if ethnic groups fully segregate in different location-sectors due to strong within-group spillovers. While I do not provide a formal proof to rule out multiplicity, I provide suggestive evidence of uniqueness at baseline parameters by initializing the algorithm from a wide range of starting values.[37] Specifically, I initiate the algorithm from 100 random starting values and from extreme Chinese population shares across counties, following Bayer and Timmins (2005), and find convergence to the same equilibrium. Note that equilibrium in my model is conditional on the initial population distribution; uniqueness here refers to the existence of at most one equilibrium given an initial population.

[37] I explore deriving sufficient conditions for uniqueness following Allen, Arkolakis and Li (2024), but the conditions are not informative in my setting (see Appendix C.1.5).

VII Identification and Estimation

To estimate the model, I use the 1957 population distribution—observed after most of the resettlement had been completed—as the initial population, and treat the 1980 data as the equilibrium outcomes.

I make a set of parametric assumptions for migration and trade costs. Bilateral migration and trade costs are both symmetric and increasing in distance. Migration costs take the form $\eta_{rn} = (d_{rn}/d_{min})^{\kappa}$, where d_{min} is the minimum within-county distance, and $\kappa \geq 0$ is the distance elasticity of migration costs. Similarly, trade costs are given by $\tau_{nr} = (d_{rn}/d_{min})^{\xi}$, where $\xi \geq 0$ is the distance elasticity of trade costs.[38]

The model is characterized by a tuple of location fundamentals $\{\phi^e_{nk}, \bar{a}^e_n\}$ and 11 structural parameters:

$$\Theta \equiv \{ \underbrace{\alpha, \sigma}_{\text{Preference}}, \underbrace{\xi, \kappa}_{\text{Trade/Migration}}, \underbrace{\theta, \gamma_A, \gamma_M, \gamma^e}_{\text{Productivity}}, \underbrace{\nu, \beta, \beta^e}_{\text{Amenity}} \}.$$

Due to data limitation, I externally set or calibrate three parameters: the elasticity of substitution across regional varieties (σ), the distance elasticity of trade costs (ξ), and the migration elasticity (ν). I set $\sigma = 8$ based on estimate from Vietnam (Balboni, 2025). Since internal trade flow data are unavailable for Malaysia, I follow Monte, Redding and Rossi-Hansberg (2018) and calibrate ξ such that $\xi(1-\sigma) = -1.29$, which implies $\xi = 0.18$, given $\sigma = 8$. Finally, I set $\nu = 3$ based on estimate from Indonesia (Bryan and Morten, 2019). Appendix C.3.2 assesses robustness of counterfactual results to alternative values.

In the remainder of the section, I first discuss how I identify and estimate the remaining eight parameters. I then discuss the estimation procedure and results.

VII.A Identification

I begin by presenting a proposition that establishes the identification of the market access terms and the agricultural expenditure share α. I then discuss how I identify the remaining parameters and location fundamentals.

Market access terms. From the equilibrium conditions, I derive four relationships involving trade and migration market access: (i) total sales equals labor payments; (ii) total income equals total expenditure; (iii) final population equals total in-migration; and (iv)

[38] Cross-county distances d_{rn} are measured by the Euclidean distance between centroids; within-county distances d_{rr} are calculated from the centroid to the nearest boundary. I allow within-county costs to exceed 1—except for the smallest county, which is normalized to 1—to account for differences in county size. This normalization is without loss of generality: higher migration costs reduce utility similarly to a decline in local amenity $\bar{a}^e_n$, and higher trade costs lowers productivity as if productive fundamentals ϕ^e_{nk} were lower.

initial population equals total out-migration.[39] These conditions yield the following system of equations:

$$\mathcal{P}_{nk}^{1-\sigma} = \frac{\alpha_k}{\Omega_{nk}} \sum_r \tau_{nr}^{1-\sigma} Y_r P_{rk}^{\sigma-1}, \tag{18}$$

$$P_{rk}^{1-\sigma} = \sum_n \tau_{nr}^{1-\sigma} Y_n \mathcal{P}_{nk}^{\sigma-1}, \tag{19}$$

$$\left(\mathcal{V}_n^e\right)^{-\nu} = \sum_r \eta_{rn}^{-v} \check{L}_r^e \left(\Pi_r^e\right)^{-v}, \tag{20}$$

$$\left(\Pi_r^e\right)^{v} = \sum_n \eta_{rn}^{-v} L_n^e \left(\mathcal{V}_n^e\right)^{\nu}, \tag{21}$$

where $\Omega_{nk} \equiv w_{nk} H_{nk}/Y_n$ is the share of income in region n generated by sector k.

Proposition 1. *Given observed data on* $\{Y_n, \Omega_{nk}, \check{L}_n^e, L_n^e\}$ *and parameter values* $\{\tau_{nr}^{1-\sigma}, \eta_{nr}^{-\nu}\}$*, there exists a unique scalar* α *and a set of values (up to scale) for* $\{\mathcal{P}_{nk}^{\sigma-1}, P_{rk}^{\sigma-1}, \left(\mathcal{V}_n^e\right)^{\nu}, \left(\Pi_r^e\right)^{v}\}$ *that satisfy equations* (18)–(21).

Proof. See Appendix C.2.1. □

This proposition shows that the market access terms are identified (up to scale) without requiring knowledge of the agglomeration parameters $\{\gamma_k, \gamma^e, \beta, \beta^e\}$, even in the presence of multiple equilibria.

Migration cost elasticity. Since migration costs affect population flows through $\eta_{rn}^{\nu} = (d_{rn}/d_{min})^{\kappa\nu}$, where the distance elasticity κ and taste dispersion ν enter as a product, I estimate $\tilde{\kappa} \equiv \kappa\nu$. Using non-linear least squares, I minimize the difference between the model-predicted and observed migration flows across districts (see Appendix C.2.2 for details). I impose a common κ across ethnic groups, as separate estimates by group yield similar values (see Appendix C.2.4).

Identification relies on the assumption that deviations between observed and predicted flows are due to classical measurement errors, uncorrelated with geography or other unobserved determinants of migration market access. As the sample size grows, these errors average out, and observed flows converge to the model predictions under the true $\tilde{\kappa}$. Appendix Figure A.9 shows that the loss function is convex, indicating a unique minimizing value of $\tilde{\kappa}$.

Migration flows in the model are defined over a 24-year period, whereas the census data report flows with an average of 12 years. To match the model's time horizon, I convert the observed 12-year migration shares into 24-year shares, assuming stable migration patterns

[39] See Appendix C.2.1 for derivations.

over time, following Artuç, Chaudhuri and McLaren (2010); Caliendo, Dvorkin and Parro (2019).[40]

Skill dispersion. The shape parameter θ governs the dispersion of Fréchet-distributed productivity across individuals, with larger θ indicating less dispersion. Let y^e_{ink} denote the earnings of individual i of ethnicity e working in sector k and residing in region n. Since earnings also follow a Fréchet distribution, it follows that:

$$\frac{\mathrm{Var}[y^e_{ink}]}{\mathbb{E}[y^e_{ink}]^2} = \frac{\Gamma(1-\frac{2}{\theta}) - \Gamma(1-\frac{1}{\theta})^2}{\Gamma(1-\frac{1}{\theta})^2}.$$

This normalized variance approaches infinity as θ approaches 2 from above and decreases monotonically toward 0 as θ increases. Thus, any observed normalized variance maps to a unique value of θ; that is, θ is identified.

Productivity spillovers. The parameters $\{\gamma_A, \gamma_M, \gamma^e\}$ govern productivity spillovers, affecting expected earnings in each sector and hence occupational choices. Rewriting the occupation choice equation (6) using trade market access terms yields:

$$\begin{aligned} \ln \bar{w}^e_n = \gamma_k \ln L_{nk} + \gamma^e \ln\left(\frac{L^e_{nk}}{L_{nk}}\right) - \frac{1}{\theta}\ln \pi^e_{nk} \\ - \left(\frac{1}{\sigma-1}\right)\ln\left(\mathcal{P}_{nk}\right)^{\sigma-1} - \left(\frac{1}{\sigma-1}\right)\ln Y_n + \underbrace{\frac{1}{\theta}\ln \phi^e_{nk}}_{\text{error term}}, \quad \forall k \in \{A, M\}. \end{aligned} \tag{22}$$

The left-hand side is the average wage of ethnic group e in region n, which I measure using household income in the data.[41] The size of sectoral employment affects wages through γ_k, while its ethnic composition affects wages through γ^e.

To identify γ^e, I subtract Equation (22) for one group from the other to remove region-sector-specific terms:

$$\ln\left(\frac{\bar{w}^c_n}{\bar{w}^m_n}\right) = \gamma^e \ln\left(\frac{L^c_{nk}}{L^m_{nk}}\right) - \frac{1}{\theta}\ln\left(\frac{\pi^c_{nk}}{\pi^m_{nk}}\right) + \underbrace{\frac{1}{\theta}\ln\left(\frac{\phi^c_{nk}}{\phi^m_{nk}}\right)}_{\text{error term}}. \tag{23}$$

This equation represents a relative (inverse) labor demand curve for sector k. The $-1/\theta$ term

[40] I first compute a 12-year migration share matrix (with each row summing to 1) and square it to obtain the 24-year matrix $\hat{m}_{jh}$. This transformation also smooths out zeros in the data, allowing the use of logs in estimation; see Appendix C.2.2.

[41] Under the Fréchet assumption and a common shape parameter across sectors, the average wage for an ethnic group in a given region is equal across sectors. This implication is supported by the data (see Appendix C.1.2).

reflects the neoclassical force driving downward-sloping demand when γ^e is not too large.

Unobserved productivity ϕ^e_{nk} enters as the error term and likely correlates positively with local population, as individuals sort into more productive locations. This selection bias tends to inflate the OLS estimates of γ_k and γ^e. In contrast, classical measurement errors in the population distribution attenuate both estimates, biasing them downward.

To address these biases, I use an instrumental variable strategy. Specifically, I instrument for local ethnic composition and sector size using exogenous variation from the resettlement program. I denote the residualized resettlement density from Section V as $Z_n^{(own)}$. I also construct another instrument, $Z_n^{(neighbor)}$, based on neighboring resettlement: the average resettlement density of neighboring counties within 5 kilometers after controlling for their expected resettlement density and own-county covariates. Since resettlement was not targeted based on location productivity (conditional on covariates), these instruments are plausibly orthogonal to ϕ^e_{nk}.

The moment conditions are given by:

$$\mathbb{E}[Z_n \ln \phi^e_{nk}] = 0,\ \forall k, e; \quad Z_n \in \{Z_n^{(own)}, Z_n^{(neighbor)}\}. \tag{24}$$

These moment conditions identify the three productivity spillover parameters: γ_A, γ_M, and γ^e. Given θ, γ^e is identified using Equation (23) by moving $\frac{1}{\theta}\ln(\pi^c_{nk}/\pi^m_{nk})$ to the left-hand side and instrumenting for $\ln(L^c_{nk}/L^m_{nk})$. Given γ^e, γ_k is then identified using Equation (22) by moving all other terms to the left-hand side and instrumenting for $\ln L_{nk}$.

One potential threat to identification is that the location fundamentals recovered in 1980 may have been affected by the resettlement program, for example, through public goods provision. A key concern is road construction, which I assess in Appendix B.3. I show that while counties with higher resettlement had slightly more local roads by 1980, the differences are economically small, and the estimated income moments remain similar when directly controlling for road coverage.

Amenity spillovers. The parameters β and β^e govern amenity spillovers and shape migration patterns. The value of residing in region n, from Equation (8), can be rewritten using migration market access as:

$$\begin{aligned}\ln \bar{w}^e_n = (-\beta + \beta^e)\ln L_n + \left(\frac{1}{\nu} - \beta^e\right)\ln L^e_n + \frac{1}{\nu}\ln\left(\mathcal{V}^e_n\right)^\nu \\ + \left(\frac{\alpha}{\sigma-1}\right)\ln P^{\sigma-1}_{nA} + \left(\frac{1-\alpha}{\sigma-1}\right)\ln P^{\sigma-1}_{nM} - \underbrace{\frac{1}{\nu}\ln \bar{a}^e_n}_{\text{error term}}.\end{aligned} \tag{25}$$

To identify β^e, I express the Chinese wage premium as:

$$\ln\left(\frac{\bar{w}_n^c}{\bar{w}_n^m}\right) = \left(\frac{1}{\nu} - \beta^e\right)\ln\left(\frac{L_n^c}{L_n^m}\right) + \frac{1}{\nu}\ln\left(\frac{(\mathcal{V}_n^c)^\nu}{(\mathcal{V}_n^m)^\nu}\right) - \underbrace{\frac{1}{\nu}\ln\left(\frac{\bar{a}_n^c}{\bar{a}_n^m}\right)}_{\text{error term}}. \tag{26}$$

This equation represents a relative (inverse) labor supply curve. When β^e is not too large, the neoclassical force $1/\nu$ predicts an upward-sloping relationship between relative wages and population share. If β^e is large, however, a higher Chinese share is an amenity for Chinese individuals, making them more willing to accept lower wages. The term $\mathcal{V}_n^e$ is a labor supply shifter that captures potential group-e migrants from other counties. The error term captures unobserved amenities that make a county more attractive to a group, likely biasing the OLS estimate of β^e upward.

To address this endogeneity, I use the same resettlement-based instruments. Assuming the program did not target areas based on amenity fundamentals, the following moment conditions identify β and β^e:

$$\mathbb{E}[Z_n \ln \bar{a}_n^e] = 0, \ \forall e, \quad Z_n \in \{Z_n^{(own)}, Z_n^{(neighbor)}\}. \tag{27}$$

Location fundamentals. I recover the exogenous location fundamentals (up to scale) as structural residuals from Equations (22) and (25). Specifically, I recover productivity fundamentals ϕ_{nk}^e from Equation (22) after estimating γ_k and γ^e, and amenity fundamentals $\bar{a}_n^e$ from Equation (25) after estimating β and β^e.

VII.B Estimation

The estimation proceeds in four steps. First, I estimate the migration cost elasticity with respect to distance, $\tilde{\kappa}$, and use it to compute the migration cost matrix $\eta_{nr}^{-\nu}$. Second, I iteratively solve for the market access terms and the agricultural expenditure share α, following Proposition 1. Third, I estimate the shape parameter of Fréchet skills, θ, by targeting the population-weighted average of normalized wage variance within (n, k, e) cells. Finally, I estimate the agglomeration parameters $\{\gamma_A, \gamma_M, \gamma^e, \beta, \beta^e\}$ using the generalized method of moments (GMM), based on the moment conditions in Equations (24) and (27).

To mitigate small-sample concerns, I exclude counties with fewer than five employed households in the 1980 census microdata and weight the estimation by household count. The final sample includes 685 counties. I bootstrap the entire procedure to obtain standard errors.[42]

[42] In each bootstrap iteration, I sample individuals with replacement from the census microdata at the district level and re-aggregate outcomes to the county level.

Table 10 reports the parameter estimates. I find strong productivity spillovers in non-agriculture: the elasticity of productivity with respect to local employment is $\gamma_M = 0.22$, in line with estimates for manufacturing in the literature (see Appendix C.2.4). In contrast, agricultural productivity declines with local employment ($\gamma_A = -0.12$), consistent with diminishing returns to labor when land is fixed.

The within-ethnic productivity spillover is estimated at $\gamma^e = 0.13$, implying that a higher Chinese share raises Chinese workers' productivity. The effect on Malay workers depends on the sector. Since $\gamma_M > \gamma^e > \gamma_A$, Equation (5) implies that a larger Chinese population increases Malay productivity in non-agriculture but reduces it in agriculture.[43]

For amenity spillovers, I estimate a small congestion elasticity with respect to total population ($\beta = -0.005$). Accounting for housing as a non-traded good, this implies a pure amenity spillover of 0.05.[44] The within-group amenity spillover is $\beta^e = 0.13$, indicating that individuals prefer living near co-ethnics. This is consistent with scale economies in ethnicity-specific amenities and social frictions in residential or consumption choices (Duranton and Puga, 2004; Davis et al., 2019).

Estimated location fundamentals correlate with observable measures of productivity and amenities. Agricultural fundamentals are negatively correlated with terrain ruggedness and positively with agricultural suitability (Appendix Table A.19, Column 1). Non-agricultural fundamentals are positively correlated with the number of manufacturing firms in 1970 (Column 2). Amenity fundamentals are positively correlated with the density of public services—such as police stations, post/telegraph offices, and schools—and negatively with distance to these amenities (Columns 3–4).[45]

VIII Counterfactuals

In this section, I use the estimated model to evaluate the aggregate impact of the resettlement program and explore how heterogeneous agglomeration forces shape the economy through policy counterfactuals. Section VIII.A simulates a counterfactual 1980 equilibrium without the resettlement program—holding fixed total Chinese and non-Chinese populations—and compares it to the observed 1980 economy to quantify the program's aggregate effects. This no-resettlement equilibrium serves as the baseline for subsequent comparisons. Section VIII.B examines how cross-ethnic barriers to productivity spillovers affect population

[43]The model assumes perfect substitutability in labor efficiency between Chinese and non-Chinese individuals. If they are imperfect substitutes, the estimated γ^e understates the true within-ethnic spillover, as a stronger spillover would be needed to rationalize the limited income gains among non-Chinese workers, who benefit from complementarity with resettled Chinese.

[44]The pure amenity spillover refers to the utility gain from local population net of housing market effects. See Appendix C.2.4 for further discussion.

[45]For the distribution of estimated fundamentals, see Appendix Figures A.12 and A.13.

distribution and aggregate productivity. Section VIII.C evaluates an industrial policy that subsidizes Malay wages in non-agriculture, motivated by actual policies implemented after the 1970s.

VIII.A Aggregate Impact of Forced Resettlement

Section V shows that counties that received resettled Chinese populations benefited economically from higher Chinese density. But what were the aggregate effects—including on the counties they left behind and those not directly involved in the program? This section evaluates the aggregate economic and welfare impacts of the resettlement program.

I simulate a counterfactual "no resettlement" equilibrium in 1980 by initializing the model with the 1947 population distribution—before resettlement—instead of the 1957 distribution, holding all parameters and location fundamentals fixed. I then compare this equilibrium to the observed 1980 economy to quantify the program's aggregate impact.[46]

Figure 5 maps how resettlement changes county outcomes. Panel A shows that resettlement shifts Chinese populations from inland and remote areas to more urban, coastal areas. Panel B shows that this shift raises Chinese income per capita (equal to output per capita in the model) in receiving counties, while income declines in counties that lose Chinese populations. Panel C shows that Malay income per capita generally declines in counties where Chinese density increases, except in major urban centers such as Kuala Lumpur, where many Malays work in non-agriculture and benefit from its strong external economies of scale despite limited cross-ethnic spillovers. At the same time, Chinese departures from rural areas raise Malay agricultural productivity through local diminishing returns to labor.

Table 11 quantifies these effects. Among Chinese, the program raises the share working in non-agriculture by 1 percentage point (1.5%) and increases their non-agricultural productivity by 1.6%. Agricultural productivity declines, however, as resettled destinations are generally less suitable for farming (Column 1). Among Malays, the *aggregate* non-agricultural share falls, as many move into agricultural land made more productive by the departure of resettled Chinese—despite the higher *local* non-agricultural shares observed in the receiving areas.[47] Malay non-agricultural productivity rises, partly due to spillovers from incoming Chinese and partly due to selection, as less productive Malays shift into agriculture (Column 2). Although the program directly targets the Chinese population, Malays experience

[46] This approach assumes that the 1947 distribution was a steady state, which would have persisted until 1957 absent resettlement. It also assumes that location fundamentals evolved after 1957 to bring the economy to the observed 1980 equilibrium. These assumptions are plausible, as resettlement accounts for nearly 80% of the population change in receiving counties from 1947 to 1957 (Section V).

[47] The decline in aggregate Malay non-agricultural share is not inconsistent with the cross-sectional results in Section V, which showed higher Malay non-agricultural shares in counties receiving more resettlement relative to less resettled counties.

notable income gains through general equilibrium effects beyond the receiving areas.

I decompose the aggregate output change as follows:

$$\sum_{n,k,e}\left(\tilde{y}_{nk}^{e}\tilde{L}_{nk}^{e}-y_{nk}^{e}L_{nk}^{e}\right)=\sum_{n,k,e}\left(\tilde{L}_{nk}^{e}-L_{nk}^{e}\right)y_{nk}^{e}+\sum_{n,k,e}\left(\tilde{y}_{nk}^{e}-y_{nk}^{e}\right)L_{nk}^{e}$$
$$+\sum_{n,k,e}\left(\tilde{y}_{nk}^{e}-y_{nk}^{e}\right)\left(\tilde{L}_{nk}^{e}-L_{nk}^{e}\right),$$

where y_{nk}^{e} is output per capita for group e in county n and sector k in the baseline, and $\tilde{y}_{nk}^{e}$ is its value in the resettled equilibrium. The first term isolates the effect of labor reallocation, holding output per capita fixed. The second term captures changes in per-capita output, holding labor allocation fixed. The third term reflects the interaction between labor reallocation and productivity changes.

Overall, the resettlement program raises aggregate output by 2%, driven primarily by labor reallocation to more productive sectors and regions (Column 3). This gain reflects that resettlement sites were located along transportation networks with better market access, and the model assigns these areas higher baseline productivity (Appendix Figure A.14). Holding productivity fixed, labor reallocation alone accounts for about two-thirds of the output gain. The program's total expenditure was approximately 133 million Malayan Dollars ($43 million) (Dhu Renick, 1965), or 0.5% of Malaysia's 1980 GDP (adjusted for inflation), implying a net output gain of around 1.5%.

The output gain corresponds to a 4.8% increase in aggregate utility (Column 3), but this figure does not capture the welfare loss from forced relocation. To benchmark the economic gain against this coercion cost, I proceed in two steps. First, I invert the model to recover the amenity fundamentals that would have sustained the 1947 population distribution through 1957. Second, I solve for the least-cost wage subsidies by location and ethnicity that would induce individuals to voluntarily relocate from their 1947 locations to match the 1957 distribution.[48]

The total subsidy received by the resettled population amounts to 17% of baseline output—exceeding the program's economic gains—suggesting a net welfare loss from the forced resettlement. This subsidy amount can be interpreted as a lower bound on the utility loss (in dollars) for two reasons. First, the forcibly resettled likely value their destinations less than the average resident in the county, biasing the required subsidy downward. Second, the calculation omits psychological costs of forced displacement, which are not reflected in the amenity fundamentals. The results underscore an intergenerational tradeoff: while

[48]Since migration depends only on relative prices across regions, I calculate the minimum (weakly positive) ad-valorem subsidies needed to achieve this voluntary migration. See Appendix C.3.1 for details.

younger cohorts benefit, the resettled generation suffers significant welfare losses.

VIII.B Reducing Cross-Ethnic Barriers to Productivity Spillovers

Section V showed that ethnic divisions shape local economic outcomes. This section examines their broader implications for population distribution, aggregate productivity, and welfare.

Figure 6 maps the effects of reducing the cross-ethnic barriers to productivity spillovers (γ^e) to half its baseline value. Panel A shows that ethnic groups become geographically more integrated: Malays move from rural to urban, Chinese-dense areas, while Chinese move in the opposite direction. This spatial reallocation increases income per capita for both groups. Chinese income per capita rises almost everywhere, especially in rural areas where Malays are the majority (Panel B). Malay income increases more in urban areas, where Chinese are concentrated (Panel C).

Table 12 summarizes the aggregate effects. Chinese non-agricultural employment falls, while Malay non-agricultural employment rises. Both groups see productivity gains, as rural Chinese and urban Malays now benefit more from stronger cross-ethnic spillovers. Aggregate output increases by 4.1%, with 89% of the gain driven by productivity improvements. Although Chinese account for only one-third of the population, they contribute nearly as much to the output gain as Malays, underscoring the economic benefits of integration.

On welfare, average Chinese utility rises by 1.2%—less than their 4.4% income gain—as they tend to move to rural areas with lower amenities (Column 1). Malay utility increases by 4.6%, exceeding their 4% income gain, due to improved amenity access in urban destinations (Column 2). Overall, welfare increases by 3.5% (Column 3).

Cross-ethnic barriers distort the spatial and sectoral allocation of human capital. Policies that reduce these frictions can facilitate urbanization and generate sizable economic and welfare gains.

VIII.C Malay Wage Subsidies in Non-Agriculture

Given the strong external economies in non-agriculture, shifting labor from agriculture to non-agriculture may raise aggregate output. A key postwar objective of Malaysia was to integrate Malays into the industrial sector, as they predominantly worked in low-productivity agriculture. This section evaluates a policy that subsidizes Malay wages in non-agriculture.

I simulate an 18% wage subsidy for Malays in non-agriculture, financed by a uniform 7.3% income tax to balance the government budget. The subsidy rate is chosen to equalize Malay and Chinese non-agricultural employment shares—one of the goals of the New Economic Policy (NEP) introduced in the 1970s, which promoted Malay participation in industry through credit access, training programs, and higher education quotas.[49]

[49]Following the 1969 racial riots, the NEP sought affirmative action to restructure society and *eliminate*

Figure 7 maps the effects of the subsidy. As non-agriculture becomes more profitable for Malays, they migrate to urban areas and crowd out Chinese workers, who switch to agriculture and relocate to rural areas (Panel A). Urban Chinese incomes decline due to increased competition from Malay workers, while rural Chinese incomes generally rise as agricultural land becomes more available (Panel B). Malay incomes increase most in major urban centers like Kuala Lumpur, where the Malay population expands substantially (Panel C).

Table 13 quantifies the aggregate effects. The share of Malays working in non-agriculture increases by 9.2 percentage points (17%), while the share of Chinese in non-agriculture declines by 8.1 percentage points (11%). Despite this ethnic reallocation, total non-agricultural employment share rises by 3.3 percentage points (5%).

Malay productivity increases by 3.4%, with a 9% gain in agriculture and a 3.5% decline in non-agriculture (Column 2). As Malays exit agriculture, diminishing returns in agriculture and selection raise average productivity among those who remain. The subsidy draws lower-productivity Malay workers into non-agriculture, but strong external economies mitigate the resulting productivity decline. For Chinese, the reverse shift—from non-agriculture to agriculture—lowers overall productivity by 2.1% (Column 1). Despite the distortions introduced by the subsidy, aggregate output rises by 1%, driven largely by productivity gains (Column 3).

Welfare effects are mixed. While aggregate welfare rises by 2% (Column 3), the gains accrue entirely to Malays, whose utility increases by 8.2% (Column 2). In contrast, Chinese utility falls by 10% (Column 1). The tax burden is shared, but only Malays benefit from the subsidy. As a result, Malay welfare increases by more than their productivity gain, while Chinese welfare declines by more than their productivity loss. Although controversial, industrial policies that target disadvantaged groups in sectors with strong agglomeration forces can raise overall output and reduce inequality.

IX Conclusion

This paper studies how social divisions shape the gains from agglomeration, leveraging a large-scale, ethnic-based resettlement program in 1950s British Malaya. The program forcibly relocated Chinese populations into villages, reshaping both economic and social structures. Despite its coercive nature—which likely limited the benefits relative to a voluntary scenario—the program generated productivity gains in receiving counties, spurring industrialization and greater division of labor. These gains, however, were unequally distributed across ethnic groups.

the identification of race with economic function. See Koon (1997) and Jomo (2017) for further discussion.

Local effects of resettlement were mediated by agglomeration externalities that varied by sector and ethnic composition. The influx of industrial labor shifted employment out of agriculture, driven by strong external economies in non-agriculture and diminishing returns in land-constrained agriculture. Denser labor markets promoted specialization and education, but segregation and cross-ethnic frictions limited spillovers to non-Chinese populations. As a result, income gains in more resettled counties accrued primarily to the Chinese community.

To assess aggregate effects, I develop and estimate a quantitative spatial model that incorporates heterogeneous agglomeration forces, migration, and occupational choices. Resettlement raised aggregate output by reallocating labor from remote, less productive areas to regions with better market access and industrial potential. However, the program ultimately reduced welfare by disregarding people's location preferences.

While this paper focuses on ethnic divisions, similar frictions can arise along other social lines, such as caste, culture, religion, and gender. In India, for example, caste norms have long constrained intergroup interactions, potentially limiting agglomeration spillovers and slowing structural transformation. These findings also speak to refugee and migrant resettlement policies, where social integration plays a key role in realizing the economic gains from migration.

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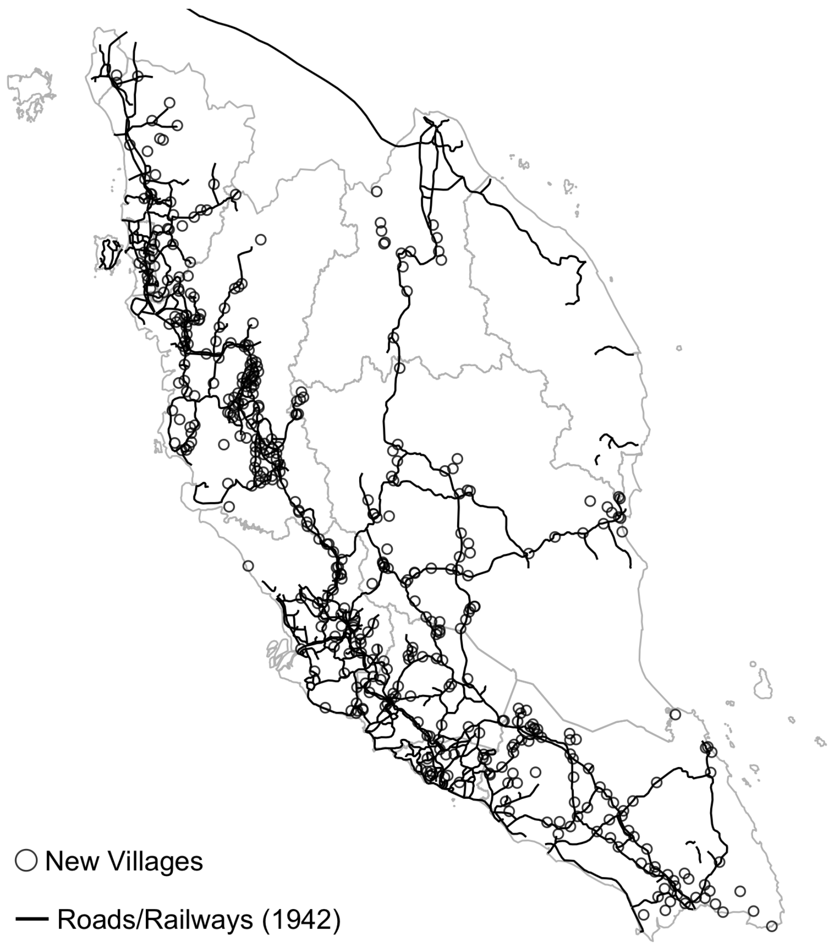
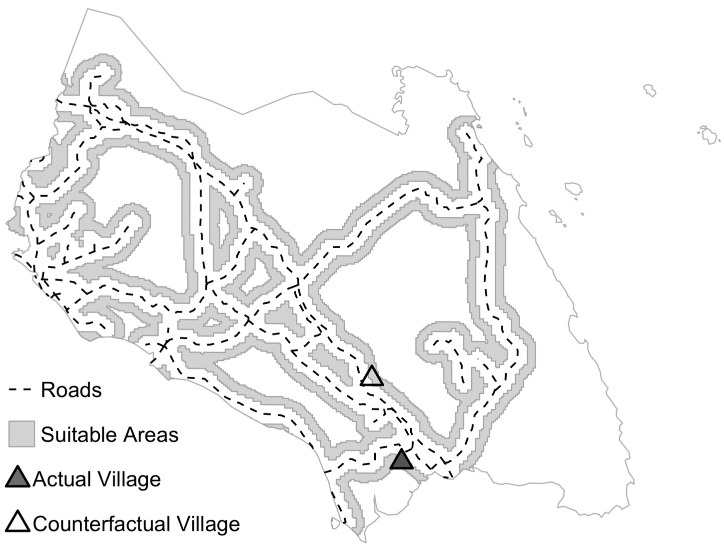
Figure 1. The New Villages and Transportation Network



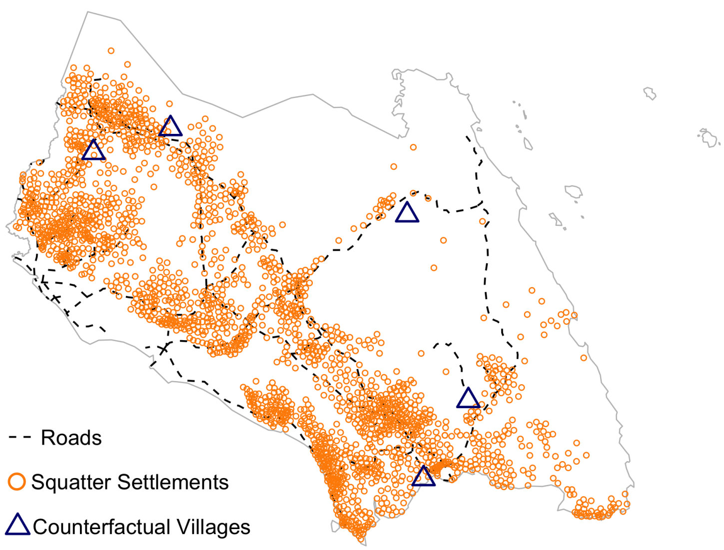
Notes: This figure shows the location of the New Villages (circles) and the roads and railway network in 1942 (lines). Gray polygons indicate state boundaries. New Village data are from the Corry report; road and railway data from the G8031 road map series (U.S. Office of Strategic Services, 1942).

Figure 2. Counterfactual Site Selection and Relocation

Panel A. Counterfactual Site Selection



Panel B. Counterfactual Relocation



Notes: This figure illustrates counterfactual site selection and relocation in the state of Johor. Panel A shows the selection of counterfactual sites. The solid triangle marks an actual New Village; dashed lines show the road and rail network; gray shaded areas indicate regions at similar distances from the actual village and equally suitable for resettlement. The hollow triangle denotes a counterfactual village, randomly drawn from these suitable areas. Panel B shows the relocation of squatters to the counterfactual sites, with orange circles marking initial squatter settlements.

Figure 3. County Resettlement Density, Expected and Residualized

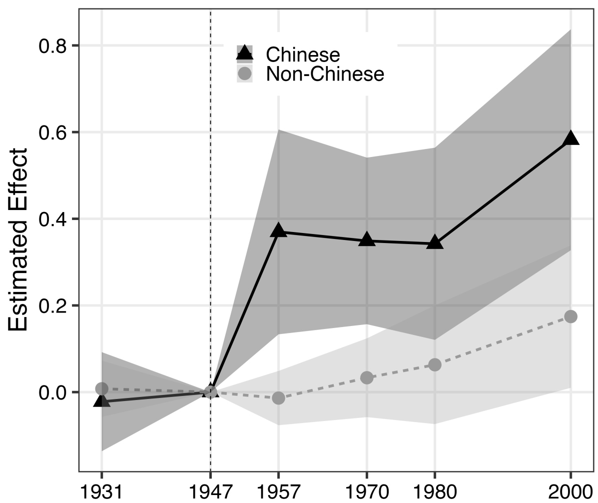
Panel A. County Resettlement Density, Expected

Panel B. County Resettlement Density, Residualized

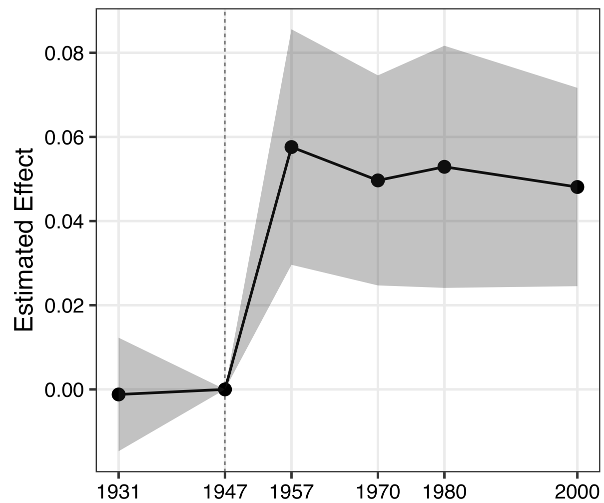
Notes: This figure maps expected and residualized county resettlement density for the 249 counties with at least one New Village. Darker shades indicate higher deciles of resettlement density. White bubbles denote New Villages, with sizes proportional to the resettled population. Panel A shows expected resettlement density, calculated using Equation (A-1). Panel B shows residualized resettlement density, controlling for state fixed effects and the main controls: expected resettlement density; log county area; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population in 1947; and the shares of land used for rubber and mining in 1943. Resettlement data from the Corry report.

Figure 4. Changes in Population Distribution from 1931 to 2000, by County Resettlement Density

Panel A. Population Growth



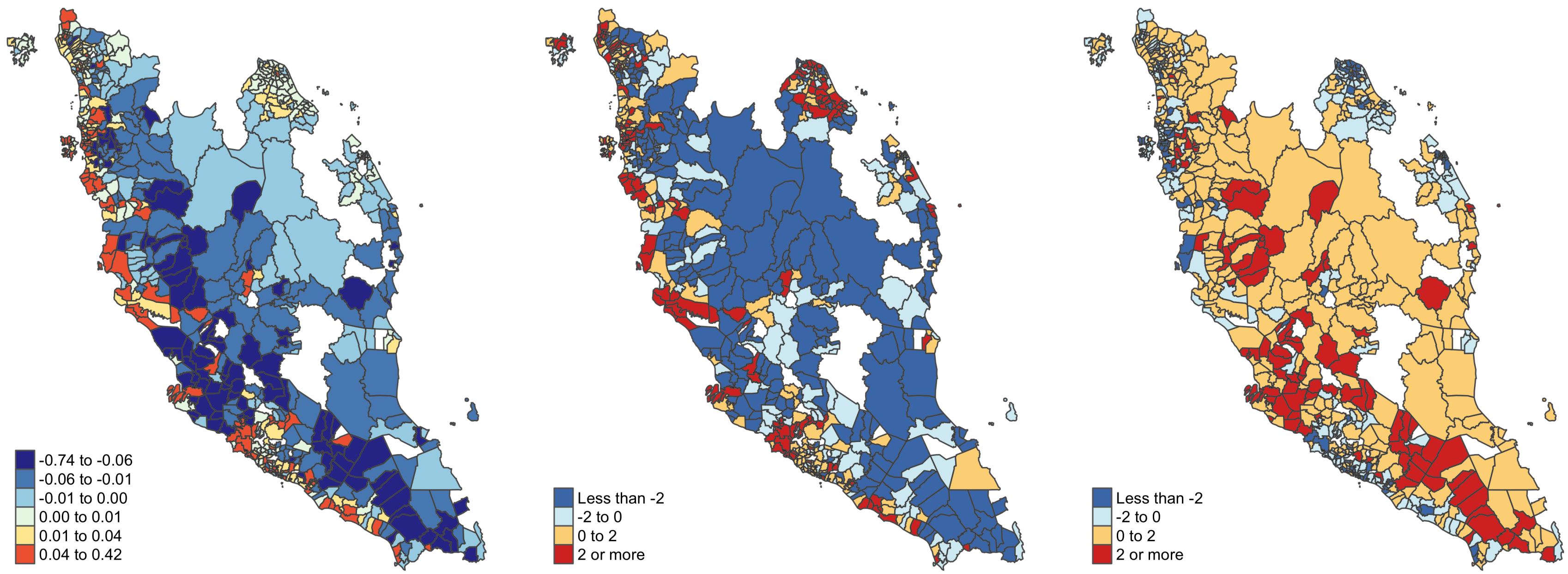
Panel B. Changes in Chinese Share



Notes: This figure shows regression estimates of county resettlement density on population growth by ethnic group (Panel A) and changes in Chinese population share (Panel B) from 1931 to 2000. All regressions include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; log county area; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population in 1947; and the shares of land used for rubber and mining in 1943. The sample includes 614 grouped counties based on 1931 boundaries. Shaded areas indicate 95% confidence interval based on Conley standard errors with a 30-kilometer distance cutoff.

Figure 5. Distributional Effects of Emergency Resettlement

Panel A. Change in Chinese Population Share | Panel B. Percent Change in Chinese Income Per Capita | Panel C. Percent Change in Malay Income Per Capita



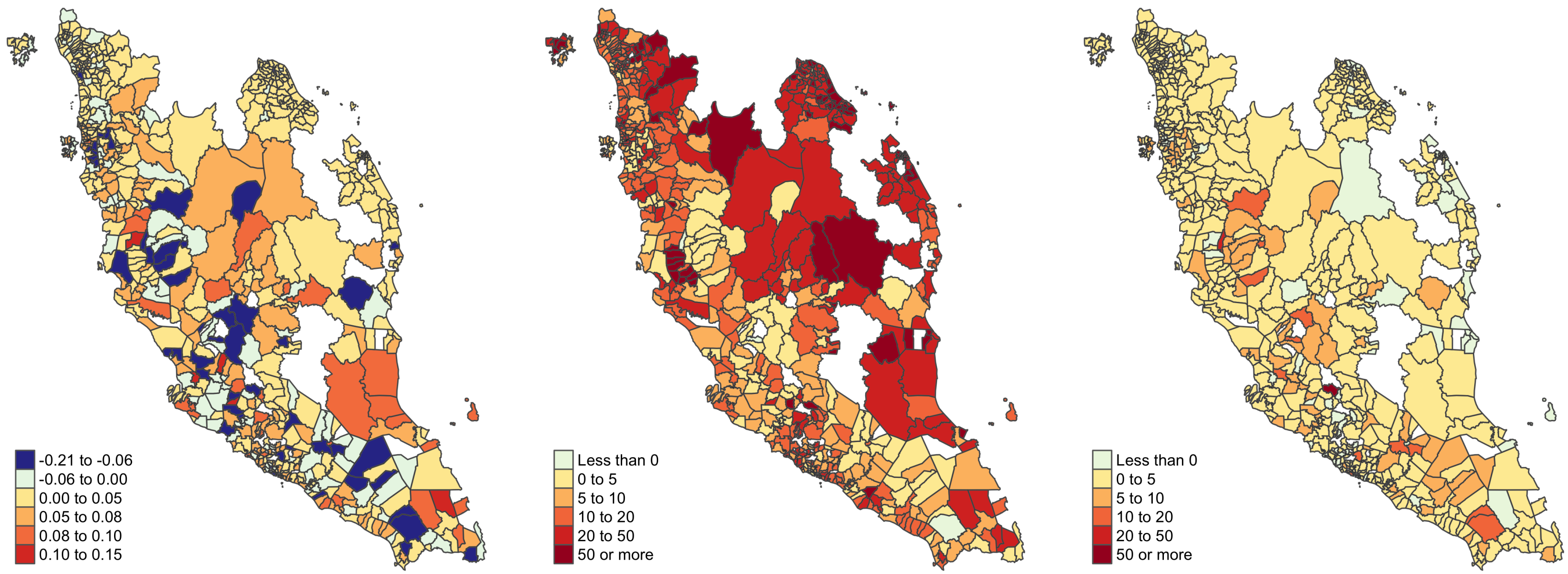
Notes: This figure shows changes in county outcomes from the "no resettlement" baseline to the observed 1980 economy. The no-resettlement baseline simulates a counterfactual 1980 equilibrium starting from the 1947 population distribution instead of the post-resettlement 1957 distribution. Panel A shows the change in Chinese population share. Panel B shows the percent change in Chinese income per capita. Panel C shows the percent change in Malay income per capita.

Figure 6. Distributional Effects of Reducing Cross-Ethnic Barriers to Productivity Spillovers

Panel A. Change in Chinese Population Share

Panel B. Percent Change in Chinese Income Per Capita

Panel C. Percent Change in Malay Income Per Capita



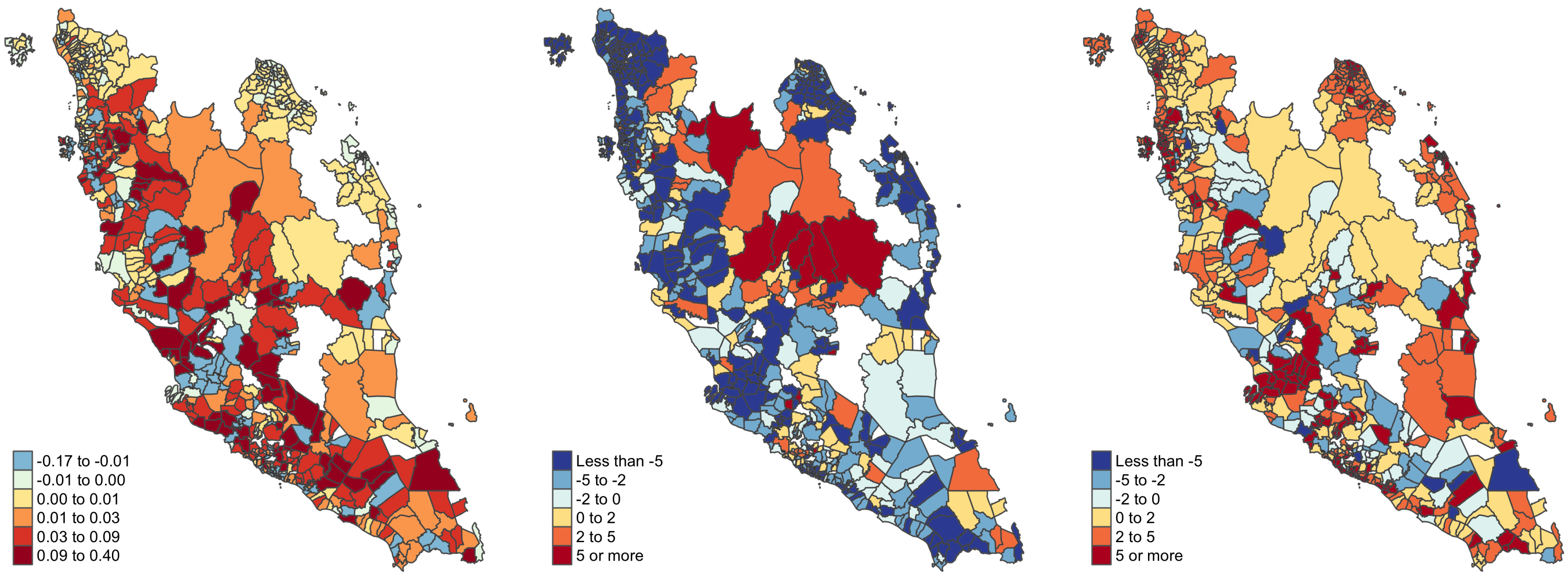
Notes: This figure shows changes in county outcomes from the "no resettlement" baseline economy to a counterfactual equilibrium in which the barrier to cross-ethnic productivity spillovers (γ^e) is reduced by half. Panel A shows the change in the Chinese population share. Panel B shows the percent change in Chinese income per capita. Panel C shows the percent change in Malay income per capita.

Figure 7. Distributional Effects of Malay Wage Subsidies in Non-Agriculture

Panel A. Change in Chinese Population Share

Panel B. Percent Change in Chinese Income Per Capita

Panel C. Percent Change in Malay Income Per Capita



Notes: This figure shows changes in county outcomes from the "no resettlement" baseline economy to a counterfactual equilibrium with an 18% wage subsidy for Malays in non-agriculture, financed by a uniform income tax. Panel A shows the change in Chinese population share. Panel B shows the percent change in Chinese income per capita. Panel C shows the percent change in Malay income per capita. All income measures are net of subsidies and taxes.

Table 1. Balance of Pre-Period Characteristics

	Geography				Public and Social Amenities				Economic Activities			
	Elev. (km)	Rugged. (m)	Rice Suitab. (t/ha)	Coconut Suitab. (t/ha)	Dist. Police (km)	Dist. Post (km)	Dist. Mosque (km)	Dist. Temple (km)	Land Use Rubber (%)	Land Use Mining (%)	Dist. Factory (km)	Dist. Cities (km)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A. Within-State Variation												
Higher Resettlement	0.02	3.86	−0.04	−0.01	−0.46	−0.35	−0.44	0.00	6.53	1.38	−0.97	−5.79
	(0.01)	(3.19)	(0.01)	(0.01)	(0.42)	(0.34)	(0.47)	(1.37)	(1.60)	(0.69)	(0.85)	(1.67)
Panel B. Baseline Covariates												
Higher Resettlement	0.01	−2.36	−0.05	−0.01	0.40	0.47	0.70	3.18	5.45	1.77	−0.69	−1.05
	(0.02)	(4.92)	(0.02)	(0.02)	(0.49)	(0.41)	(1.07)	(1.96)	(1.36)	(1.30)	(1.01)	(1.90)
Panel C: Expected Resettlement												
Higher Resettlement	0.02	−7.35	−0.05	−0.02	0.43	0.74	0.16	2.09	2.59	1.50	−0.99	0.52
	(0.03)	(5.84)	(0.03)	(0.03)	(0.61)	(0.54)	(1.00)	(2.21)	(2.52)	(1.18)	(1.10)	(2.44)
Mean	0.09	62.77	1.21	1.12	9.35	11.33	31.72	66.13	23.56	1.03	26.23	87.69
Standard Deviation	0.15	74.19	0.23	0.21	8.19	8.68	47.67	47.40	29.72	6.59	18.08	69.82
# Counties	777	777	777	777	777	777	777	777	777	777	777	777

Notes: This table shows the relationship between county characteristics and county resettlement density. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Columns 1–4 examine geographic characteristics: elevation (Column 1), ruggedness (Column 2), padi rice suitability (Column 3), and coconut (Column 4). Columns 5–8 examine access to amenities as of 1945: distance to the nearest police station (Column 5), post or telegraph office (Column 6), mosque (Column 7), and Chinese temple (Column 8). Columns 9–12 examine pre-period economic activity: land use share for rubber in 1943 (Column 9), land use share for mining in 1943 (Column 10), distance to industrial facilities in 1945 (Column 11), and distance to major cities (Column 12). Panel A include state fixed effects. Panel B adds baseline controls (excluding land use shares for rubber and mining): an indicator for any resettlement; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; and log county area. Panel C additionally controls for expected resettlement density. The unit of observation is the county. See Appendix Table A.1 for the data sources. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table 2. Post-Resettlement Population Distribution, by County Resettlement Density

	1957 (1)	1980 (2)	2000 (3)
Panel A. Log Total Population			
Higher Resettlement	0.094	0.110	0.179
	(0.034)	(0.062)	(0.075)
Mean of Outcome	8.31	8.83	9.14
Panel B. Chinese Population Share			
Higher Resettlement	0.048	0.050	0.041
	(0.012)	(0.011)	(0.011)
Mean of Outcome	0.22	0.19	0.14
# Counties	777	777	777

Notes: This table shows the relationship between county resettlement density and population distribution from 1957 to 2000. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Panel A reports effects on log total population in 1957 (Column 1), 1980 (Column 2), and 2000 (Column 3). Panel B reports effects on Chinese population share in these years. All regressions are estimated by OLS and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the county. Data from the tabulated Census of Population. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table 3. Sectoral Employment in 1980–1991, by County Resettlement Density

	Agriculture (1)	Non-Agriculture (2)	Difference (2) − (1) (3)
Panel A. Total Employment			
Higher Resettlement	0.111	0.291	0.180
	(0.037)	(0.129)	(0.139)
# County-Years	1,554	1,554	
Panel B. Chinese Employment			
Higher Resettlement	0.272	0.351	0.079
	(0.059)	(0.180)	(0.176)
# County-Years	1,516	1,502	
Panel C. Non-Chinese Employment			
Higher Resettlement	−0.006	0.244	0.250
	(0.045)	(0.107)	(0.124)
# County-Years	1,516	1,502	

Notes: This table shows the relationship between county resettlement density and sectoral employment in 1980–1991. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Panel A reports effects on total employment in the agricultural sector ("Agriculture", Column 1), the non-agricultural sector ("Non-Agriculture", Column 2), and the difference between the two estimates (Column 3). Panels B and C report effects on Chinese and non-Chinese employment, respectively. The agricultural sector includes agriculture, hunting, forestry, fishing, mining, and quarrying. The non-agricultural sector includes manufacturing, utility, construction, wholesale and retail trade, transport and communication, and finance, business and other services. All regressions pool data from 1980 and 1991, are estimated using the Poisson pseudo-maximum-likelihood (PPML) estimator, and include state-by-year fixed effects and the main controls interacted with year: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the county-year. Data from the Census of Population in 1980 and 1991. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table 4. Household Income in 1980, by County Resettlement Density

	Chinese Households (1)	Non-Chinese Households (2)	Difference (1) − (2) (3)
Panel A. Log Earnings			
Higher Resettlement	0.110	0.039	0.071
	(0.049)	(0.031)	(0.035)
Mean of Outcome	8.55	8.15	
# Households	9,634	20,549	
Panel B. Log Earnings, Agriculture			
Higher Resettlement	0.067	−0.007	0.073
	(0.041)	(0.037)	(0.042)
Mean of Outcome	8.32	7.94	
# Households	2,197	9,359	
Panel C. Log Earnings, Non-Agriculture			
Higher Resettlement	0.126	0.051	0.074
	(0.045)	(0.030)	(0.028)
Mean of Outcome	8.63	8.33	
# Households	7,437	11,190	

Notes: This table shows the relationship between county resettlement density and household income in 1980. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Panel A reports effects on log household earnings for Chinese households (Column 1), non-Chinese households (Column 2), and the difference between the two estimates (Column 3). Panel B restricts the sample to households whose head is employed in the agricultural sector (agriculture and mining). Panel C restricts the sample to households whose head is employed in the non-agricultural sector. All regressions are estimated by OLS and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the household. Data from the 2% individual-level Census of Population microdata in 1980. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table 5. Participation and Specialization in the Labor Market in 1980–1991, by County Resettlement Density

	Chinese Individuals (1)	Non-Chinese Individuals (2)	Difference (1) − (2) (3)
Panel A. Labor Force Participation			
Higher Resettlement	0.015	0.002	0.014
	(0.008)	(0.007)	(0.010)
Mean of Outcome	0.578	0.569	
# Individuals	38,390	71,234	
Panel B. Industry Specialization Index			
Higher Resettlement	0.017	−0.006	0.023
	(0.008)	(0.009)	(0.009)
Mean of Outcome	0.306	0.266	
# Counties	752	776	
Panel C. Occupation Specialization Index			
Higher Resettlement	0.016	−0.006	0.022
	(0.008)	(0.008)	(0.009)
Mean of Outcome	0.255	0.257	
# Counties	752	776	
Panel D. Share Employed in Managerial Occupations			
Higher Resettlement	0.012	−0.001	0.012
	(0.006)	(0.000)	(0.006)
Mean of Outcome	0.056	0.025	
# Counties	752	776	

Notes: This table shows the relationship between county resettlement density and labor market outcomes in 1980 or 1991. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Each panel examines a different labor market outcome: labor force participation rate in 1980 (Panel A); concentration of employment across industries in 1991, measured by the Herfindahl-Hirschman Index (HHI) (Panel B); concentration of employment across occupations in 1991, also measured by the HHI (Panel C); and the share of workers employed in managerial occupations (legislators, senior officials, and managers) in 1991 (Panel D). Column 1 shows estimates for Chinese individuals, Column 2 for non-Chinese individuals, and Column 3 the difference between the two. All regressions are estimated by OLS and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the individual for Panel A and the county for Panels B–D. Data from the 2% Census of Population microdata in 1980 and tabulations in 1991. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table 6. Educational Attainment in 1980, by County Resettlement Density

	Chinese Individuals				Non-Chinese Individuals				
	All (1)	Age 20–35 (2)	Age 36–50 (3)	Age >50 (4)	All (5)	Age 20–35 (6)	Age 36–50 (7)	Age >50 (8)	Difference (1) − (5) (9)
Panel A. Years of Schooling									
Higher Resettlement	0.436	0.438	0.338	0.069	0.105	0.127	0.100	−0.102	0.330
	(0.231)	(0.238)	(0.193)	(0.135)	(0.123)	(0.113)	(0.131)	(0.097)	(0.162)
Mean of Outcome	5.39	7.36	4.60	2.05	5.12	7.20	3.67	1.78	
Panel B. Primary Education									
Higher Resettlement	0.037	0.031	0.030	0.007	0.018	0.016	0.025	−0.004	0.019
	(0.018)	(0.014)	(0.017)	(0.016)	(0.012)	(0.011)	(0.014)	(0.012)	(0.010)
Mean of Outcome	0.56	0.79	0.46	0.20	0.51	0.75	0.33	0.12	
Panel C. Secondary Education									
Higher Resettlement	0.041	0.047	0.038	0.001	0.013	0.014	0.006	0.000	0.028
	(0.023)	(0.028)	(0.018)	(0.007)	(0.012)	(0.012)	(0.011)	(0.007)	(0.018)
Mean of Outcome	0.29	0.44	0.19	0.06	0.26	0.44	0.10	0.03	
# Individuals	31,507	15,597	8,843	7,067	57,345	30,087	15,056	12,202	

Notes: This table shows the relationship between county resettlement density and educational attainment in 1980. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Each panel reports effects on a different education outcome: years of schooling (Panel A); completion of primary education (Panel B); and completion of secondary education (Panel C). Columns 1–4 report estimates for Chinese individuals: column 1 pools all Chinese aged 20 and above, while columns 2–4 report estimates by age cohort. Columns 5–8 report the corresponding estimates for non-Chinese individuals. Column 9 reports the overall difference between Chinese and non-Chinese individuals (column 1 minus column 5). All regressions are estimated by OLS and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the individual. The sample includes individuals aged 20 or above from the 2% Census of Population microdata in 1980. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table 7. Manufacturing Activity in 1970, by County Resettlement Density and Pre-Period Industry Share of Chinese Employment

	Number of Manufacturing Establishments (1)	Share of Employer Establishments (2)
Higher Resettlement	0.022	0.001
	(0.125)	(0.020)
Higher Resettlement × Chinese Industries	0.217	0.020
	(0.082)	(0.009)
# County-Industries	15,540	2,142

Notes: This table shows how the relationship between county resettlement density and manufacturing activity in 1970 varies by industries with high pre-period Chinese employment. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. "Chinese Industries" is an indicator for industries where more than 80% of employment in 1947 was Chinese. These include all manufacturing industries except food products, wood products, textiles, and other miscellaneous manufacturing (see Appendix Figure A.7). Column 1 reports the effect on the number of manufacturing establishments, estimated using the Poisson pseudo-maximum-likelihood (PPML) estimator. Column 2 reports OLS estimates for the share of establishments with at least one full-time employee, weighted by the number of establishments in the county-industry. All regressions include 2-digit industry fixed effects, state fixed effects, and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the county-industry. Data from the Directory of Manufacturing in 1970 and the tabulated Population Census in 1947. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table 8. Firm Ownership and Revenue in 2011–2015, by County Resettlement Density

	Chinese Owned Firms (1)	Non-Chinese Owned Firms (2)	Difference (1) − (2) (3)
Panel A. Number of Firms			
Higher Resettlement	0.518	0.306	0.212
	(0.301)	(0.097)	(0.128)
# County-Industry-Years	93,240	93,240	
Panel B. Log of Average Firm Revenue			
Higher Resettlement	0.189	0.082	0.107
	(0.078)	(0.104)	(0.126)
# County-Industry-Years	12,608	9,468	

Notes: This table shows the relationship between county resettlement density and firm activity in 2011–2015. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Panel A reports effects on the number of firms, estimated using Poisson pseudo-maximum-likelihood (PPML). Panel B reports OLS estimates of log average firm revenue, weighted by the number of establishments by ownership in the county-industry-year. Column 1 shows estimates for Chinese-owned firms, Column 2 for non-Chinese owned firms, and Column 3 the difference between the two. Firm ownership is based on the ethnicity of ultimate owners, as described in Appendix ??. All regressions include two-digit NAICS industry-by-year fixed effects and their fully saturated interactions with state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the county-industry-year. Data from the Orbis Historical Disk. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table 9. Heterogeneous Effects of County Resettlement Density on 1980 Household Income, by Ethnic Segregation and Social Divisions

	Log Household Earnings		
	Chinese Households (1)	Non-Chinese Households (2)	Difference (1) − (2) (3)
Panel A. Ethnic Dissimilarity Index			
Higher Resettlement	0.150	0.147	0.003
	(0.055)	(0.063)	(0.053)
Higher Resettlement × Dissimilarity Index	−0.281	−0.478	0.196
	(0.156)	(0.225)	(0.161)
Panel B. Distance Segregation Index			
Higher Resettlement	0.165	0.115	0.050
	(0.053)	(0.041)	(0.039)
Higher Resettlement × Normalized Mosque-to-Temple Distance	−0.008	−0.026	0.018
	(0.015)	(0.011)	(0.010)
Panel C. Share of Chinese Fluent in Malay			
Higher Resettlement	0.092	−0.022	0.114
	(0.055)	(0.039)	(0.041)
Higher Resettlement × Above 10% Malay-Fluent Chinese	0.018	0.086	−0.068
	(0.038)	(0.032)	(0.026)
# Households	9,634	20,549	

Notes: This table shows how the effect of county resettlement density on household income in 1980 varies measures of ethnic segregation and social divisions. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Panel A interacts "Higher Resettlement" with the 1947 district-level ethnic dissimilarity index (Chinese vs. non-Chinese); Panel B with the average 1945 distance between mosques and non-Muslim temples, relative to the distance between mosques; and Panel C with an indicator for counties where at least 10% of Chinese adults were fluent in Malay in 1980. Columns 1 and 2 report estimates for Chinese and non-Chinese households, respectively; column 3 reports the difference. All regressions are estimated by OLS and include the main controls: state fixed effects, expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the household. Data from the Census of Population and the Ministry of Education. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table 10. Parameter Estimates

Parameter (1)	Description (2)	Value (3)	SE (4)
Panel A. Estimated Parameters			
κ	Distance elasticity of migration costs	0.517	(0.003)
α	Expenditure share on agriculture	0.310	(0.001)
θ	Skill dispersion	3.327	(0.035)
γ_A	Productivity spillover w.r.t. size, agr.	-0.120	(0.067)
γ_M	Productivity spillover w.r.t. size, non-agr.	0.223	(0.052)
γ^e	Productivity spillover w.r.t. ethnic share	0.132	(0.039)
β	Amenity spillover w.r.t. size	-0.005	(0.044)
β^e	Amenity spillover w.r.t. ethnic share	0.125	(0.091)
Panel B. External Parameters			
σ	Elasticity of substitution	8.00	
ν	Migration elasticity	3.00	
ξ	Distance elasticity of trade costs	0.18	

Notes: This tables reports parameter estimates from the model. Panel A reports parameters estimated from the data, as described in Section VII. Panel B reports three parameters that are assumed or calibrated using external moments: σ and ν from Balboni (2025), and ξ from Monte, Redding and Rossi-Hansberg (2018). Column 1 lists the parameter symbols, Column 2 provides descriptions, Column 3 reports estimates, and Column 4 shows bootstrap standard errors in parentheses.

Table 11. Aggregate Impact of the Emergency Resettlement

	Chinese (1)	Malays (2)	Total (3)
Changes in Outcomes Relative to Baseline:			
Non-Agricultural Employment Share	1.03	−0.95	−0.28
Output per Capita	1.54	2.37	2.01
For Agriculture	−0.13	1.28	0.38
For Non-Agriculture	1.61	3.58	2.84
Aggregate Output (% Baseline Output)	0.67	1.34	2.01
From Reallocation of Labor	0.45	1.03	1.48
From Changes in Productivity	−0.19	0.08	−0.10
From Joint Changes	0.40	0.23	0.63
Average Utility	1.21	6.68	4.82

Notes: This table shows changes in economic outcomes from the "no resettlement" baseline to the observed 1980 economy. The baseline equilibrium uses the 1947 population distribution as the initial condition, as opposed to the post-resettlement 1957 distribution. Results are reported for Chinese (Column 1), Malays (Column 2), and the overall economy (Column 3). The first panel reports percentage point changes in the share of non-agricultural employment. The second panel reports percent changes in output per capita—overall and by sector. The third panel reports changes in aggregate output as a share of baseline output, decomposed into contributions from labor reallocation, productivity changes, and their interaction. The final panel reports percent changes in average utility.

Table 12. Aggregate Impact of Reducing Cross-Ethnic Frictions

	Chinese (1)	Malays (2)	Total (3)
Changes in Outcomes Relative to Baseline:			
Non-Agricultural Employment Share	−1.72	0.81	−0.05
Output per Capita	4.35	3.99	4.14
For Agriculture	4.52	2.70	3.98
For Non-Agriculture	4.89	4.22	4.24
Aggregate Output (% Baseline Output)	1.89	2.26	4.14
From Reallocation of Labor	−0.50	0.52	0.02
From Changes in Productivity	2.09	1.58	3.67
From Joint Changes	0.29	0.16	0.45
Average Utility	1.24	4.63	3.47

Notes: This table shows changes in economic outcomes from the baseline economy to a counterfactual equilibrium where the cross-ethnic barrier to productivity spillovers γ^e is reduced by half. The baseline equilibrium uses the 1947 population distribution as the initial condition, as opposed to the post-resettlement 1957 distribution. Results are reported for Chinese (Column 1), Malays (Column 2), and the overall economy (Column 3). The first panel reports percentage point changes in the share of non-agricultural employment. The second panel reports percent changes in output per capita—overall and by sector. The third panel reports changes in aggregate output as a share of baseline output, decomposed into contributions from labor reallocation, productivity changes, and their interaction. The final panel reports percent changes in average utility.

Table 13. Aggregate Impact of Malay Wage Subsidies in Non-Agriculture

	Chinese (1)	Malays (2)	Total (3)
Changes in Outcomes Relative to Baseline:			
Non-Agricultural Employment Share	−8.14	9.17	3.29
Output per Capita	−2.12	3.36	0.98
For Agriculture	1.39	9.07	10.67
For Non-Agriculture	−0.60	−3.46	−4.20
Aggregate Output (% Baseline Output)	−0.92	1.90	0.98
From Reallocation of Labor	−0.86	1.21	0.35
From Changes in Productivity	−0.20	1.30	1.10
From Joint Changes	0.14	−0.60	−0.46
Average Utility	−10.03	8.24	2.03

Notes: This table shows changes in economic outcomes from the baseline economy to a counterfactual equilibrium with an 18% wage subsidy for Malays in non-agriculture, financed by a uniform 7.3% income tax to balance the government budget. The baseline equilibrium uses the 1947 population distribution as the initial condition, as opposed to the post-resettlement 1957 distribution. Results are reported for Chinese (Column 1), Malays (Column 2), and the overall economy (Column 3). The first panel reports percentage point changes in the share of non-agricultural employment. The second panel reports percent changes in output per capita—overall and by sector. The third panel reports changes in aggregate output as a share of baseline output, decomposed into contributions from labor reallocation, productivity changes, and their interaction. The final panel reports percent changes in average utility.

Online Appendices

Coercive Growth: Forced Resettlement and Ethnicity-Based Agglomeration

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A Data Appendix

This Appendix section describes data sources and construction details not covered in the main text. Appendix Table A.1 lists each data source along with the variables extracted.

A.1 Supplemental Data

The Malayan Christian Council Survey. The Malayan Christian Council surveyed the New Villages in 1958, listing more than 500 villages, including about 100 small villages established after 1954 that the Corry report does not cover. For villages documented in both sources, population counts are broadly consistent. The survey also records the number of families (in addition to the number of persons) and, for most villages, the main occupation. I use these data to assess demographic balance in the resettled population across areas with varying resettlement density (Appendix Section B.2).

Schools. I draw on two sources to measure school access. For historical data on Chinese schools, I digitize the 1959 Directory of Singapore and Malaya ("Xing Ma Tong Jian") (Ju, 1959), which lists the Chinese schools in the Federation of Malaya at the time, including their names and addresses. I geocode each school to obtain precise locations, which I use to construct county-level measures of school access.

For modern data, I use administrative records from Malaysia's Ministry of Education, which list all primary and secondary schools as of 2022. The dataset includes school names, the number of teachers and students, and geographic coordinates. I identify Chinese vernacular schools based on whether their names contain any Chinese characters.

Buildings. I use data from the Global Human Settlement Layer (GHSL) to measure built-up volumes starting in 1975. The data combine surface and height measurements at 100-meter resolution, derived from Sentinel-2 and Landsat satellite imagery.

Orbis data and firm ownership by ethnicity. I classify firms in the Orbis database as Chinese-owned or non-Chinese. Orbis provides annual ownership "Links" datasets that report direct and indirect ownership shares at the subsidiary-shareholder level (see Bureau van Dijk (2019) for details). I trace each firm's ownership structure from 2001 to 2021 to identify its ultimate owner, using the global ultimate owner (GUO) indicators: GUO50 and GUO25, which flag entities holding over 50% or 25% control, respectively. I define the ultimate owner as follows:

(i). If any GUO with at least 50% ownership is listed, I use that entity.

(ii). If no GUO exceeds 50% but some exceeds 25%, I use the 25%-plus owner(s).

(iii). If no GUO is listed, I use the largest immediate shareholder with over 25% ownership.[50]

I classify an ultimate owner as a natural person if Orbis lists its entity type as "I" (individual or family). Some firms lack a person owner in certain years due to dispersed ownership or missing data. To reduce measurement error, I use each firm's full ownership history from 2001 to 2021. I classify a firm as Chinese-owned if any of its ultimate person owners at any point during this period has a name containing any Chinese characters (e.g., "Lee," "Tan," "Lim," "Chen").

A.2 Measuring Ethnic Segregation and Divisions

This Appendix section describes how I measure ethnic segregation using two complementary approaches.

A.2.1 Dissimilarity Index

I begin with the standard dissimilarity index Duncan and Duncan (1955), widely used in studies of racial and ethnic segregation. Since the most granular population data are at the county level, I compute the index at the district level, where a district contains about twelve counties on average. I use the 1947 population data to measure pre-resettlement segregation.

Formally, the dissimilarity index for district r is defined as:

$$DI_r = \frac{1}{2} \sum_n \left| \frac{L_n^c}{\sum_n L_n^c} - \frac{L_n^m}{\sum_n L_n^m} \right|,$$

where n indexes counties in district r, L_n^c is the Chinese population in county n, and L_n^m is the non-Chinese population in county n.

The index ranges from 0 to 1 and measures the share of individuals from either group (Chinese or non-Chinese) who would need to relocate across counties for all counties in the district to have identical ethnic composition.

While intuitive and easy to interpret, this index has some limitations. Most importantly, it suffers from the "checkerboard problem": it ignores the spatial relationship of counties. A district where Chinese-majority counties are geographically clustered would yield the same dissimilarity as one where these counties are more dispersed. The index is also sensitive to the size and shape of geographic units. See White (1983) for further discussion.

A.2.2 Distance Segregation Index

To complement the index of dissimilarity, I construct a distance-based measure of segregation based on the spatial distribution of religious sites prior to resettlement. Unlike the dissimi-

[50] Ownership thresholds in the 20–25% range are commonly used to define control (La Porta, Lopez-de Silanes and Shleifer, 1999; Claessens, Djankov and Lang, 2000).

larity index, this measure takes into account spatial proximity, and thus is less prone to the limitations discussed above. Because nearly all Malays were Muslim and most other ethnic groups were not, the location of mosques and non-Muslim temples was correlated with the distribution of ethnic groups. Indeed, the number of mosques in a county increases by about 0.59% for every 1% increase in the 1947 Malay population. I digitized 1945 topographic maps and geocoded all mosques and non-Muslim temples across British Malaya.

The index follows Harari (2024) and is related to the proximity index introduced by White (1983). It measures the spatial separation between mosques and temples, normalized by the average distance between any two religious sites.

Formally, the distance segregation index for county n is constructed in three steps:

(i). For each mosque, calculate its distance to the nearest non-Muslim temple, d_i^{mt}; for each temple, calculate the distance to the nearest mosque, d_i^{tm}. Also, for each religious site, calculate the distance to the nearest other religious site of any kind, d_i.[51]

(ii). Average these distances to the county level:

$$D_n^{mt} = \frac{\sum_i d_i^{mt} \cdot \mathbf{1}\{i \in S_n^m\}}{\sum_i \mathbf{1}\{i \in S_n^m\}};$$
$$D_n^{tm} = \frac{\sum_i d_i^{tm} \cdot \mathbf{1}\{i \in S_n^t\}}{\sum_i \mathbf{1}\{i \in S_n^t\}};$$
$$\bar{D}_n = \frac{\sum_i d_i \cdot \mathbf{1}\{i \in S_n^m \cup S_n^t\}}{\sum_i \mathbf{1}\{i \in S_n^m \cup S_n^t\}},$$

where S_n^m and S_n^t are the sets of mosques and temples in county n.

(iii). Define the distance segregation index as:

$$DSI_n = \frac{D_n^{\text{cross}}}{\bar{D}_n},$$

where D_n^{cross} is the weighted average distance between mosques and non-Muslim temples in the county:

$$D_n^{\text{cross}} = \frac{N_n^m D_n^{mt} + N_n^t D_n^{tm}}{N_n^m + N_n^t},$$

and N_n^m and N_n^t are the number of mosques and temples in county n, respectively.

[51] I allow the nearest religious site to lie outside the county. This differs from Harari (2024), who calculates neighborhood distances within cities. Since segregation patterns likely extended beyond county boundaries, this approach ensures that separation is still measured in counties that contain only one type of religious site (e.g., only mosques or only temples).

The distance segregation index captures the relative spatial separation between mosques and temples, scaled by the average distance between religious sites in the county. A value higher than one indicates greater-than-random separation between religious groups. In areas with greater ethnic tension, it may have been more difficult for one group to establish religious sites near the other. As such, the index may reflect both spatial and social dimensions of segregation.

A.3 Data Imputation for Structural Estimation

While the 1980 population census 2% microdata is representative, some counties lack ethnicity-by-sector income data. To improve geographic coverage for structural estimation, I impute missing average incomes using the following two-step procedure.

First, if a county-sector-ethnicity cell is missing but the corresponding district-sector-ethnicity cell is observed, I impute income using the district-level average. This assumes that the average income for an ethnic group in a sector is similar across counties within the same district (districts average about twelve counties). Appendix Figure A.10 supports this assumption: household earnings for an ethnic group in a sector are centered around the district average.

Second, if a county-sector-ethnicity cell is missing and no district-level value is available, but income data exist for the same county and ethnicity in the other sector, I impute income using that sector's average. This assumes sectoral earnings are similar within a county-ethnicity cell, consistent with the assumption of Fréchet-distributed productivity. Appendix Figure A.11 supports this assumption by showing that household earnings in both sectors are centered around the same county-ethnicity mean.

B Empirical Results

This Appendix section first formalizes the identification assumptions outlined in Section IV, then presents supplementary analyses and robustness checks.

B.1 Identification Assumptions

To illustrate the assumptions, I express county resettlement density in terms of smaller geographic units at which resettlement occurred:

$$ResettleDensity_c = f_c(g_1, g_2) \equiv \operatorname{asinh}\left(\frac{\sum_{i \in c} g_{1i} \times g_{2i}}{area_c}\right),$$

where $i = 1, \ldots, I$ indexes resettlement "sites," each potentially hosting at most one New Village, with a total of I sites in the state. The indicator g_{1i} denotes whether site i was selected for resettlement, and g_{2i} is the number of people resettled to that site. $area_c$ denotes the area of county c.[52]

Based on historical accounts, I make two assumptions regarding the selection of sites ($g_1 \equiv \{g_{1i}\}_{i=1}^I$) and the relocation of population to sites ($g_2 \equiv \{g_{2i}\}_{i=1}^I$). First, I assume that site selection is orthogonal to county-level unobservables $\varepsilon \equiv \{\varepsilon_c\}_{c=1}^C$ (where C is the number of counties in the state), conditional on a vector of site-level characteristics $w_1 \equiv \{w_{1i}\}_{i=1}^I$, including distance to the transportation network, land-use type, and the decile of the county's squatter population. Second, conditional on selected sites g_1 and a vector of characteristics $w_2 \equiv \{w_{2i}\}_{i=1}^I$—including distances to initial squatter settlements and their populations—the number of people resettled to each site is also orthogonal to ε. These assumptions are formalized as follows.

Assumption 1. (Resettlement Exogeneity)

(i) (Site selection) $g_1 \perp\!\!\!\perp \varepsilon \mid w_1$: Conditional on distance to transportation, land-use patterns, and the decile of county squatter population, site selection was exogenous.[53]

(ii) (Number resettled) $g_2 \perp\!\!\!\perp \varepsilon \mid (g_1, w_2)$: Conditional on selected sites and the initial squatter distribution, the number resettled to a site was exogenous.

Under Assumption 1, the potential source of omitted variable bias is the conditional expectation of resettlement density $\mathbb{E}[f_c(g_1, g_2)|w]$, where $w \equiv (w_1, w_2)$. As shown by Borusyak and Hull (2023), the coefficient β is identified if this expected density is either controlled for or used to re-center the realized resettlement density.

To measure $\mathbb{E}[f_c(g_1, g_2)|w]$, I leverage institutional knowledge and impose two additional assumptions on the distributions of g_1 and g_2, denoted by $G_1(\cdot)$ and $G_2(\cdot)$, respectively.

Assumption 2. (Resettlement Design)

(i) (Equally suitable sites) $G_1(g_1|w_1)$ is uniform: All sites were equally likely to be selected conditional on their distance to transportation, land-use patterns, and the decile of county squatter population.

(ii) (Minimizing dislocation) $\mathbb{E}[f_c(g_1, g_2)|g_1, w] = f_c(g_1, \bar{g}_2(g_1, w))$: Conditional on selected village sites and the initial distribution of Chinese squatters, the average relocation followed a gravity-based model, where

$$\bar{g}_2(g_1, w) = \sum_{j=1}^{J} n_{j \to i} = \sum_{j=1}^{J} n_j \times \frac{d_{ji}^{-\psi}}{\sum_{s=1}^{I} d_{js}^{-\psi}}.$$

[52]The variables g_{1i} and g_{2i} are interdependent, as no resettlement occurs at a site unless it was selected.

[53]A weaker assumption of mean independence between g_1 and ε, conditional on w, suffices for identification.

Here, n_j is the population of Chinese squatters at origin j, d_{ji} is the distance between j and site i, and ψ is the elasticity of relocation costs with respect to distance.

Assumption 2(i) states that the British considered observationally similar sites as equally suitable, while 2(ii) captures their stated objective of minimizing dislocation, albeit subject to idiosyncratic constraints. Together, these imply:

$$\begin{aligned}
\mathbb{E}\left[f_c(g_1, g_2) \mid w\right] &= \int_{G_1}\int_{G_2} f_c\left(g_1, g_2\right) dG_2\left(g_2|g_1, w\right) dG_1\left(g_1|w\right) \\
&= \int_{G_1} f_c\left(g_1, \bar{g}_2\right) dG_1\left(g_1|w\right),
\end{aligned}$$

where the first equality follows from the law of iterated expectation and the second equality follows from Assumption 2(ii).

I approximate this conditional expectation using a permutation procedure, indexed by $i = 1, \ldots, S$, and implemented independently for each state.

(i). Randomly (and uniformly) permute counterfactual New Village sites $g_1^{(s)}$, conditional on w_1.

(ii). Calculate gravity-based counterfactual resettled populations $\bar{g}_2^{(s)}$.

(iii). Calculate counterfactual county resettlement density as $f_c(g_1^{(s)}, \bar{g}_2^{(s)})$.

The expected resettlement density is then approximated by averaging the counterfactual county resettlement density across permutations:

$$\overline{ResettleDensity}_c \equiv \frac{1}{S}\sum_{s=1}^{S} f_c(g_1^{(s)}, \bar{g}_2^{(s)}). \tag{A-1}$$

Empirically, actual county resettlement densities center around the expected densities (Appendix Figure A.3), suggesting that the gravity-based model captures the major component of resettlement patterns. While not required for identification, site-level resettled populations also align closely with model predictions.

B.2 Supplementary Analysis

Characteristics of resettled Chinese. As discussed in Section II, about 60 percent of the resettled individuals were employed in food or rubber cultivation—an agricultural share slightly higher than that of the average Chinese. To further understand the characteristics of the resettled Chinese, I turn to the Second Malaysian Family Life Survey, conducted in the late 1980s, which includes "New Village" as a type of place in the migration records.

There, I classify resettled Chinese as Chinese individuals not born in a New Village but who reported having "migrated" there before 1960.

Appendix Table A.8, Panel A, shows that resettled Chinese were more agricultural and less educated than the average Chinese of the same gender in similar age in the same state: they were 15 percentage points more likely to employ in agriculture as their first job (Column 1), 13 percentage points less likely to finish primary education (Column 2), and 15 percentage points less likely to finish secondary education (Column 3).[54]

Panel B shows that resettled Chinese were still 16 percentage points less agricultural than the average Malays of similar age in the same state (Column 1). Educational attainment of the resettled Chinese was not statistically different than the average Malays (Columns 2–3).

Balance of compositions of the resettled population. Section IV.B shows that counties with higher resettlement are balanced on key pre-existing conditions after controlling for main covariates and expected resettlement density. A remaining concern is that the composition of the inflowing population might differ across counties with varying resettlement density, perhaps because some counties were closer to areas with different unobserved characteristics. While controlling for expected resettlement density helps address this, I further examine this concern using the Malayan Christian Council Survey, which provides additional information on the resettled population.

Appendix Table A.13 shows that resettled villagers in counties with higher resettlement density had similar demographic and economic compositions. Average family size was comparable (Column 1), with households in higher-resettlement counties averaging 0.1 additional members. This suggests little difference in household socioeconomic status, which typically correlates with household size. The survey also reports the main occupation for most villages, and I find no evidence of systematic differences in occupational structure across counties with varying resettlement density (Columns 2–3).

Built-up volume. Higher population density in more resettled areas was accompanied by a substantial increase in build-up volume (Appendix Table A.4). By 1975, counties with higher resettlement had 17% more buildings (Column 1). The larger percentage increase in buildings relative to population (11%, Table 2) suggests a relatively elastic housing supply. At a finer scale, 1990 satellite imagery shows dense building clusters around New Villages, in contrast to the more uniform settlement patterns in surrounding areas before the Emergency (Appendix Figure A.5).

[54]On average, about 36% of the resettled Chinese reported to be employed in agriculture as their first job. This share is much lower than what historical accounts suggest, likely due to a selective set of more industrial and richer individuals that survived until being surveyed in the late 1980s.

Internal migration and fertility change. Post-1960 population growth in counties with higher resettlement was driven by internal migration rather than fertility. The 1980 census microdata indicates whether a person was an internal migrant (from another village or town in Malaysia), an external migrant (from outside Malaysia), or a non-migrant, as well as how long they had lived in their current locality.

Appendix Table A.5 shows that, by 1980, individuals were slightly more likely (though not statistically significantly so) to be internal migrants in counties with higher resettlement density, particularly those who moved to their 1980 locality after 1960 (Panels A and B). However, households in these counties did not have more children (Panel C), nor did they have larger household sizes. If anything, non-Chinese women had slightly lower fertility rates in counties with higher resettlement.

B.3 Alternative Mechanisms

This appendix expands on Section V.E by discussing additional potential explanations for the observed effects of resettlement in the receiving counties.

Selective migration after the Emergency. After mobility restrictions were lifted in 1960, subsequent voluntary migration into counties with higher resettlement may have been selective. Those who moved might be wealthier or more educated than local incumbents, given the presence of fixed migration costs. Appendix Table A.9 shows that when the sample is restricted to non-migrant households (those without any migrant household members), the positive income gains for Chinese households remain similar, albeit slightly attenuated (Column 1). Interestingly, the gains for non-Chinese are even smaller in the non-migrant sample (Column 2), suggesting that income improvements among them may have been largely driven by selective in-migration of those better positioned to benefit from higher Chinese densities.

Land ownership. Resettled families were allocated house lots and, if previously farmers, agricultural land. Land ownership may improve labor productivity through various mechanisms, such as encouraging investment in land (e.g., Besley (1995); Banerjee, Gertler and Ghatak (2002)), improving access to credit (Field and Torero, 2006), among others. These benefits could be passed to the next generation.

Appendix Table A.8 shows that, on average, resettled Chinese households owned roughly 3 acres of land by the late 1980s, consistent with the allocation of a one-sixth-acre house lot and two acres of farmland (Sandhu, 1964*b*). However, this was substantially less than the land owned by the average Chinese or Malays in the same state: resettled Chinese owned roughly 105 fewer acres than the average Chinese household and 47 fewer acres than the average Malay household (Column 4). While land ownership may have benefited resettled households and their descendants—relative to a counterfactual with no titles—local incumbents were so much wealthier in land holdings that land ownership alone is unlikely to explain the improved outcomes in more resettled counties.[55]

Education. Educational improvements may partly reflect better school access (supply) or shifting preferences toward education (demand). On the supply side, the primary Chinese schools commonly established in the New Villages may have lowered the cost of education for resettled households and nearby incumbents. To assess this channel, I compile a comprehensive list of Chinese vernacular schools in 1959 and all primary and secondary schools in Malaysia in 2022. Appendix Table A.10 shows that counties with higher resettlement density did not have greater access to Chinese vernacular schools in either 1959 (Column 1), near the end of the Emergency, or in 2022 (Column 2). If anything, access to non-Chinese schools in modern times is slightly better in these counties (Column 3), though the effect is small: counties with higher resettlement are, on average, 87 meters closer to the nearest non-Chinese school (Panel A, Column 3).

On the demand side, the *uprooted hypothesis* posits that forced migration may have increased the incentives of resettled individuals and their descendants to invest in human capital (Becker et al., 2020; Michalopoulos et al., 2025). Since the 1980 census does not distinguish descendants of the resettled from other Chinese, I cannot isolate this channel from broader agglomeration effects. Nevertheless, Appendix Table A.11 shows that the income effects of higher resettlement vary systematically by ethnicity across all education levels. While individuals with higher education benefited more—consistent with higher returns to specialization in these counties—the ethnic gap in income gains was even larger among the more educated, suggesting that education alone does not fully explain the much higher productivity among Chinese in these counties.

Roads. During the Emergency, the British colonial government built roads to connect remote villages to the main transport network when no road or river access was available. These early infrastructure investments may have had persistent effects through path dependence. While road access would not necessarily favor one ethnic group over another, I examine its relationship with county resettlement density in the two decades following resettlement.

Appendix Table A.12 shows that in 1961, shortly after the Emergency, counties with higher resettlement density had slightly greater density of local roads but no better access to main roads (Panel D, Column 1), consistent with historical accounts. Over the next two decades, road access improved modestly in these counties, possibly reflecting input sharing as a channel of agglomeration economies, though the differences are not statistically significant (Column 3). Overall, the differences are small—about 400 meters closer to minor roads and

[55] Historical accounts also indicate that many resettled households received land titles only years later, and some were unaware of their land rights (Strauch, 1981, pp. 63–72).

12 meters per square kilometer higher road density—and the main results are robustness to directly controlling for 1983 road access (Appendix Table A.16).

Direct effects of counterinsurgency. A potential concern is that counterinsurgency itself may have had lasting economic impacts. Since the empirical analysis focuses on areas that received resettlement, however, several contextual factors suggest this channel is unlikely to be a main driver of the results.

The Briggs Plan was largely successful in suppressing insurgent activity, which peaked between 1950 and 1952 and declined sharply afterward as the communists lost their support networks due to resettlement (Short, 1975). Most counterinsurgency operations occurred in jungle areas where communists were based, whereas resettlement sites were typically located along major roads and closer to towns.

While insurgents occasionally attacked estates to disrupt the colonial economy, these areas were more often the origins of resettlement than its destinations. If anything, counties that received more resettlement due to their proximity to communist risks may have experienced greater disruption of physical and social capital, which would bias the results against finding positive effects on industrialization and Chinese incomes.

Differential scale elasticity across ethnic groups. The resettlement program shifted both population size and ethnic composition. While the reduced-form estimates capture their joint effect, the model imposes a parametric structure on agglomeration externalities that allows me to separately identify the effect of size from that of ethnic share. Specifically, I assume a common agglomeration elasticity with respect to sectoral employment size across ethnic groups, as the data lack sufficient power to flexibly estimate group-specific scale elasticities.

One potential interpretation of the results is that Chinese workers had a higher scale elasticity and thus benefited more from density. While theoretically possible, it is unclear what microfoundations would justify such differences. For instance, why would individuals of different ethnicity—but with similar education and employed in the same sector—systematically experience different returns to labor pooling, input sharing, or knowledge spillovers? Moreover, differences in scale elasticity alone are unlikely to explain the heterogeneous effects of resettlement across regions with varying degrees of segregation and ethnic divisions (Table 9). A more plausible explanation for the observed patterns seems to be that cross-ethnic spillovers were weaker than within-ethnic spillovers.

B.4 Robustness

This Appendix section examines the robustness of the main results to alternative definitions of expected resettlement density, different covariate controls, and sample restrictions.

B.4.1 Alternative Specifications of Expected Resettlement Density

Appendix Table A.14 begins by showing estimates without controlling for expected resettlement density (row 2). Most results change only slightly, but the effect on agricultural employment is 33% smaller. This likely reflects omitted variable bias: counties with similar baseline covariates but higher expected resettlement density tended to be closer to more urbanized and industrialized regions before resettlement.

Next, I explore robustness to alternative specifications of expected resettlement density by adding different constructions to the baseline specification. The baseline assumes the British prioritized siting near rivers when no roads were within 5 kilometers. Results are similar when assuming a preference for roads over rivers up to 10 kilometers (row 3).

While the baseline allows counterfactual villages to be arbitrarily close—which is plausible given that some observed villages are only 200 meters apart—I also consider imposing a 1-kilometer minimum spacing between counterfactual villages, as most are at least 1 kilometer apart. Results are similar (row 4).

The expected number of squatters resettled to each counterfactual site is also robust to alternative squatter definitions. The baseline defines squatters as Chinese communities living within 5 kilometers of the forest. Estimates are similar when using cutoffs that are half or twice the baseline distance (rows 5–6).

The baseline resettlement cost elasticity with respect to distance is calibrated to 0.65 using observed villager populations. A higher elasticity implies that counterfactual resettlement density more closely mirrors the original squatter density. Results are robust to alternative elasticity values that are 20 percent lower and higher (rows 7–8).

Finally, the baseline population shifter is defined as the inverse hyperbolic sine of the number of resettled persons per unit area. As county resettlement density is used to shift log population density in the structural estimation, this log-like transformation improves the power of the first stage. Results are similar when using a log transformation, restricting the sample to resettled counties (row 9).[56]

B.4.2 Additional Controls for Determinants of Resettlement Density

Although expected resettlement density helps absorb omitted variable bias from broader road and settlement patterns, this appendix further tests robustness by more flexibly controlling for road networks and population distributions—key determinants of resettlement density.

Appendix Table A.15 begins by augmenting the baseline specification with cubic polynomials of county total population and Chinese share in 1947 (row 2). The estimates remain

[56] Since regressions include an indicator for any resettlement, identification comes only from variation among resettled counties. Non-resettled counties are included to improve precision in estimating the effects of covariates.

similar.

Next, I examine robustness to potential spillovers from road and population patterns beyond county borders, within 3 to 5 kilometers. This range is particularly relevant because it was believed that the communists targeted areas within two miles of population clusters for access to food and information. In response, the British may have strategically sited New Villages either farther away for security or closer for monitoring and to minimize dislocation.

Rows 3 and 4 add controls for distance to the nearest road and road density in neighboring areas within 3 km and 5 km, respectively. Rows 5 and 6 add controls for total population, Chinese share, and indicators for the presence of towns (clusters of 1,000–5,000 people) and cities (over 5,000 people) within the same distance bands.

The estimates remain largely stable when these neighboring controls are added individually or together (row 7), suggesting that selection based on initial spatial patterns of roads and population does not drive the main results, or is already captured by the baseline controls.

B.4.3 Additional Controls for Initial Economic Activities

Section IV.B shows that key pre-period characteristics are largely balanced in the baseline specification. This appendix further tests robustness by adding controls for initial economic conditions, as well as geographic characteristics that may have shaped those conditions.

Appendix Table A.16 begins by controlling for topographical features—elevation, ruggedness, and slope (row 2). These terrain characteristics are closely related to drainage and agricultural suitability, both key factors in site selection.

Although the balance table (Table 1) shows that point estimates are economically small, some covariates, such as rice and coconut suitability, become statistically insignificant due to larger standard errors rather than smaller coefficients. To address this, I directly control for the suitability and land use shares of paddy rice (row 3) and coconut (row 4).

Because the British prioritized state-owned land for resettlement, which might initially exhibit different land use patterns, I also assess robustness to controlling for land use patterns observed in 1943 maps. The estimates are similar when adding controls for land use shares in categories outside of rice and coconut: pineapple, mangrove, forest, swamp, and town (row 5).

While section IV.B shows that industrial location fundamentals are balanced, results are similar when directly controlling for distance to prewar industrial facilities (row 6) and to major cities (row 7).

Finally, to assess whether the results are driven by road construction in areas where New Villages were established, I control for road access measured in 1983 (row 8). These include average distance to and density of major roads and local roads. The estimates remain stable, suggesting that road construction during or after the Emergency is not the main driver.

B.4.4 Sample Restrictions

Counties in the baseline sample vary in size, with some large, sparsely populated counties inland and smaller, denser counties along the coast. While the main specification controls for county area, Appendix Table A.17 shows that results are robust to excluding the largest and smallest counties (row 2), counties with extreme resettlement density (row 3), or the most densely populated prewar towns (row 4). Finally, since individual- and household-level outcomes come from the 2% microdata of the 1980 Population Census, which covers only about two-thirds of the baseline counties with sampled Chinese, I restrict the sample to these counties and find similar results (row 5).

C Theoretical Model and Estimation

This section first provides additional details on the spatial general equilibrium model. It then discusses the identification and estimation of model parameters and compares the estimates to those in the literature. Finally, it shows how I derive a lower bound for the utility loss from forced resettlement, and assesses robustness of the counterfactual results to alternative parameter values.

C.1 Model Details

C.1.1 Sectoral Labor Supply

I now derive key equations pertaining to sectoral labor supply. Individuals draw their efficiency units independently across sectors of agriculture and manufacturing $\Lambda^e = (\Lambda^e_A, \Lambda^e_M)$ from the joint distribution:

$$F^e_n(\Lambda_A, \Lambda_M) = \prod_{k=A,M} F^e_{nk}(\Lambda_k),$$

where the marginal probability distribution is Fréchet:

$$F^e_{nk}(\Lambda_k) = \exp\left(-\phi^e_{nk}\Lambda_k^{-\theta}\right).$$

After knowing their efficiency units, they choose the sector that pays higher earnings. Let λ_{ink} be the net efficiency unit of individual i in industry k and region n, and w_{nk} be the wage per efficiency unit for the region-industry. The earnings of individual i of ethnicity e

in industry k, location n is thus:

$$\begin{aligned} y^e_{ink} &= w_{nk}\lambda^e_{ink} \\ &= w_{nk}\Lambda^e_{ink} f(L^c_{nk}, L^m_{nk}) \\ &= w^e_{nk}\Lambda^e_{ink}, \end{aligned}$$

where

$$w^e_{nk} \equiv w_{nk} f(L^c_{nk}, L^m_{nk}).$$

Since y^e_{ink} equals a constant w^e_{nk} multiplied by a Fréchet random variable Λ^e_{ink}, it is also Fréchet distributed with shape θ and scale $\phi^e_{nk}(w^e_{nk})^\theta$. The expected earnings for ethnicity e in industry k and region n is thus $\Gamma_\theta \left(\phi^e_{nk}(w^e_{nk})^\theta\right)^{1/\theta}$.

For an individual of ethnicity e in region n, the probability of working in industry k is:

$$\pi^e_{nk} \equiv \mathbb{P}(y^e_{ink} = \max_s y^e_{ins}) = \frac{\phi^e_{nk}(w^e_{nk})^\theta}{\sum_s \phi^e_{ns}(w^e_{ns})^\theta} = \phi^e_{nk}\left(\frac{w^e_{nk}}{\bar{w}^e_n}\right)^\theta, \tag{A-2}$$

where

$$\bar{w}^e_n \equiv \left(\phi^e_{nA}\left(w^e_{nA}\right)^\theta + \phi^e_{nM}\left(w^e_{nM}\right)^\theta\right)^{1/\theta}.$$

People of ethnicity e choose the sector that pays more, and this process continues until the (e-specific) earning equalize across sectors. In equilibrium, the average wage for ethnic group e in region n is given by:

$$\mathbb{E}[\max_k y^e_{ink}] = \Gamma_\theta\left(\sum_k \phi^e_k(w^e_{nk})^\theta\right)^{1/\theta} = \Gamma_\theta \bar{w}^e_n.$$

Moreover, due to the Fréchet property, ethnic group e in region n attain, on average, the same earning across the two sectors.

It follows that $\mathbb{E}[\max_k y^e_{ink}]/w_{nk}$, the average efficiency unit of group-e in region n and sector k, is given by $\Gamma_\theta \bar{w}^e_n w^{-1}_{nk}$, which can be rewritten in terms of occupation shares using Equation (A-2) as:

$$\Gamma_\theta\left(\phi^e_{nk}\right)^{1/\theta}\left(\pi^e_{nk}\right)^{-1/\theta} f(L^c_{nk}, L^m_{nk}).$$

The neoclassical force $(\pi^e_{nk})^{-1/\theta}$ implies that, due to selection, a higher share of labor supply tends to lower the average skill in the sector. In contrast, the externality term $f(L_{nk}^c, L_{nk}^m)$ tends to raise average productivity by increasing the effective skills through agglomeration.

The aggregate sectoral earnings of ethnicity e in industry k and region n equal the ethnic population in region n, multiplied by the share working in sector k and by their average earnings in that sector:

$$w_{nk} H_{nk}^e = L_n^e \pi_{nk}^e \Gamma_\theta \bar{w}_n^e.$$

This implies that the aggregate human capital supply in industry k, region n is:

$$\begin{aligned} H_{nk} &= \Gamma_\theta \sum_e L_n^e \phi_{nk}^e (w_{nk}^e)^\theta w_{nk}^{-1} (\bar{w}_n^e)^{1-\theta} \\ &= \Gamma_\theta \sum_e L_n^e \phi_{nk}^e w_{nk}^{-1} w_{nk}^e (w_{nk}^e)^{\theta-1} (\bar{w}_n^e)^{1-\theta} \\ &= \Gamma_\theta \sum_e L_n^e \phi_{nk}^e \left(L_{nk}\right)^{\gamma_k} \left(\frac{L_{nk}^e}{L_{nk}}\right)^{\gamma^e} \left(\frac{w_{nk}^e}{\bar{w}_n^e}\right)^{\theta-1}. \end{aligned}$$

C.1.2 Justifying Fréchet

Assuming Fréchet distributed productivity draws provides computational tractability, as it leads to a closed-form expressions for the share of group-e individuals in region n who choose a given industry. This Appendix section provides empirical support for this assumption.

With a common shape parameter across sectors, Fréchet productivity draws imply that average wages for an ethnic group within a county equalize across sectors in equilibrium. Appendix Figure A.11 tests this prediction by plotting, separately for Chinese and non-Chinese, the distributions of log household earnings—demeaned by county-ethnicity average—for those working in agriculture and non-agriculture. For both ethnic groups, earnings are centered around the county-ethnicity mean across sectors, consistent with the Fréchet-based prediction.

C.1.3 Migration

Individuals of group e draw an idiosyncratic taste shock for each location and decide where to migrate before knowing their efficiency units. The taste shock u_n^e is drawn from the following location-specific Fréchet distribution:

$$F_n^e(a) = \exp\left(-\bar{a}_n^e a^{-\nu}\right),$$

where the scale parameter $\bar{a}_n^e$ captures the average attractiveness of location n for group e, and the shape parameter ν captures the dispersion of taste, assumed to be the same across groups and locations.

The value of relocating from r to n for ethnicity e is:

$$V_{rn}^{e}=\eta_{rn}^{-1}a_{n}^{e}\Gamma_{\theta}\bar{w}_{n}^{e}P_{n}^{-1}$$

where η_{rn} is the migration cost and the amenity term a_{n}^{e} depends on the local population:

$$a_{n}^{e}=u_{n}^{e}\left(L_{n}\right)^{\beta}\left(\frac{L_{n}^{e}}{L_{n}}\right)^{\beta^{e}}.$$

As V_{rn}^{e} is a Fréchet random variable u_{n}^{e} multiplied by a constant $\eta_{rn}^{-1}L_{n}^{\beta}(L_{n}^{e}/L_{n})^{\beta^{e}}\Gamma_{\theta}\bar{w}_{n}^{e}P_{n}^{-1}$, it is itself Fréchet distributed. This implies that the probability of relocating from r to n for ethnicity e is:

$$m_{rn}^{e}\equiv\mathbb{P}\left(V_{rn}^{e}=\max_{l}V_{rl}^{e}\right)=\frac{\bar{a}_{n}^{e}\left(\eta_{rn}^{-1}\left(L_{n}\right)^{\beta}\left(L_{n}^{e}/L_{n}\right)^{\beta^{e}}\bar{w}_{n}^{e}P_{n}^{-1}\right)^{\nu}}{\sum_{l=1}^{N}\bar{a}_{l}^{e}\left(\eta_{rl}^{-1}\left(L_{l}\right)^{\beta}\left(L_{l}^{e}/L_{l}\right)^{\beta^{e}}\bar{w}_{l}^{e}P_{l}^{-1}\right)^{\nu}}.$$

C.1.4 Iterative Procedure for Solving the Equilibrium

Given the model parameters and the inferred location fundamentals, I solve for equilibrium quantities and prices using an iterative approach with three nested loops. The outer loop solves for population by ethnic group $\{L_{n}^{e}\}$; the second loop, given $\{L_{n}^{e}\}$, solves for sector-specific wages per efficiency unit $\{w_{nk}\}$; and the inner loop, taking $\{L_{n}^{e},w_{nk}\}$ as given, solves for occupation shares $\{\pi_{nk}^{e}\}$, prices, and incomes.

The iterative algorithm starts with an initial guess for the equilibrium population distribution $\{L_{n}^{e}\}$, followed by the steps below.

1. Solve for wages $\{w_{nk}\}$:
    (a) Set an initial guess for wages $\{w_{nk}\}$.
    (b) Solve for occupational choices $\{\pi_{nk}^{e}\}$:
        i. Set an initial guess for $\{\pi_{nA}^{e}\}$ and calculate $\pi_{nM}^{e}=1-\pi_{nA}^{e}$.
        ii. Calculate sectoral employment $L_{nk}^{e}=L_{n}^{e}\pi_{nk}^{e}$.
        iii. Calculate the average wage by ethnic group:

$$\bar{w}_{n}^{e}=\left(\phi_{nA}^{e}(w_{nA}^{e})^{\theta}+\phi_{nM}^{e}(w_{nM}^{e})^{\theta}\right)^{1/\theta},$$

where

$$w^e_{nk} = w_{nk}(L_{nk})^{\gamma_k} \left(\frac{L^e_{nk}}{L_{nk}}\right)^{\gamma^e}.$$

iv. Calculate the implied occupational shares:

$$\tilde{\pi}^e_{nk} \equiv \phi^e_{nk} \left(\frac{w^e_{nk}}{\bar{w}^e_n}\right)^{\theta}.$$

v. Update the occupational choices iteratively until convergence, using:

$$\pi^e_{nk,new} \equiv \iota \pi^e_{nk} + (1-\iota)\tilde{\pi}^e_{nk},$$

where $\iota \in (0,1)$ is the relaxation parameter in the Gauss-Seidel update. A lower ι accelerates the process but is more prone to overshooting and instability. I set $\iota = 0.95$ in practice.

(c) Calculate prices $\{p_{nrk}\}$ with $p_{nnk} = w_{nk}\tau_{nn}$ and $p_{nrk} = p_{nnk}\left(\frac{\tau_{nr}}{\tau_{nn}}\right)$, where τ_{nn} is the within-county trade cost, which can be greater than 1.

(d) Calculate labor efficiency $\{H_{nk}\}$ with $H_{nk} = \sum_e H^e_{nk}$, where $H^e_{nk} = \Gamma_\theta L^e_{nk}\left(\frac{\bar{w}^e_n}{w_{nk}}\right)$.

(e) Solve for regional income $\{Y_n\}$, such that $\sum_n Y_n = 1$.

i. Calculate total income of n by summing its trade flow expenditures over k and r:

$$Y_n = \sum_r \sum_k Y_r \underbrace{\alpha_k \left(\frac{p^{1-\sigma}_{nrk}}{\sum_l p^{1-\sigma}_{lrk}}\right)}_{\equiv \tilde{p}_{nrk}} = \sum_r \sum_k \tilde{p}_{nrk} Y_r = \sum_r \tilde{p}_{nr} Y_r,$$

where $\tilde{p}_{nr} \equiv \tilde{p}_{nrA} + \tilde{p}_{nrM}$. In matrix form, this can be written as:

$$Y = \tilde{P}Y \iff (I - \tilde{P})Y = \mathbf{0},$$

where

$$\tilde{P} = \begin{bmatrix} \tilde{p}_{11} & \cdots & \tilde{p}_{1N} \\ \vdots & \ddots & \vdots \\ \tilde{p}_{N1} & \cdots & \tilde{p}_{NN} \end{bmatrix}.$$

ii. Since this system has rank $N-1$, I impose $\sum_n Y_n = 1$ as a numeraire to pin down the level of Y. By dropping the last equation from above and replacing it with $\sum_n Y_n = 1$, I obtain:

$$\begin{bmatrix} 1-\tilde{p}_{11} & \cdots & -\tilde{p}_{1N} \\ \vdots & \ddots & \vdots \\ 1-\tilde{p}_{N-1,1} & & -\tilde{p}_{N-1,N} \\ 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} Y_1 \\ \vdots \\ Y_{N-1} \\ Y_N \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

(f) Calculate the implied wages: $\tilde{w}_{nk} = Y_{nk}/H_{nk}$, where $Y_{nk} = \sum_r \tilde{p}_{nrk} Y_r$.

(g) Update wages iteratively until convergence:

$$w_{nk,new} \equiv \iota w_{nk} + (1-\iota)\tilde{w}_{nk}.$$

2. Calculate migration shares:

$$m_{rn}^e = \frac{\left(\eta_{rn}^{-1} V_n^e\right)^\nu}{\sum_{l=1}^N \left(\eta_{rl}^{-1} V_l^e\right)^\nu},$$

where

$$V_n^e = \left(\bar{a}_n^e\right)^{1/\nu} L_n^\beta \left(\frac{L_n^e}{L_n}\right)^{\beta^e} \bar{w}_n^e P_n^{-1}.$$

3. Calculate the implied population distribution: $\tilde{L}_n^e = \sum_r \check{L}_r^e m_{rn}^e$.

4. Update population iteratively until convergence:

$$L_{n,new}^e \equiv \iota L_n^e + (1-\iota)\tilde{L}_n^e.$$

C.1.5 Sufficient Conditions for Uniqueness of Equilibrium

This Appendix section applies Theorem 1 from Allen, Arkolakis and Li (2024) to derive sufficient conditions for the uniqueness of equilibrium. I rewrite the system of equations that characterizes the equilibrium in terms of 15 unknowns

$$\{w_{nA}, w_{nM}, P_{nA}, P_{nM}, H_{nA}, H_{nM}, \bar{w}_n^c, \bar{w}_n^m, L_{nA}, L_{nM}, L_n^c, L_n^m, L_n, \Pi_n^c, \Pi_n^m\}$$

and 15 equations:

$$
\begin{aligned}
w_{nk}^{\sigma} H_{nk} &= \sum_r \alpha_k \tau_{nr}^{1-\sigma} \left(w_{rA} H_{rA} + w_{rM} H_{rM} \right) P_{rk}^{\sigma-1} \\
P_{nk}^{1-\sigma} &= \sum_r \tau_{rn}^{1-\sigma} w_{rk}^{1-\sigma} \\
w_{nk} H_{nk} &= \sum_e \Gamma_\theta \left(\phi_{nk}^e \right)^{\frac{1}{1-\gamma^e\theta}} \bar{w}_n^e \left(w_{nk} \right)^{\frac{\theta}{1-\gamma^e\theta}} \left(L_n^e \right)^{\frac{1}{1-\gamma^e\theta}} \left(L_{nk} \right)^{\frac{\theta(\gamma_k-\gamma^e)}{1-\gamma^e\theta}} \\
\left(\bar{w}_n^e \right)^{\theta} &= \sum_k \left(\phi_{nk}^e \right)^{\frac{1+\gamma^e\theta}{1-\gamma^e\theta}} \left(w_{nk} \right)^{\frac{\theta}{1-\gamma^e\theta}} \left(L_n^e \right)^{\frac{\gamma^e\theta}{1-\gamma^e\theta}} \left(L_{nk} \right)^{\frac{\theta(\gamma_k-\gamma^e)}{1-\gamma^e\theta}} \\
L_{nk}^{1-(\gamma_k-\gamma^e)\theta} \left(w_{nk} \right)^{-\theta} &= \sum_e \left(\phi_{nk}^e \right)^{\frac{1+\gamma^e\theta}{1-\gamma^e\theta}} \left(\bar{w}_n^e \right)^{-\theta} \left(w_{nk} \right)^{\frac{\gamma^e\theta^2}{1-\gamma^e\theta}} \left(L_n^e \right)^{\frac{1}{1-\gamma^e\theta}} \left(L_{nk} \right)^{\frac{\gamma^e\theta^2(\gamma_k-\gamma^e)}{1-\gamma^e\theta}} \\
\left(L_n \right)^{(\beta^e-\beta)\nu} \left(L_n^e \right)^{1-\beta^e\nu} \left(\bar{w}_n^e \right)^{-\nu} P_{nA}^{\nu\alpha} P_{nM}^{\nu(1-\alpha)} &= \sum_r \bar{a}_n^e \check{L}_r^e \eta_{rn}^{-\nu} \left(\Pi_r^e \right)^{-\nu} \\
\left(\Pi_n^e \right)^{\nu} &= \sum_r \eta_{nr}^{-\nu} \bar{a}_r^e \left(L_r \right)^{(\beta-\beta^e)\nu} \left(L_r^e \right)^{\beta^e\nu} \left(\bar{w}_r^e \right)^{\nu} P_{rA}^{-\nu\alpha} P_{rM}^{-\nu(1-\alpha)} \\
L_n &= \sum_e L_n^e.
\end{aligned}
$$

The equilibrium contains a set of $\mathcal{N} = \{1, \ldots, N\}$ locations and a set of $\mathcal{H} = 1, \ldots, H$ economic interactions (or endogenous variables), where $H = 15$. The $H \times H$ matrices B and Γ, as in Allen, Arkolakis and Li (2024), are given by

$$
B = \begin{bmatrix}
1 & 1 & \sigma-1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
1 & 1 & \cdot & \sigma-1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
1-\sigma & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & 1-\sigma & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\frac{\theta}{1-\gamma^e\theta} & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \frac{\theta(\gamma_A-\gamma^e)}{1-\gamma^e\theta} & \cdot & \frac{1}{1-\gamma^e\theta} & \frac{1}{1-\gamma^e\theta} & \cdot & \cdot & \cdot \\
\cdot & \frac{\theta}{1-\gamma^e\theta} & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \frac{\theta(\gamma_M-\gamma^e)}{1-\gamma^e\theta} & \frac{1}{1-\gamma^e\theta} & \frac{1}{1-\gamma^e\theta} & \cdot & \cdot & \cdot \\
\frac{\theta}{1-\gamma^e\theta} & \frac{\theta}{1-\gamma^e\theta} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \frac{\theta(\gamma_A-\gamma^e)}{1-\gamma^e\theta} & \frac{\theta(\gamma_M-\gamma^e)}{1-\gamma^e\theta} & \frac{\gamma^e\theta}{1-\gamma^e\theta} & \cdot & \cdot & \cdot & \cdot \\
\frac{\theta}{1-\gamma^e\theta} & \frac{\theta}{1-\gamma^e\theta} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \frac{\theta(\gamma_A-\gamma^e)}{1-\gamma^e\theta} & \frac{\theta(\gamma_M-\gamma^e)}{1-\gamma^e\theta} & \cdot & \frac{\gamma^e\theta}{1-\gamma^e\theta} & \cdot & \cdot & \cdot \\
\frac{\gamma^e\theta^2}{1-\gamma^e\theta} & \cdot & \cdot & \cdot & \cdot & \cdot & -\theta & -\theta & \frac{\gamma^e\theta(\gamma_A-\gamma^e)}{1-\gamma^e\theta} & \cdot & \frac{1}{1-\gamma^e\theta} & \frac{1}{1-\gamma^e\theta} & \cdot & \cdot & \cdot \\
\cdot & \frac{\gamma^e\theta^2}{1-\gamma^e\theta} & \cdot & \cdot & \cdot & \cdot & -\theta & -\theta & \cdot & \frac{\gamma^e\theta(\gamma_M-\gamma^e)}{1-\gamma^e\theta} & \frac{1}{1-\gamma^e\theta} & \frac{1}{1-\gamma^e\theta} & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & -\nu & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & -\nu \\
\cdot & \cdot & -\nu\alpha & \nu(\alpha-1) & \cdot & \cdot & \nu & \cdot & \cdot & \cdot & \beta^e\nu & \cdot & (\beta-\beta^e)\nu & \cdot & \cdot \\
\cdot & \cdot & -\nu\alpha & \nu(\alpha-1) & \cdot & \cdot & \cdot & \nu & \cdot & \cdot & \cdot & \beta^e\nu & (\beta-\beta^e)\nu & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & 1 & \cdot & \cdot & \cdot
\end{bmatrix}
$$

$$
\Gamma = \begin{bmatrix}
\sigma & . & . & . & 1 & . & . & . & . & . & . & . & . & . & . \\
. & \sigma & . & . & . & 1 & . & . & . & . & . & . & . & . & . \\
. & . & 1-\sigma & . & . & . & . & . & . & . & . & . & . & . & . \\
. & . & . & 1-\sigma & . & . & . & . & . & . & . & . & . & . & . \\
1 & . & . & . & 1 & . & . & . & . & . & . & . & . & . & . \\
. & 1 & . & . & . & 1 & . & . & . & . & . & . & . & . & . \\
. & . & . & . & . & . & \theta & . & . & . & . & . & . & . & . \\
. & . & . & . & . & . & . & \theta & . & . & . & . & . & . & . \\
-\theta & . & . & . & . & . & . & . & 1-(\gamma_A-\gamma^e)\theta & . & . & . & . & . & . \\
. & -\theta & . & . & . & . & . & . & . & 1-(\gamma_M-\gamma^e)\theta & . & . & . & . & . \\
. & . & \nu\alpha & \nu(1-\alpha) & . & . & -\nu & . & . & . & 1-\beta^e\nu & . & (\beta^e-\beta)\nu & . & . \\
. & . & \nu\alpha & \nu(1-\alpha) & . & . & . & -\nu & . & . & . & 1-\beta^e\nu & (\beta^e-\beta)\nu & . & . \\
. & . & . & . & . & . & . & . & . & . & . & . & . & \nu & . \\
. & . & . & . & . & . & . & . & . & . & . & . & . & . & \nu \\
. & . & . & . & . & . & . & . & . & . & . & . & 1 & . & .
\end{bmatrix}
$$

Next, I calculate $A = B\Gamma^{-1}$ and its spectral radius, denoted by $\rho(A)$ (i.e. its largest eigenvalue in absolute value). According to Theorem 1 in Allen, Arkolakis and Li (2024), a sufficient condition for uniqueness is that $\rho(A) < 1$. Although my baseline parameter values (Table 10) does not imply a spectral radius of A that is smaller than one, this is only a sufficient condition, so the equilibrium may still be unique. As noted in Remark 5 of Allen, Arkolakis and Li (2024), changing the system of equations through a change of variables may reduce the spectral radius, leading to a different sufficient condition that is more likely to hold.

C.2 Structural Estimation

This Appendix provides additional details on how model parameters are calibrated and estimated.

C.2.1 Identification of Market Access Terms

I provide a formal proof of Proposition 1, which establishes that the market access terms are identified up to a scale. The proof begins by deriving four underlying conditions involving the trade and migration market access terms from the equilibrium conditions (15)–(17).

(i). Total sales equals payments to labor: $w_{nk}H_{nk} = \sum_r X_{nrk}$. Using Equation (12), this can be written as:

$$
\mathcal{P}_{nk}^{1-\sigma} = \frac{\alpha_k}{\Omega_{nk}} \sum_r \tau_{nr}^{1-\sigma} Y_r P_{rk}^{\sigma-1},
$$

where $\Omega_{nk} \equiv w_{nk}H_{nk}/Y_n$ is the share of income in region n generated from sector k.

(ii). Total income equals total expenditure: $Y_r \alpha_k = \sum_n X_{nrk}$. This can be written as:

$$P_{rk}^{1-\sigma} = \sum_n \tau_{nr}^{1-\sigma} Y_n \mathcal{P}_{nk}^{\sigma-1}.$$

(iii). Final population equals total in-migrations: $L_n^e = \sum_{r=1}^N L_{rn}^e$. Using Equation (9), this can be written as:

$$(\mathcal{V}_n^e)^{-\nu} = \sum_r \eta_{rn}^{-v} \check{L}_r^e (\Pi_r^e)^{-v}$$

(iv). Initial population equals total out-migrations: $\check{L}_r^e = \sum_{n=1}^N L_{rn}^e$. This can be written as:

$$(\Pi_r^e)^v = \sum_n \eta_{rn}^{-v} L_n^e (\mathcal{V}_n^e)^\nu.$$

Putting these together, the derivation above yields a system of four equations:

$$\mathcal{P}_{nk}^{1-\sigma} = \frac{\alpha_k}{\Omega_{nk}} \sum_r \tau_{nr}^{1-\sigma} Y_r P_{rk}^{\sigma-1}, \tag{A-3}$$

$$P_{rk}^{1-\sigma} = \sum_n \tau_{nr}^{1-\sigma} Y_n \mathcal{P}_{nk}^{\sigma-1}, \tag{A-4}$$

$$(\mathcal{V}_n^e)^{-\nu} = \sum_r \eta_{rn}^{-v} \check{L}_r^e (\Pi_r^e)^{-v}, \tag{A-5}$$

$$(\Pi_r^e)^v = \sum_n \eta_{rn}^{-v} L_n^e (\mathcal{V}_n^e)^\nu, \tag{A-6}$$

Given data on total income $\{Y_n\}$ and sectoral income shares $\{\Omega_{nk}\}$, the agricultural expenditure share α is identified. Since each region spends the same proportion of income on agricultural goods, the economy as a whole must also spend that same share in aggregate:

$$\alpha = \frac{\sum_n w_{nA} H_{nA}}{\sum_n w_{nA} H_{nA} + w_{nM} H_{nM}} = \frac{\sum_n Y_n \Omega_{nA}}{\bar{Y}} = \sum_n Y_n \Omega_{nA}.$$

The four equations (A-3)–(A-6) can be separated into two sets: one for the trade market access and the other for migration market access. The equations for trade market access are:

$$\mathcal{P}_{nk}^{1-\sigma} = \sum_r \frac{\alpha_k}{\Omega_{nk}} \tau_{nr}^{1-\sigma} Y_r P_{rk}^{\sigma-1},$$
$$P_{nk}^{1-\sigma} = \sum_r \tau_{rn}^{1-\sigma} Y_r \mathcal{P}_{rk}^{\sigma-1}.$$

The migration market access equations are:

$$(\mathcal{V}_n^e)^{-\nu} = \sum_r \eta_{rn}^{-v} \check{L}_r^e \left(\Pi_r^e\right)^{-v},$$
$$(\Pi_n^e)^{v} = \sum_r \eta_{nr}^{-v} L_r^e \left(\mathcal{V}_r^e\right)^{\nu}.$$

I can rewrite the first set of equations as:

$$x_{nk}^{-1} = \sum_r K_{nrk}^{A} y_{rk},$$
$$y_{nk}^{-1} = \sum_r K_{nr}^{B} x_{rk},$$

where $x_{nk} \equiv \mathcal{P}_{nk}^{\sigma-1}$ and $y_{nk} \equiv P_{nk}^{\sigma-1}$.

Using Allen, Arkolakis and Li (2024), I compute matrices B_P and Γ_P as:

$$B_P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \quad \Gamma_P = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Thus, we have:

$$A_P \equiv |B_p \Gamma_P^{-1}| = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The spectral radius of A_P, which is the largest absolute value of its eigenvalues, is 1. Based on Theorem 1, part ii.b of Allen, Arkolakis and Li (2024), this guarantees the existence of a unique solution for $\{\mathcal{P}_{nk}^{\sigma-1}, P_{nk}^{\sigma-1}\}$ up to a scale.

Similarly, the second set of equations can be rewritten as:

$$x_{ne}^{-1} = \sum_r K_{nre}^{C} y_{re}^{-1},$$
$$y_{ne} = \sum_r K_{nre}^{D} x_{re},$$

where $x_{ne} \equiv (\mathcal{V}_n^e)^\nu$ and $y_{ne} \equiv (\Pi_n^e)^\nu$.

The corresponding matrices B_V and Γ_V are:

$$B_V = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}; \quad \Gamma_V = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus, we have:

$$A_V \equiv \left| B_V \Gamma_V^{-1} \right| = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Since the spectral radius of A_V is also 1, by the same argument, there exists a unique solution for $\{(\mathcal{V}_n^e)^\nu, (\Pi_n^e)^\nu\}$ up to a scale.

C.2.2 Migration Cost Elasticity

I estimate the product of migration cost elasticity κ and taste dispersion ν, denoted $\tilde{\kappa} \equiv \kappa\nu$, using non-linear least squares. The estimation proceeds as follows.

(i). Guess an initial $\tilde{\kappa}$ and calculate the corresponding migration costs $\eta_{rn}^\nu = (d_{rn}/d_{min})^{\tilde{\kappa}}$.

(ii). Use the initial and final population data $\{\check{L}_r^e, L_n^e\}$ to solve for the migration market access terms $(\mathcal{V}_n^e)^\nu, (\Pi_n^e)^\nu$ as per Proposition 1.

(iii). Calculate the implied bilateral migration flows:

$$L_{rn} = \sum_e L_{rn}^e = \sum_e d_{rn}^{-\tilde{\kappa}} \times \frac{\check{L}_r^e}{(\Pi_r^e)^\nu} \times \frac{L_n^e}{(\mathcal{V}_n^e)^{-\nu}}.$$

(iv). Aggregate the model-implied migration flows to the district level and compute bilateral migration shares:

$$m_{jh} = \frac{\sum_{r\in j(r)} \sum_{n\in h(n)} L_{rn}}{\sum_{r\in j(r)} \sum_n L_{rn}},$$

where $j(r)$ and $h(n)$ denote the districts that counties r and n belong.

(v). Calculate the loss function as the sum of squared differences between the model-predicted and observed (log) migration shares:

$$loss \equiv \frac{1}{N_d^2} \sum_{j,h} (\ln m_{jh} - \ln \hat{m}_{jh})^2, \tag{A-7}$$

where N_d is the total number of districts, and $\hat{m}_{jh}$ denotes the observed migration shares.

(vi). Search over the space of $\tilde{\kappa}$ to minimize the loss function.

C.2.3 Calibration

I externally set or calibrate three parameters: the substitution elasticity σ, the distance elasticity of trade costs ξ, and migration elasticity ν. This Appendix section provides additional details on these calibrated values.

Substitution elasticity. For σ, the literature typically reports values between 4 and 9. For example, Peters (2022) estimates 5.02 for post-war Germany, Balboni (2025) estimates 7.92 for Vietnam in 2009, and Donaldson and Hornbeck (2016) estimate 9.22 for 19th century U.S. I set $\sigma = 8$ in the baseline, following Balboni (2025), and consider values of 4, 6, and 10 for robustness.

Distance elasticity of trade costs. I calibrate ξ using a moment from Monte, Redding and Rossi-Hansberg (2018), who estimate it from trade shipments within the United States during 1993–2017. Since trade frictions in 1980 Malaysia may be larger than in the modern U.S., I consider values of ξ that are 1.5 and 2 times the baseline. A higher ξ implies that trade costs rise more steeply with distance, as would be expected under weaker transport infrastructure.

Migration elasticity. I set $\nu = 3$ in baseline, based on (Bryan and Morten, 2019), who estimate 3.2 for Indonesia during 1995–2012 and 2.7 for the United States (1990–2010). Estimates of ν are rare, especially in developing countries. Peters (2022) estimate 2.1 for post-war Germany, and Tombe and Zhu (2019) estimate 1.5 for China in the early 2000s. I consider values of 1.5, 2.5, and 3.5 for robustness, holding fixed $\tilde{\kappa} = \kappa\nu = 1.56$, as $\tilde{\kappa}$ is estimated from observed migration flows. The range of ν from 1.5 to 3.5 implies a distance elasticity of migration costs κ spanning 0.45 and 1.04, which falls within the 0.37–1.09 range reported in previous studies (Bryan and Morten, 2019; Peters, 2022).

C.2.4 Comparing Estimates to the Literature

Distance elasticity of migration costs. I estimate $\kappa = 0.52$, given $\nu = 3$, pooling all ethnic groups. Allowing for heterogeneity by ethnicity yields similar values: $\kappa = 0.54$ for Chinese and $\kappa = 0.51$ for non-Chinese. These values align with the range of existing estimates in the literature. For example, Bryan and Morten (2019) find an elasticity of 0.37 in Indonesia between 1995 and 2012, while Peters (2022) reports an elasticity of 1.09 in post-war Germany in 1955.[57]

Skill dispersion. The estimated Fréchet shape parameter, $\theta = 3.3$, lies within the range found in the literature. For instance, Lagakos and Waugh (2013) estimate a θ of 5.3 for agriculture and 2.7 for the non-agricultural sector in the U.S. between 1996 and 2010. Similarly, Hsieh et al. (2019) report values between 1.5 and 2.6 for the U.S. from 1960 to 2012.[58]

Productivity spillovers. I estimate that labor productivity increases externally with local employment in the non-agricultural sector, with an elasticity of $\gamma_M = 0.22$, while the agricultural sector shows a smaller and negative elasticity of $\gamma_A = -0.12$. My estimate for non-agriculture is similar to the 0.2 elasticity reported by Kline and Moretti (2014) and falls within the 0.11–0.31 range for manufacturing sectors in Bartelme et al. (2025), though it is lower than the 1.25–3.1 range found by Greenstone, Hornbeck and Moretti (2010). While few studies estimate agglomeration elasticities in agriculture, the smaller estimate is consistent with the general understanding that agglomeration effects in agriculture tend to be weaker than in industrial sectors.[59] In addition, because the model does not incorporate land inputs in agricultural production, the negative elasticity also reflects diminishing returns to labor given a fixed amount of land.

I estimate a sizable productivity spillover elasticity with respect to ethnic composition, $\gamma^e = 0.13$. This suggests that, holding county population constant, an increase in the Chinese employment share raises the productivity of local Chinese workers. The effect on Malay workers is more nuanced and depends on the sector. Since $\gamma^e < \gamma_M$, Equation (5) indicates that Malays in non-agricultural sectors benefit from an increase in the Chinese population. However, because $\gamma^e > \gamma_A$, an increase in the Chinese population reduces Malays' agricultural productivity. These predictions are consistent with empirical evidence showing that Malays in non-agricultural sectors in more resettled areas experienced marginal income gains, while those in agriculture did not.

Although there are no direct comparisons for ethnicity-based spillovers in the literature, similar externalities have been examined using other demographic characteristics, such as education and occupation. For instance, Moretti (2004) estimates wage elasticities of 0.14 for

[57]Bryan and Morten (2019) estimate migration costs non-parametrically, rather than assuming proportionality to distance. I translate their Figure 3 into my setting, where $1 - \eta_{nr}^{-1} \approx -0.5 + 0.147 \ln d_{nr}$. This implies that their distance elasticity varies with distance, unlike the constant elasticity assumed in my model. For comparison, I use the average log distance of 7.5 in their setting, resulting in $\partial \ln \eta_{nr} / \partial \ln d_{nr} \approx 0.37$.

[58]One reason their estimates may be lower is that wage variance in their model reflects differences in (endogenous) educational attainment in addition to idiosyncratic productivity draws.

[59]See Melo, Graham and Noland (2009), Combes and Gobillon (2015), and Ahlfeldt and Pietrostefani (2019) for a review of density-productivity elasticity, typically between 0.02 and 0.09 in developed countries. Estimates for developing countries are less common but tend to be above 0.1.

college graduates and 0.21 for high school graduates with respect to college share in a city.[60] Rossi-Hansberg, Sarte and Schwartzman (2023) estimate wage elasticities with respect to the share of workers in "cognitive non-routine" occupations, finding substantial elasticities of 1.3 for workers in these occupations and 0.84 for those in non-cognitive roles.

Amenity spillovers. I estimate the amenity spillover elasticity with respect to local population size at $\beta = -0.005$. This small value suggests that congestion forces—such as increased traffic or higher housing prices—are relatively weak. As discussed in Bryan and Morten (2019), extending the model to include housing as a non-traded good implies that the amenity spillover can be decomposed as $\beta = \beta_a - \delta\beta_r$, where β_a represents the pure amenity spillover, β_r is the inverse of housing supply elasticity, and δ is the share of income spent on housing. Using the resettlement shocks as a demand shifter and housing prices from the MFLS-2 data, I estimate $\beta_r \approx 0.3$, corresponding to a housing supply elasticity of 3.3 (Appendix Table A.20). This elasticity is higher than U.S. estimates, which range from 1 to 3 (Gyourko, Saiz and Summers, 2008; Saiz, 2010).[61] In 1980, housing expenditure accounted for 17.6% of total spending, implying a pure amenity spillover of $\beta_a = \beta + \delta\beta_r = 0.05$.[62]

There are few estimates of β in low-income countries. Bryan and Morten (2019) report a value of 0.04, though with limited precision. Allen and Donaldson (2022) estimate both contemporaneous and historical amenity spillovers in the United States from 1800 to 2000, finding a -0.26 contemporaneous spillover and a 0.31 historical spillover (based on population 50 years prior). Since my model does not differentiate between contemporaneous and historical effects, my estimate reasonably falls between these two values.

My baseline estimate of the amenity spillover elasticity with respect to ethnic composition is $\beta^e = 0.13$. The positive β^e suggests that an increase in the population of an ethnic group raises the utility of people from that same group more than those from the other group. The stronger within-ethnic amenity spillover is consistent with the economies of scale in the provision of urban amenities, such as restaurants or entertainment, as discussed in Duranton and Puga (2004). It also aligns with the presence of social frictions, as reflected in consumption segregation documented in Davis et al. (2019).[63]

[60] Moretti (2004) finds that a 1 percentage point increase in the share of college-educated workers leads to a 1.3% wage increase. I convert this to an elasticity, assuming an average college share of 0.25 in 1990. Diamond (2016) finds higher elasticities—0.31 for college graduates and 0.93 for non-college workers—though these estimates include substitution effects between high- and low-skilled workers.

[61] In Indonesia, Bryan and Morten (2019) estimate a value of 4, though with limited statistical power.

[62] The expenditure category is "gross rent, fuel, and power." In 1973, the same category accounted for 14.9% of expenditures. See Department of Statistics Malaysia (1980).

[63] There are no direct comparisons for ethnicity-based amenity spillovers. The closest comparison comes from Fajgelbaum and Gaubert (2020), who use Diamond's (2016) estimates of amenity spillovers by college share. Similar to productivity spillovers, the authors calibrate four constant amenity spillover elasticities: $(\gamma^A_{UU}, \gamma^A_{SU}, \gamma^A_{US}, \gamma^A_{SS}) = (-0.43, 0.18, -1.24, 0.77)$, where γ^A_{SU} denotes the marginal amenity spillover of a

C.3 Counterfactuals

This Appendix section first explains how I derive a lower bound on the utility loss from forced resettlement. It then assesses the robustness of the counterfactual results to alternative parameter values.

C.3.1 Utility Loss from Forced Resettlement

I begin by recovering the location amenity fundamentals that rationalize the 1947 population distribution as a steady state that, absent resettlement, would have persisted through 1957. Let $\tilde{a}_n^e$ denote these 1947 amenity fundamentals. The share of group-e individuals who migrate from region r to n during 1947–1957 is given by:

$$\tilde{m}_{rn}^e = \frac{\tilde{a}_n^e \left(\eta_{rn}^{-1} \left(L_{n,47}\right)^{\beta} \left(L_{n,47}^e / L_{n,47}\right)^{\beta^e} \bar{w}_n^e P_n^{-1}\right)^{\nu}}{\sum_{l=1}^{N} \tilde{a}_l^e \left(\eta_{rl}^{-1} \left(L_{l,47}\right)^{\beta} \left(L_{l,47}^e / L_{l,47}\right)^{\beta^e} \bar{w}_l^e P_l^{-1}\right)^{\nu}},$$

where $\{L_{n,47}^e\}$ denotes the population distribution in 1947.

Using the balance of migration flows, $L_{n,47}^e = \sum_r L_{r,47}^e \tilde{m}_{rn}^e$, we obtain:

$$L_{n,47}^e = \sum_r L_{r,47}^e \frac{\tilde{a}_n^e \left(\eta_{rn}^{-1} \left(L_{n,47}\right)^{\beta} \left(L_{n,47}^e / L_{n,47}\right)^{\beta^e} \bar{w}_n^e P_n^{-1}\right)^{\nu}}{\sum_{l=1}^{N} \tilde{a}_l^e \left(\eta_{rl}^{-1} \left(L_l\right)^{\beta} \left(L_{l,47}^e / L_{l,47}\right)^{\beta^e} \bar{w}_l^e P_l^{-1}\right)^{\nu}},$$

which can be arranged as

$$\left(\tilde{a}_n^e\right)^{-1} = \frac{1}{L_{n,47}^e} \sum_r L_{r,47}^e \frac{\eta_{rn}^{-\nu} \left(\left(L_{n,47}\right)^{\beta} \left(L_{n,47}^e / L_{n,47}\right)^{\beta^e} \bar{w}_n^e P_n^{-1}\right)^{\nu}}{\sum_{l=1}^{N} \tilde{a}_l^e \eta_{rl}^{-\nu} \left(\left(L_{l,47}\right)^{\beta} \left(L_{l,47}^e / L_{l,47}\right)^{\beta^e} \bar{w}_l^e P_l^{-1}\right)^{\nu}}. \tag{A-8}$$

This system provides $N-1$ equations, allowing me to solve for $\tilde{a}_n^e$ up to a scale.

Using these recovered amenity fundamentals, I then solve for the ethnicity- and place-specific wage subsidies required to voluntary relocate Chinese and Malays from the 1947 population distribution to the 1957 resettled distribution.

Let ϵ_n^e denote the ad-valorem subsidy for group e in region n. Again, using the balance of migration flows, $L_{n,57}^e = \sum_r L_{r,47}^e m_{rn}^e$, we obtain:

$$L_{n,57}^e = \sum_r L_{r,47}^e \frac{\tilde{a}_n^e \left(\eta_{rn}^{-1} \left(L_{n,57}\right)^{\beta} \left(L_{n,57}^e / L_{n,57}\right)^{\beta^e} \left(1+\epsilon_n^e\right) \bar{w}_n^e P_n^{-1}\right)^{\nu}}{\sum_{l=1}^{N} \tilde{a}_l^e \left(\eta_{rl}^{-1} \left(L_{l,57}\right)^{\beta} \left(L_{l,57}^e / L_{l,57}\right)^{\beta^e} \left(1+\epsilon_l^e\right) \bar{w}_l^e P_l^{-1}\right)^{\nu}}.$$

college graduate (S) on the utility of a non-college graduate (U)), and so on.

Rearranging gives:

$$(1+\epsilon_n^e)^{-\nu} = \frac{1}{L_{n,57}^e} \sum_r L_{r,47}^e \frac{\tilde{a}_n^e \eta_{rn}^{-\nu} \left((L_{n,57})^{\beta} \left(L_{n,57}^e / L_{n,57} \right)^{\beta^e} \bar{w}_n^e P_n^{-1} \right)^{\nu}}{\sum_{l=1}^N \tilde{a}_l^e \eta_{rl}^{-\nu} \left((L_{l,57})^{\beta} \left(L_{l,57}^e / L_{l,57} \right)^{\beta^e} (1+\epsilon_l^e)\, \bar{w}_l^e P_l^{-1} \right)^{\nu}}. \tag{A-9}$$

Since migration depends only on relative wages across regions, Equation (A-9) implies that ϵ_n^e is determined only up to a scale. To calculate the least-cost, weakly positive subsidies, I scale the solution vector so that the minimum subsidy across locations is zero. Specifically, let $\{\epsilon_n^e\}$ be any solution to Equation (A-9). Then, the minimum wage subsidies for group e, denoted $\{\varepsilon_n^e\}$, are given by:

$$\varepsilon_n^e \equiv \frac{1+\epsilon_n^e}{1+\min_n(\epsilon_n^e)} - 1.$$

Finally, I calculate the total real subsidies received by the resettled population as:

$$\sum_e \sum_n \varepsilon_n^e L_{n,resettled}^e \frac{y_n^e}{P_n},$$

where $y_n^e = \sum_k w_{nk} H_{nk} / L_n^e$ denotes average output per capita for group e in region n, and $L_{n,resettled}^e$ is the number of resettled persons from group e in region n, which I approximate by assuming that 90% of the resettled population was Chinese based on historical accounts.

C.3.2 Robustness to Parameter Values

I externally set or calibrate three parameters: migration elasticity ν, substitution elasticity σ, and the distance elasticity of trade costs ξ. While the remaining parameters are estimated, three of them are estimated with relatively large standard errors: the agricultural productivity spillover γ_A, the amenity spillover with respect to population size β, and the amenity spillover with respect to own-ethnic share β^e. This Appendix section examines how the main counterfactual results vary with alternative values of these parameters.

For the calibrated parameters, I consider values within ranges commonly reported in the literature (see Appendix Section C.2.3): $\nu \in \{1.5,\ 2.5,\ 3.5\}$, $\sigma \in \{4,\ 6,\ 8,\ 10\}$, and $\xi \in \{0.18,\ 0.28,\ 0.37\}$. For the estimated parameters γ_A and β^e, few estimates exist in the literature, so I consider values 50% above and below the baseline: $\gamma_A \in \{-0.18,\ -0.12,\ -0.06\}$ and $\beta^e \in \{0.06,\ 0.13,\ 0.19\}$. For β, while Bryan and Morten (2019) estimate a positive value of 0.04 with large uncertainty, the literature typically finds a negative value, reflecting congestion. I consider $\beta \in \{-0.06,\ -0.01,\ 0.04\}$, where -0.01 is the baseline.

Aggregate impact of resettlement. Appendix Table A.21 reports how the aggregate output gains from resettlement vary with alternative values of the calibrated parameters. Gains are larger when migration elasticity ν is low, as the program overcomes migration inertia and reallocates workers to more productive regions. Gains also increase with higher substitution elasticity σ, which facilitates greater concentration and specialization.

The effect of the trade cost distance elasticity ξ is more nuanced and depends on σ: when regional goods are less substitutable (low σ), higher trade frictions ξ amplify gains because the resettlement relocates people to places with greater market access and lower price index. When goods are highly substitutable (high σ), spatial concentration from resettlement is beneficial, but higher trade frictions reduce gains by limiting the benefits from that concentration. Overall, output gains range from 0.03% to 4% of baseline output, with the highest gains occurring under low ν and high σ.

Appendix Table A.22 examines robustness to alternative values of the estimated parameters. Gains are larger when agriculture exhibits weaker decreasing returns to scale (larger γ_A), as this reduces the productivity loss from clustering squatters in rural receiving counties. Gains also increase with stronger amenity spillovers from population size (higher β), which amply the benefits from forced agglomeration. In contrast, stronger residential homophily (higher β^e) reduces the gains from resettlement by further discouraging Malays from co-locating with incoming Chinese workers in urban areas. Across these parameter combinations, output gains range from 1.67% to 2.22% of baseline output.

Reducing cross-ethnic barriers to productivity spillovers. Appendix Table A.23 reports the output gains from halving cross-ethnic spillover barriers (γ^e). The gains exhibit an inverse-U shape in migration elasticity: they are smaller when ν is low (limited reallocation) or high (baseline already somewhat integrated), and peak at moderate ν. The effects of σ and ξ are modest, though gains are slightly higher when σ is low. When goods are less substitutable, demand is more evenly distributed across regional varieties, and cross-ethnic spillovers raise productivity broadly across space as ethnic groups integrate. Overall, output gains are stable across parameter values, ranging from 3.47% to 4.31% of baseline output.

Appendix Table A.24 examines robustness to alternative values of the estimated parameters. Gains vary little with the external economies of agriculture (γ_A), though they are slightly larger when agriculture exhibits weaker decreasing returns to scale (higher γ_A). This is likely due to greater population concentration when cross-ethnic spillovers are stronger. Indeed, gains also increase when congestion forces are weaker (higher β). In contrast, stronger residential homophily (higher β^e) leads to greater ethnic segregation, reducing the gains from cross-ethnic spillovers. Across these parameter combinations, output gains range from 2.76% to 4.45% of baseline output.

Malay wage subsidies in non-agriculture. Appendix Table A.25 reports the output gains from an 18% wage subsidy for Malays in non-agriculture. Gains increase monotonically with migration elasticity ν, as higher ν facilitates reallocation of Malays into urban areas. The effects of σ and ξ are non-monotonic and interact with other parameters. Overall, output gains range from 0.46% to 1.39% of baseline output.

Appendix Table A.26 examines robustness to alternative values of the estimated parameters. Gains are larger when agriculture exhibits stronger decreasing returns to scale (lower γ_A), as this further pushes Malays into non-agriculture, amplifying agglomeration spillovers in that sector. Gains also increase when congestion forces are weaker (higher β), as this encourages Malays to agglomerate in urban areas. In contrast, stronger residential homophily (higher β^e) reduces the gains from this policy by limiting the potential for spillovers from urban Chinese to Malays. Across these parameter combinations, output gains range from 0.65% to 1.38% of baseline output.

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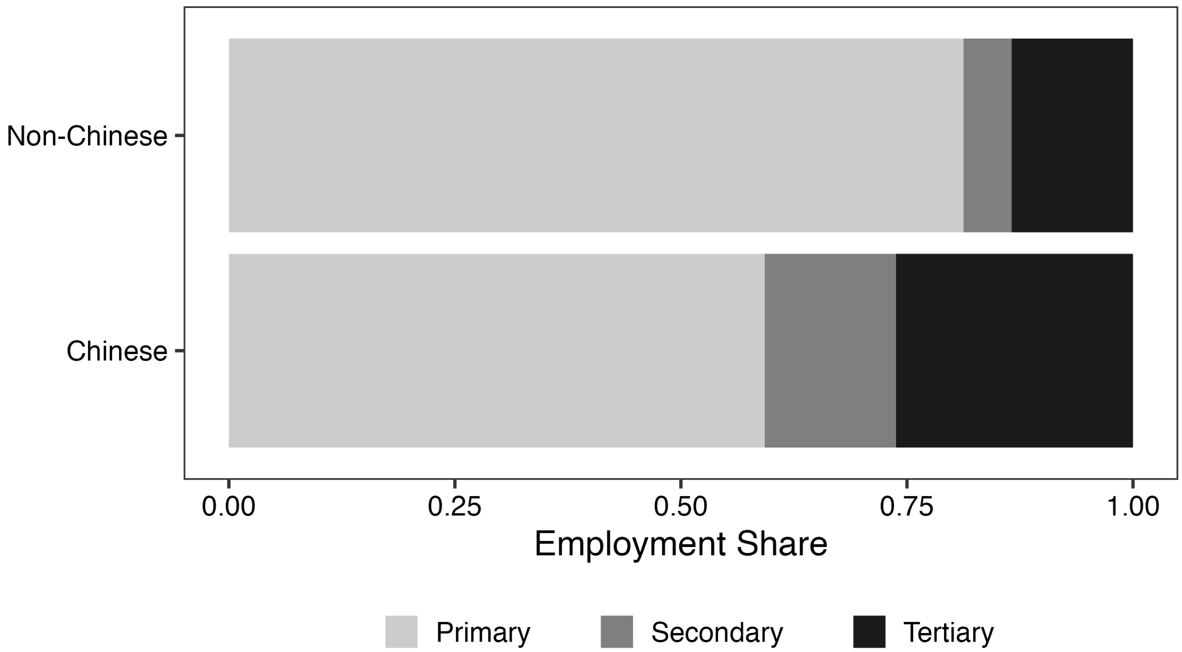
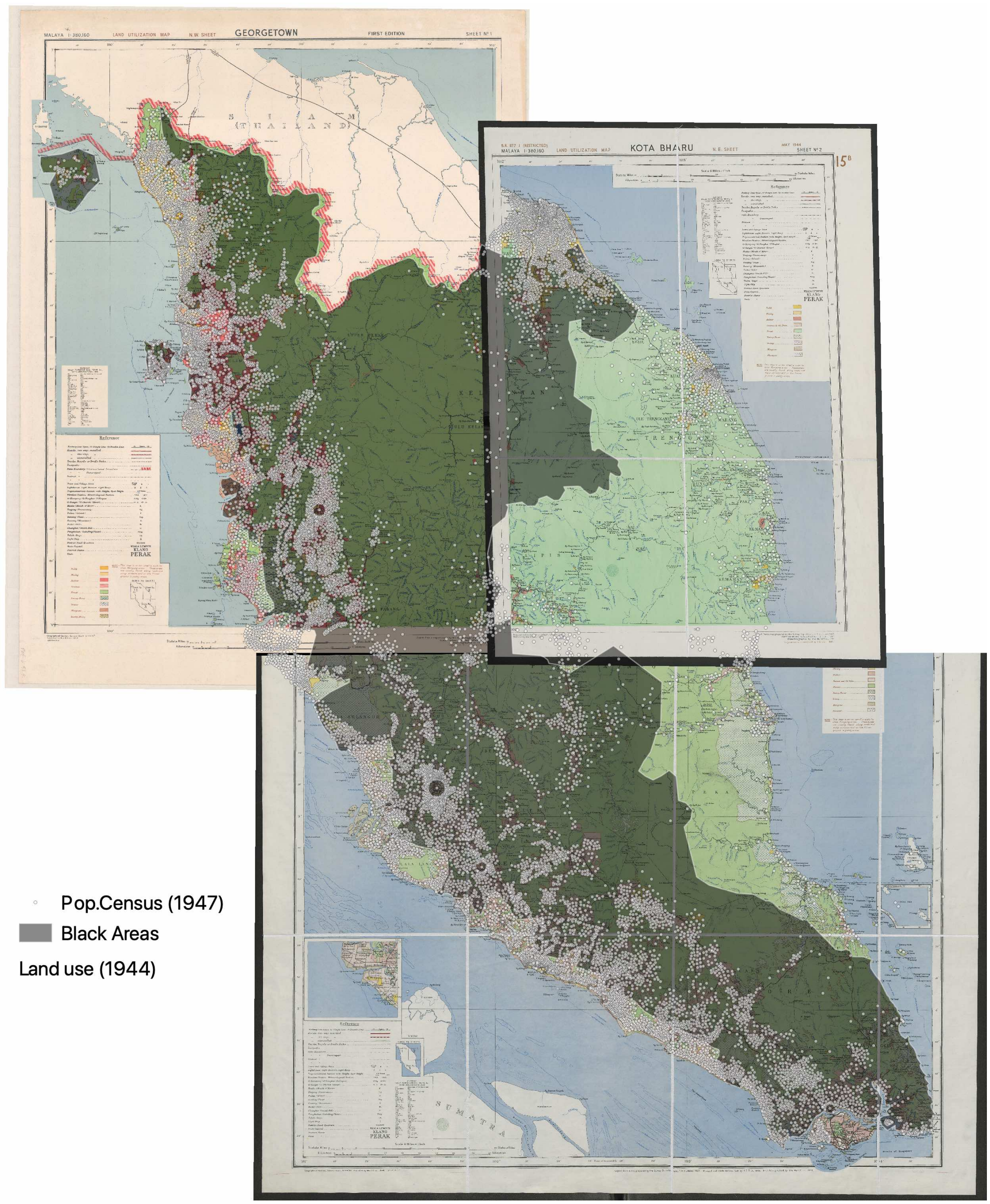
D Appendix Figures

Figure A.1. Employment Share in 1947, by Ethnic Group



Notes: This figure shows the distribution of employment across primary, secondary, and tertiary sectors for Chinese and non-Chinese populations in 1947. The primary sector (also referred to as "agriculture" in the main text) includes agriculture, hunting, forestry, fishing and mining. The secondary sector includes manufacturing, utilities, and construction. The tertiary sector includes storage, transport, communication, commerce, finance, business, and other services. Data from the 1947 Census of Population (Del Tufo, 1947).

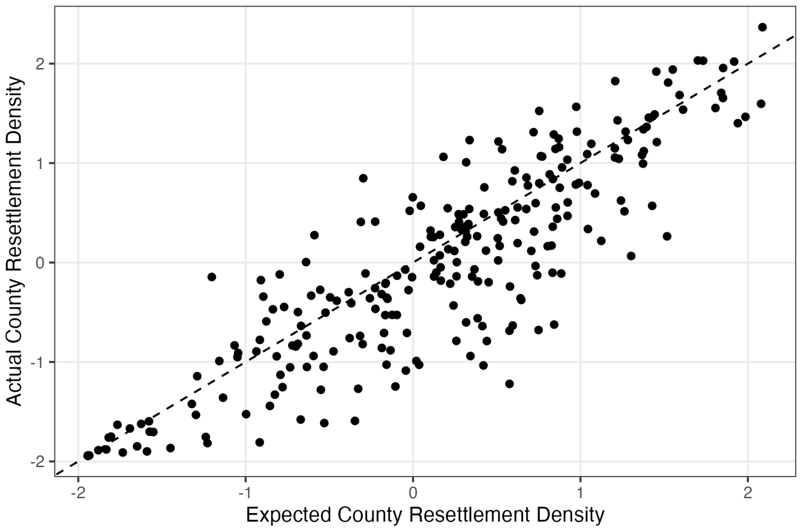
Figure A.2. Distribution of Squatter Settlements in 1947



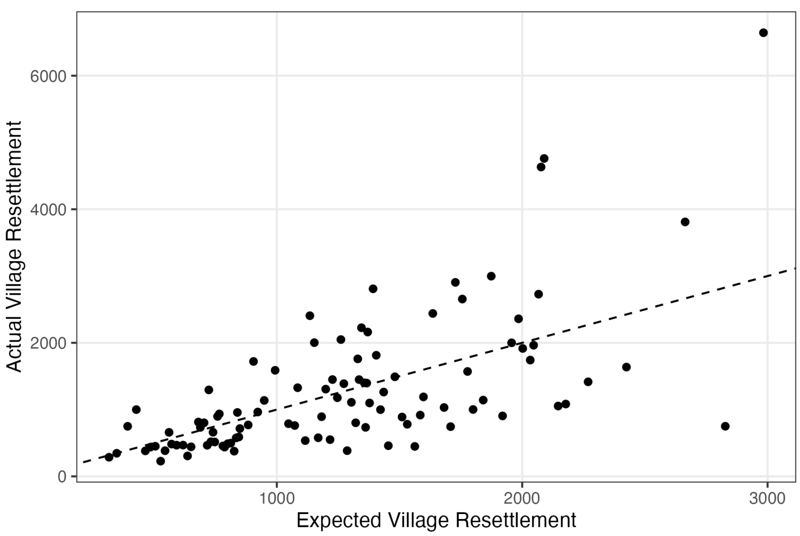
Notes: This figure maps the distribution of squatter settlements based on the intersection of three historical sources. Gray dots indicate population clusters from the 1947 Census of Population. Dark-shaded areas represent "Black areas" under Emergency regulations due to communist insurgency. Green-shaded areas are forests, based on land utilization maps from 1943 (War Office, 1943).

Figure A.3. Comparison of Actual and Predicted Resettlement, by County and Village

Panel A. County Resettlement Density, Compared to Expected Resettlement Density



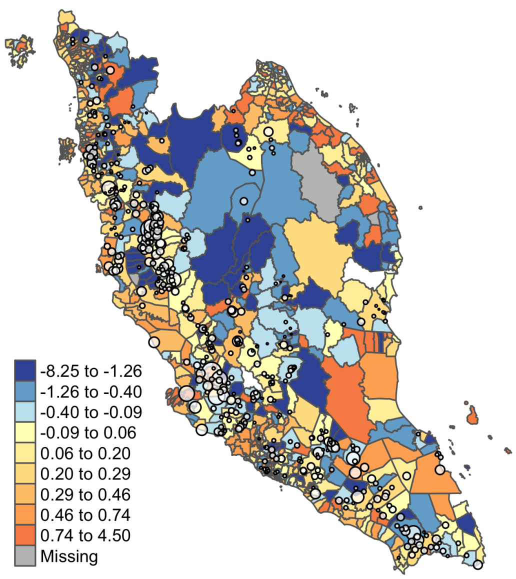
Panel B. Village Resettled Population, Compared to Expected Resettled Population



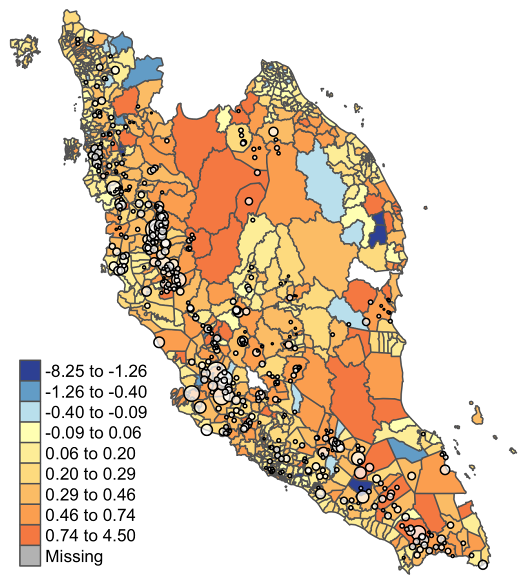
Notes: This figure compares actual resettlement outcomes with predictions from the gravity model described in Appendix B.1. Panel A plots actual county resettlement density against expected density, calculated from Equation (A-1). Panel B compares the actual resettled population in each village with the counterfactual population predicted by the dislocation-minimizing plan in Equation (2), also conditional on village locations. Data from the Corry report.

Figure A.4. County Population Growth from 1947 to 1957, by Ethnic Group

Panel A. Log Change in Population, Chinese

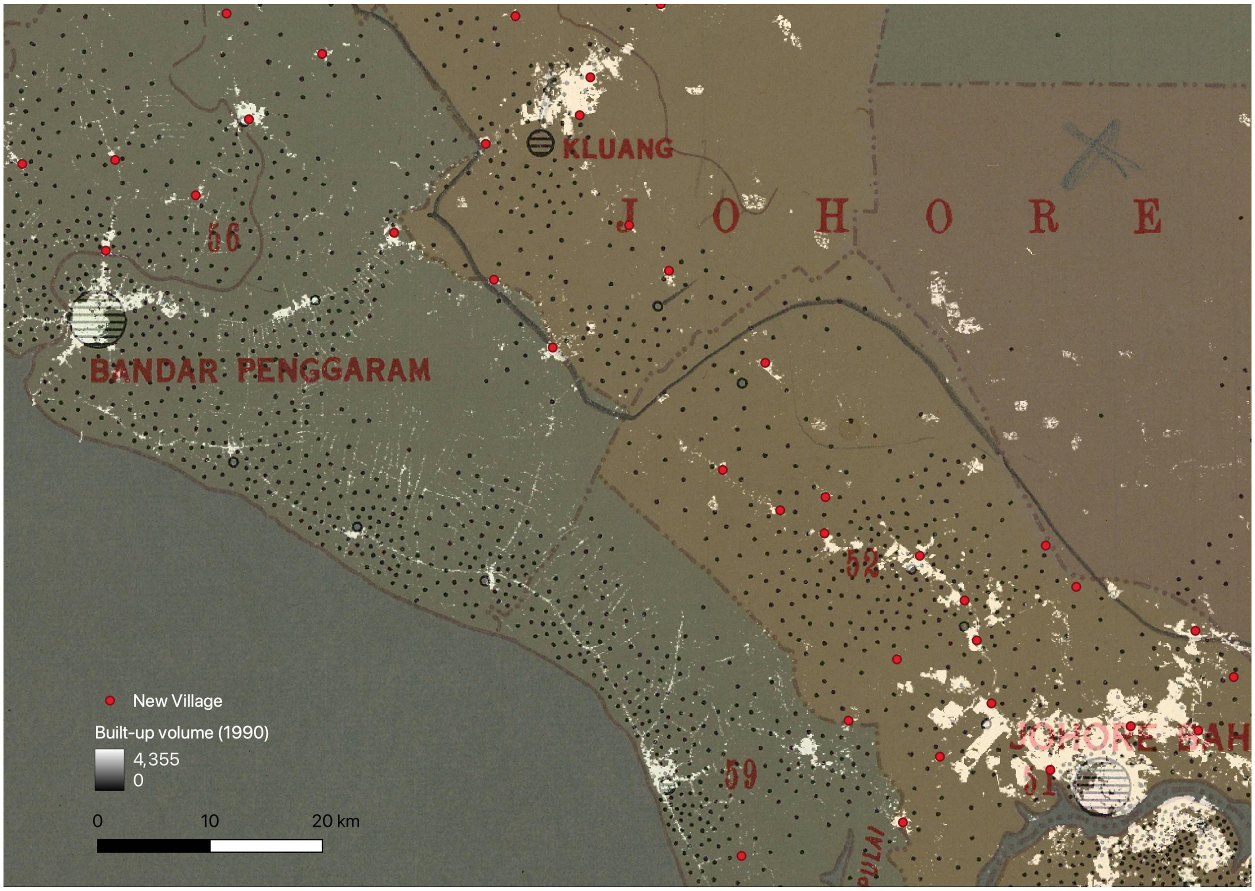


Panel B. Log Change in Population, Non-Chinese



Notes: This figure maps county population growth from 1947 to 1957 by ethnic group. Panel A shows log changes in Chinese population. Panel B shows log changes in non-Chinese population. White bubbles denote New Villages, with size proportional to the log resettled population. Counties with missing data are shaded in gray. Data from the tabulated Census of Population and the Corry report.

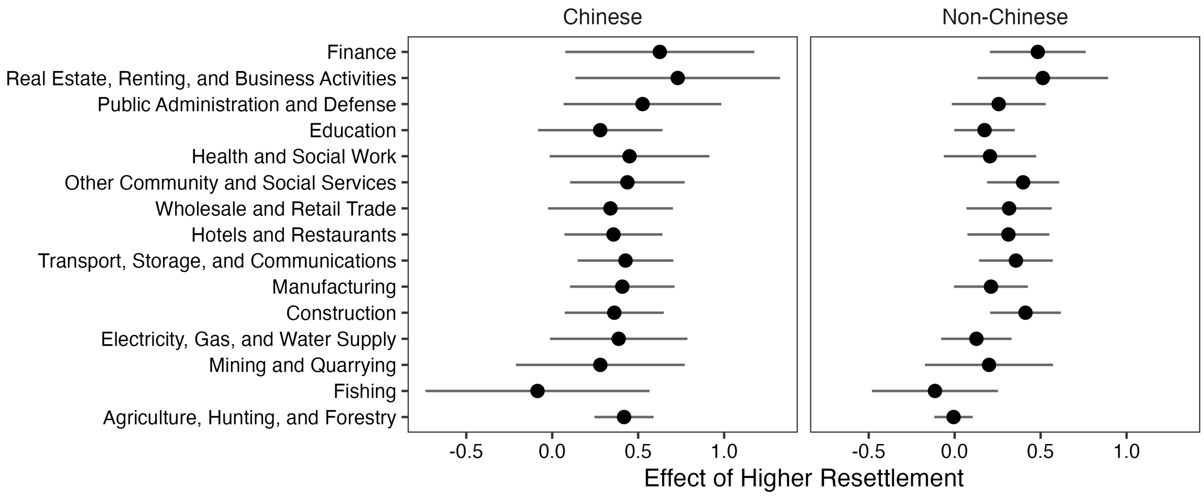
Figure A.5. Built-up Volume in 1990, Johor



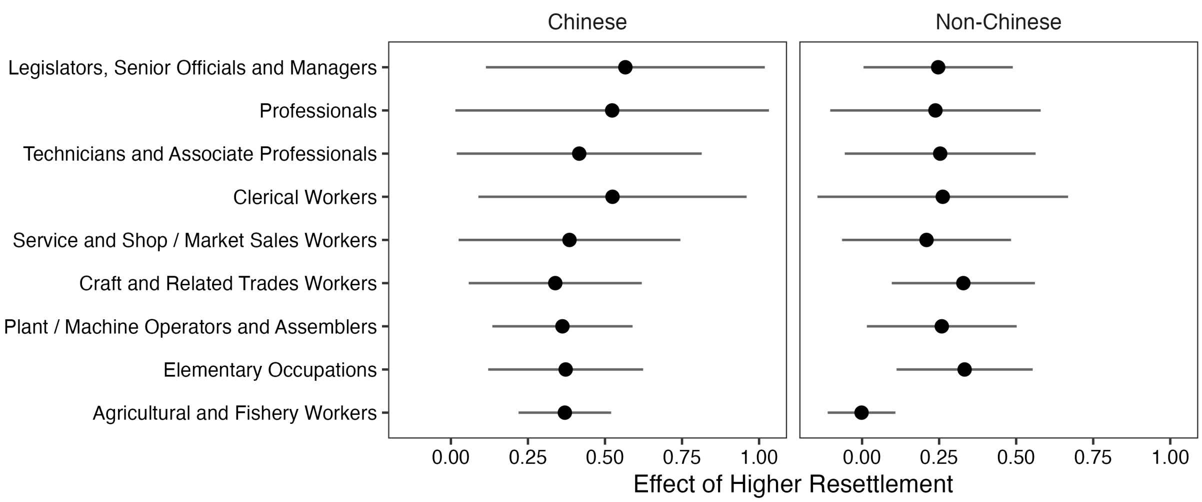
Notes: This figure maps built-up volumes in 1990 for a region in Johor. Built-up volumes are calculated from 100-meter resolution surface and height data based on Sentinel-2 and Landsat satellite imagery, with higher volumes shaded in white. Red dots indicate the locations of New Villages, and black dots represent population clusters from the 1947 Census. Built-up volume data from the GHSL project; New Village locations from the Corry report.

Figure A.6. Effect of Higher Resettlement on Employment Size in 1991, by Industry and Occupation

Panel A. By Industry

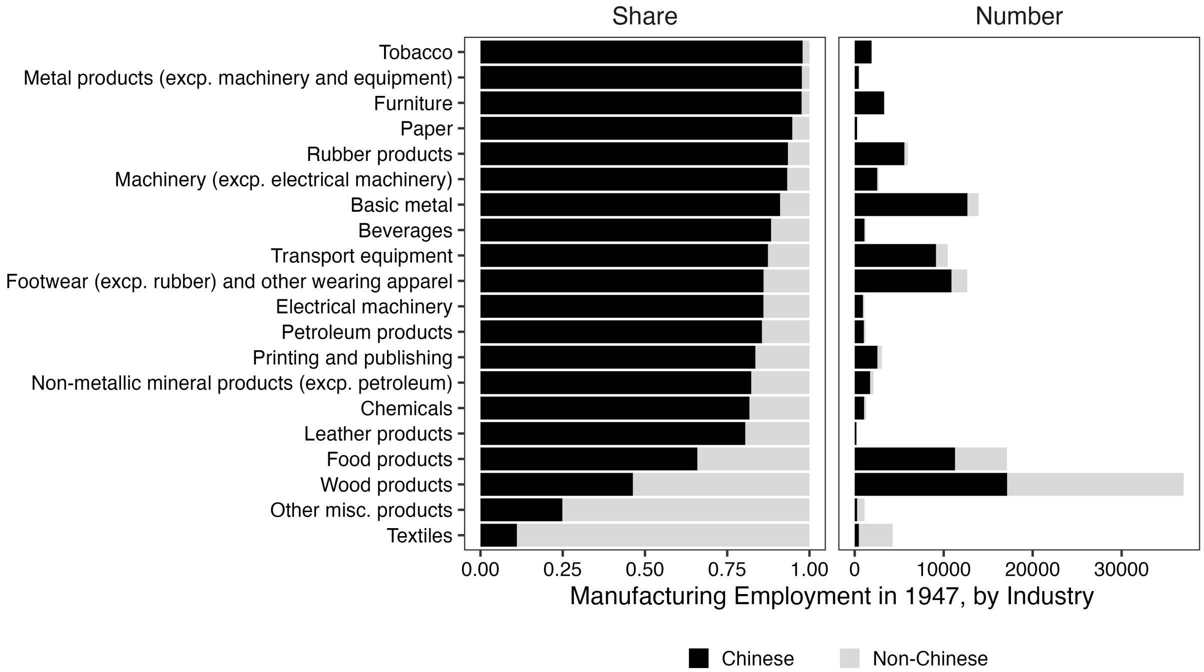


Panel B. By Occupation



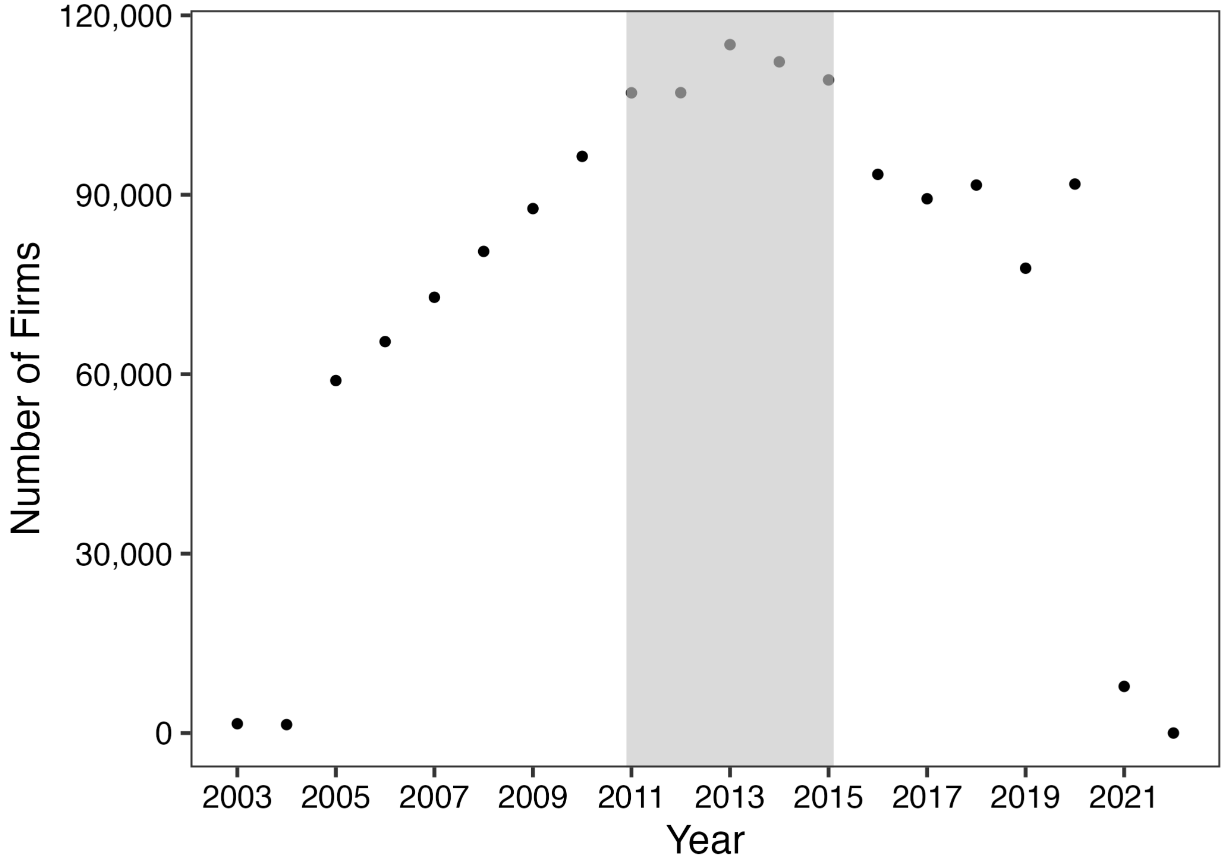
Notes: This figure shows the relationship between county resettlement density and employment size by industry (Panel A) and occupation (Panel B) in 1991, separately for Chinese and non-Chinese workers. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. All regressions are estimated using the Poisson pseudo-maximum-likelihood (PPML) estimator and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; log county area; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; and the shares of land used for rubber and mining in 1943. The unit of observation is the county. Data from the tabulated Census of Population in 1991. Error bars show 95% confidence intervals based on Conley standard errors with a 30-kilometer cutoff.

Figure A.7. Chinese Manufacturing Employment Share and Number in 1947, by Industry



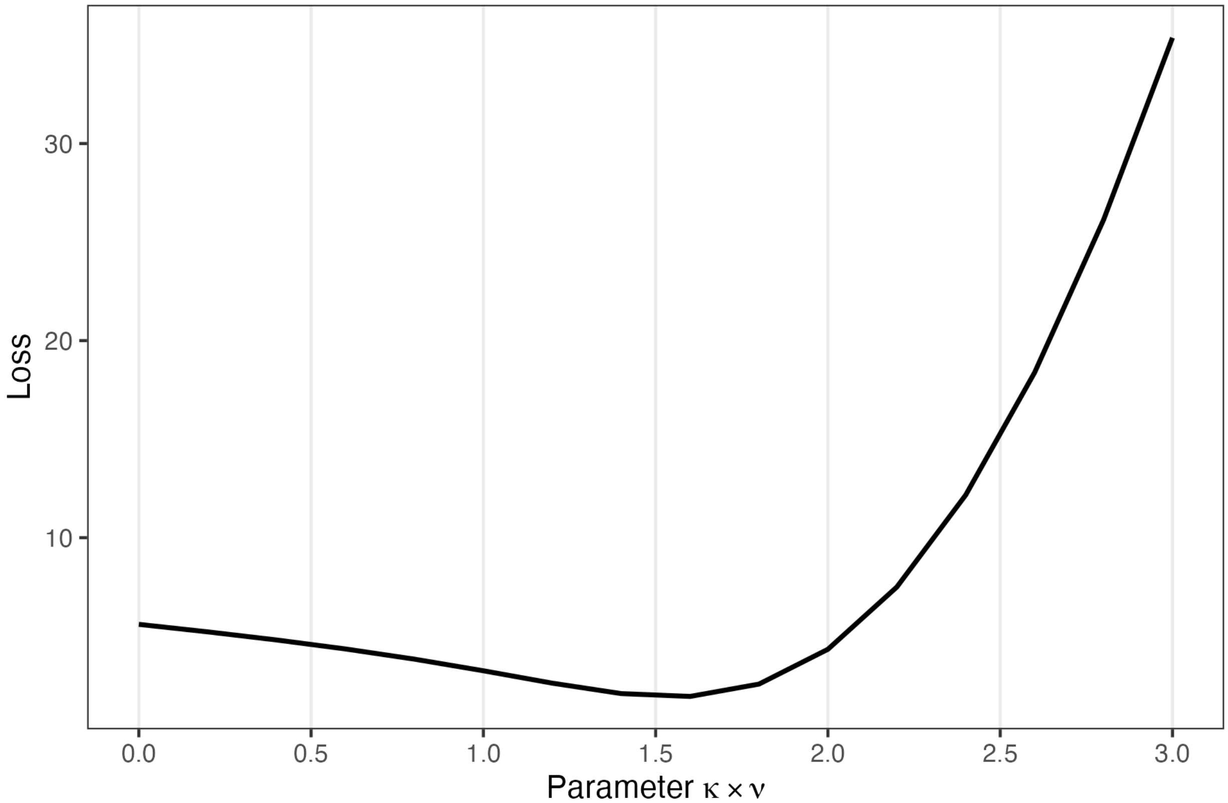
Notes: This figure shows the distribution of Chinese and non-Chinese employment across manufacturing industries in 1947. The left panel shows the share of employment by ethnicity within each industry. The right panel shows the total number of workers by ethnicity. Black bars denote Chinese employment, and gray bars denote non-Chinese employment. Data from the 1947 Census of Population (Del Tufo, 1947).

Figure A.8. Number of Active Malaysian Firms in Orbis, 2003–2022



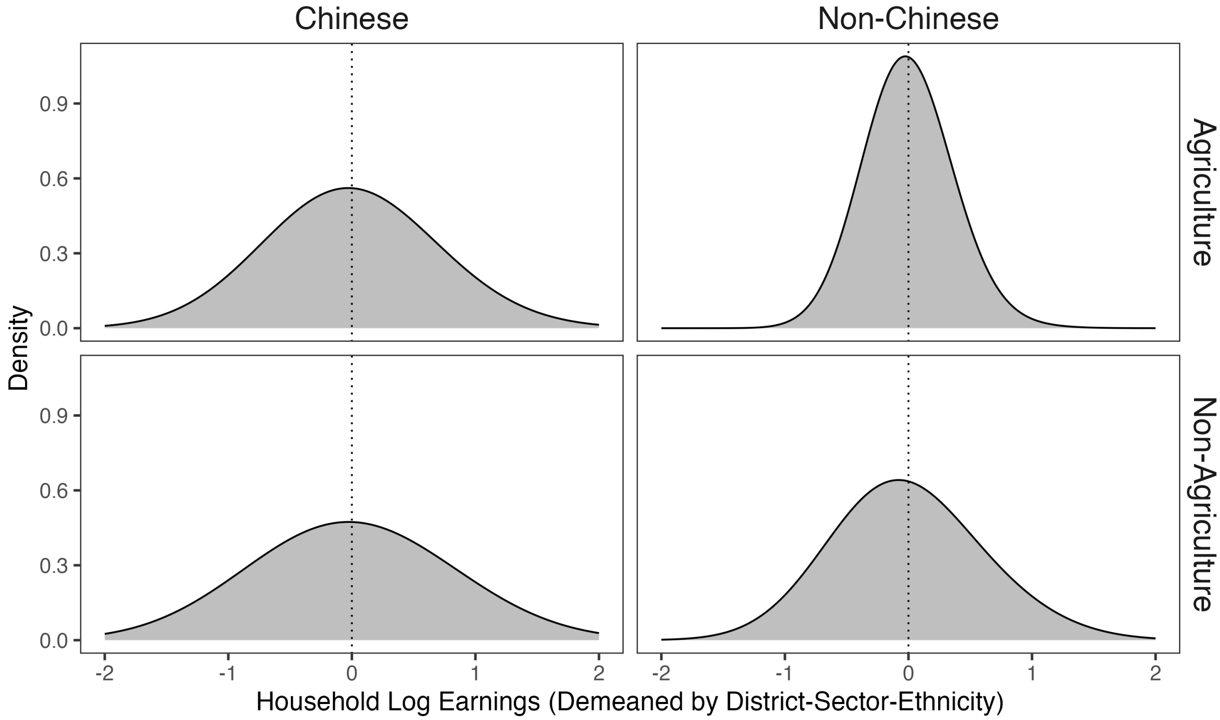
Notes: This figure shows the number of firms in the Orbis database from 2003 to 2022 that meet the main sample criteria: positive revenue, a Malaysian ultimate person owner, a non-missing NAICS industry code, and a location in Peninsular Malaysia. The analysis focuses on the 2011–2015 period, shaded in gray. The classification of ultimate person owners is described in Appendix A.1. Data from the Orbis Historical Disk, accessed in 2023.

Figure A.9. Convexity of the Loss Function in Estimating Migration Costs



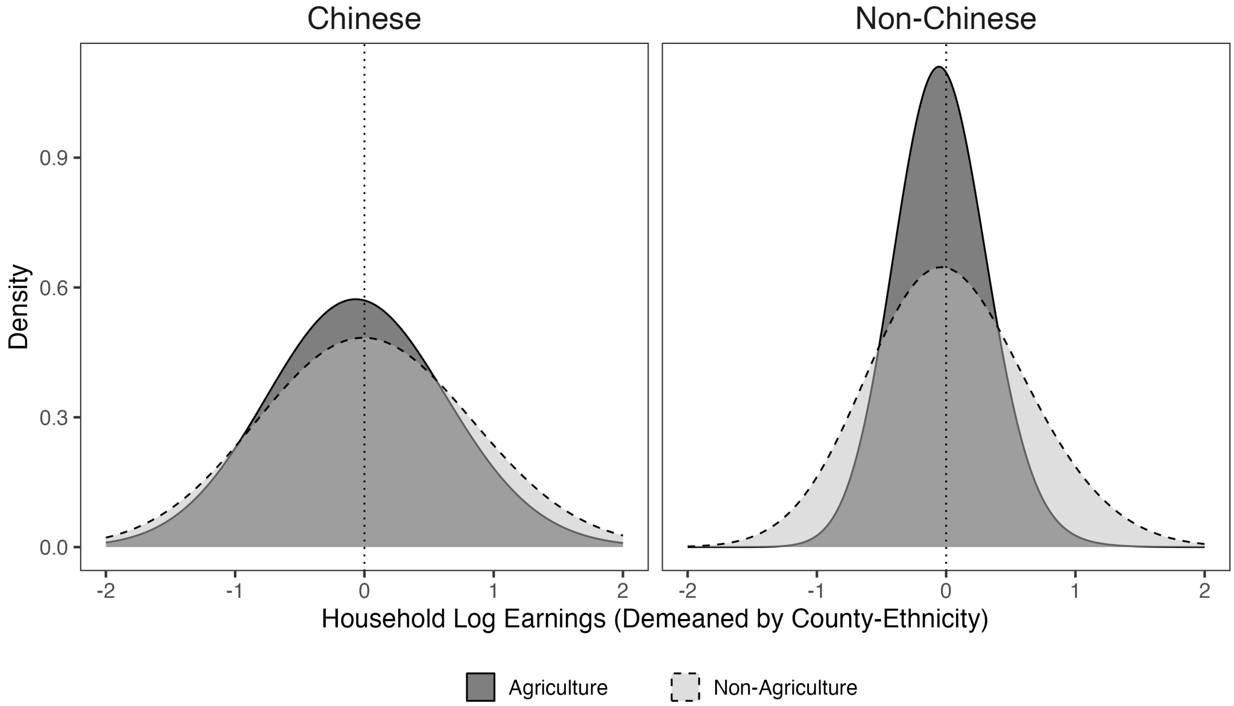
Notes: This figure shows the convexity of the loss function used to estimate migration costs. The y-axis plots the loss from Equation (A-7). The x-axis shows the product of κ and ν, the parameter being estimated. Data from the tabulated Census of Population in 1980.

Figure A.10. Distribution of Demeaned Household Log Earnings, Relative to District-Sector-Ethnicity Average



Notes: This figure shows the distribution of household log earnings after demeaning by the district-sector-ethnicity average, separately for Chinese and non-Chinese households. The top panel shows households in agriculture, and the bottom panel shows households in non-agriculture. The vertical line marks the group mean of zero. Data from the 2% Census of Population microdata in 1980.

Figure A.11. Distribution of Demeaned Household Log Earnings, Relative to County-Ethnicity Average



Notes: This figure shows the distribution of household log earnings after demeaning by the county-ethnicity average, separately for Chinese and non-Chinese households. Within each group, distributions are plotted for households in agriculture (solid line) and non-agriculture (dashed line). The vertical line marks the group mean of zero. Data from the 2% Census of Population microdata in 1980.

Figure A.12. Estimated Production Fundamentals in 1980, by Sector

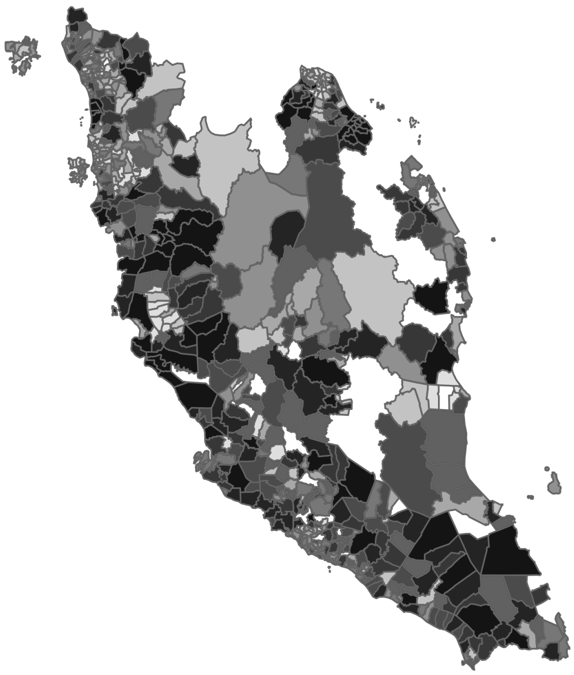
Panel A. Agriculture

Panel B. Non-Agriculture

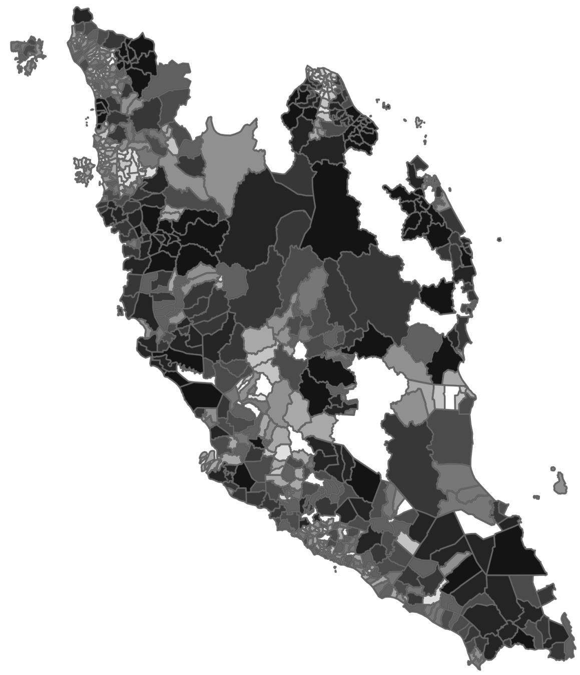
Notes: This figure maps the estimated log production fundamentals by county, with darker shades indicating higher productivity deciles. Panel A shows agricultural fundamentals. Panel B shows non-agricultural fundamentals. Values represent ethnic-population-weighted averages within each county-sector. Counties not included in the sample are shown in white.

Figure A.13. Estimated Amenity Fundamentals in 1980, by Ethnic Group

Panel A. Chinese

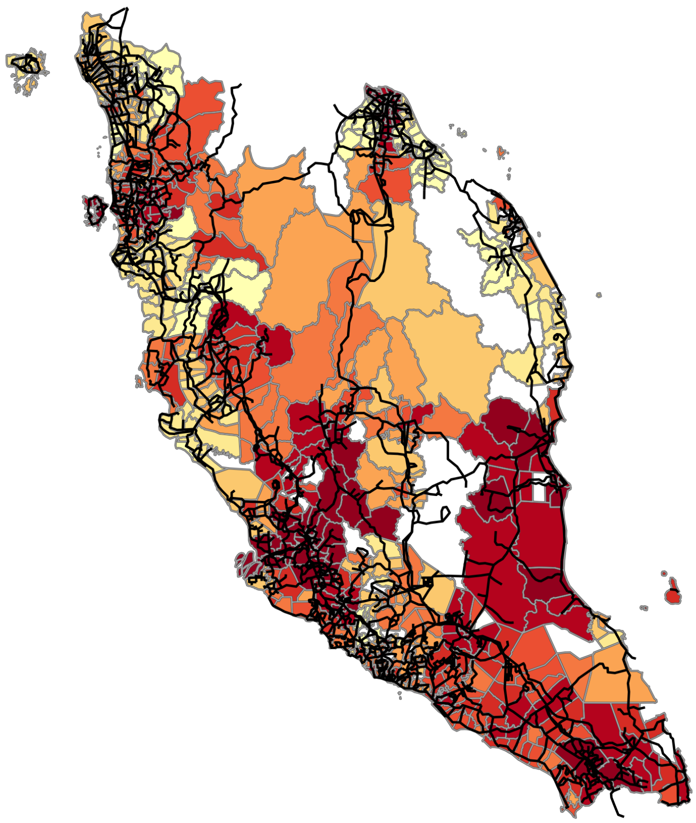


Panel B. Non-Chinese



Notes: This figure maps the estimated log amenity fundamentals by county, with darker shades indicating higher amenity deciles. Panel A shows amenity fundamentals for the Chinese population; Panel B for the non-Chinese population. Counties not included in the sample are shown in white.

Figure A.14. County Productivity and Transport Network in 1980 Equilibrium



Notes: This figure maps county productivity (real wages) in the 1980 equilibrium, averaged across ethnic groups and sectors. Darker red shades indicate higher productivity. Black lines show the road and railway network in 1983. Transport data from G8031 road map series.

E Appendix Tables

Table A.1. Data Sources

Data Source	Publication Year(s)	Variables
Population Census (Tabulated)	1911, 1921, 1931, 1947, 1957, 1970, 1980, 1991, 2000	Population by ethnicity (1911, 1921); population by county and ethnicity (1931–2000); employment by industry and ethnicity (1947); population map (1947); migration flows by district and ethnicity (1980); employment by county, ethnicity, and industry/occupation (1991)
Population Census (Microdata)	1980	Household assets (house, vehicle, phone, etc.); household size; number of children born; educational attainment; years of schooling; duration of residence in present locality; migration status; employment status and industry; language spoken
Second Malaysian Family Life Survey	1989	Household assets; annual earnings; characteristics of resettled Chinese: first job, schooling, land ownership
Directory of Manufacturing	1970	Establishments by county and industry; share employing full-time workers
Orbis Historical Disk	2003–2022	Firms by ownership, county, and industry; average annual revenue
Ministry of Education Malaysia	2022	Number of schools, distance to nearest school, teacher-student ratio, by school type (Chinese/non-Chinese)
Directory of Singapore and Malaya	1959	Distance to Chinese schools; number of Chinese schools
G8031 Road Maps	1942, 1961, 1983	Distance to roads and railroads
HIND 1076 Topographical Maps	1945	Distance to rail station, police station, post/telegraph office, hospital, mosque, non-Muslim temple
GSGS 4474 Land Use Maps	1943	Land share in rubber, mining, paddy rice, coconut, pineapple, mangrove, forest, swamp, town
US National Archives, RG226	1944	Distance to industrial facilities
Global Human Settlement Layer	1975, 1990, 2005	Built-up volumes
Shuttle Radar Topography Mission (SRTM)	2000	Elevation
Nunn and Puga (2012)	2012	Terrain ruggedness
DIVA-GIS	2011	Distance to rivers
FAO GAEZ v4	2022	Rice suitability; coconut suitability
Galor and Özak (2016)	2016	Caloric suitability index
A General Survey of New Villages (Corry, 1954)	1954	Resettled population by county
The National Archives in the UK (CO 1030/1)	1955	Black Areas
Author's Calculations	N/A	Distance to coastline and major cities; squatter population distribution; ethnic dissimilarity index (1947); distance segregation index (1945)

Table A.2. Population of Peninsula Malaya by Ethnic Group, 1911–1957

	Chinese		Malays		Indians and Others	
Year	Number (1)	Percent (2)	Number (3)	Percent (4)	Number (5)	Percent (6)
1911	692,228	30%	1,367,245	59%	275,928	12%
1921	855,863	29%	1,568,588	54%	482,270	17%
1931	1,284,094	34%	1,863,723	49%	638,445	17%
1947	1,882,700	39%	2,395,686	49%	594,448	12%
1957	2,328,480	37%	3,126,773	50%	819,055	13%

Notes: This table shows the population and share by ethnic group in Peninsula Malaya from 1911 to 1957. Columns 1 and 2 report the number of Chinese and its share in total population of a given year. Columns 3 and 4 report the same figures for Malays. Columns 5 and 6 report the same figures for Indians and other ethnic groups. Data from the Census of Population 1911–1957 (Vlieland, 1931; Del Tufo, 1947; Purcell, 1947; Fell, 1960).

Table A.3. Predicting Log Household Income in 1988

	Log Household Earning, 1988	
	(1)	(2)
Vehicle	0.567	0.608
	(0.288)	(0.319)
Motorcycle	0.002	0.098
	(0.157)	(0.159)
Bicycle	−0.113	−0.055
	(0.154)	(0.159)
Phone	1.006	0.907
	(0.400)	(0.398)
Refrigerator	0.424	0.253
	(0.176)	(0.179)
Television	0.182	0.117
	(0.142)	(0.144)
Household Size	0.050	0.054
	(0.010)	(0.010)
R^2	0.28	0.34
Fixed Effects	State	District
# Households	1,413	1,413

Notes: This table shows a linear model predicting (log) household income based on asset ownership and household size. The independent variables include household size and indicators for ownership of various household assets—vehicle, motorcycle, bicycle, phone, refrigerator, and television—as well as pairwise interactions between these asset indicators (not shown here due to space constraints). Column 1 includes state fixed effects. Column 2 includes state-district fixed effects. Data from the Second Malaysian Family Life Survey (1988–1989). Standard errors robust to heteroskedasticity reported in parentheses.

Table A.4. Build-up Volumes, by County Resettlement Density

	Build-up Volumes, by Year:		
	1975	1990	2005
	(1)	(2)	(3)
Higher Resettlement	0.169	0.185	0.105
	(0.084)	(0.070)	(0.062)
# Counties	777	777	777

Notes: This table shows the relationship between build-up volumes from 1975 to 2005 and county resettlement density. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Columns 1–3 report the effect of resettlement density on county build-up volumes in 1975 (Column 1), 1990 (Column 2), and 2005 (Column 3). All regressions are estimated using the Poisson pseudo-maximum-likelihood (PPML) estimator and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the county. Data from the Global Human Settlement Layer (GHSL) project. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.5. Migration and Fertility in 1980, by County Resettlement Density

	Chinese Individuals (1)	Non-Chinese Individuals (2)	Difference (1) − (2) (3)
Panel A. Internal Migrant			
Higher Resettlement	0.051	−0.005	0.056
	(0.028)	(0.017)	(0.023)
Mean of Outcome	0.42	0.49	
# Individuals	31,507	57,345	
Panel B. Internal Migrant After 1960			
Higher Resettlement	0.055	0.004	0.050
	(0.032)	(0.019)	(0.026)
Mean of Outcome	0.31	0.41	
# Individuals	31,387	57,117	
Panel C. Number of Children Born			
Higher Resettlement	−0.017	−0.144	0.128
	(0.118)	(0.062)	(0.114)
Mean of Outcome	4.05	4.24	
# Women	12,118	23,546	
Panel D. Household Size			
Higher Resettlement	0.078	−0.060	0.138
	(0.106)	(0.048)	(0.125)
Mean of Outcome	5.81	5.20	
# Households	9,820	21,238	

Notes: This table shows the relationship between county resettlement density and migration and fertility outcomes in 1980. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Each panel examines a different outcome: Panel A reports whether a person is an internal migrant (moved to current locality from another village or town within Malaysia); Panel B reports whether a person is an internal migrant who moved to current locality within the past 20 years (or after 1960); Panel C reports the number of children ever born; and Panel D reports household size. Column 1 shows estimates for Chinese individuals, column 2 for non-Chinese individuals, and column 3 shows the difference between estimates in columns 1 and 2. All regressions are estimated using OLS and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the individual for Panels A–C and the household for Panel D. The sample is restricted to those age 20 and older. Data from the 2% individual-level Census of Population microdata in 1980. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.6. Secondary and Tertiary Employment in 1980–1991, by County Resettlement Density

	Secondary Industries (1)	Tertiary Industries (2)	Difference (2) − (1) (3)
Panel A. Total Employment			
Higher Resettlement	0.300	0.282	−0.018
	(0.124)	(0.139)	(0.064)
# County-Years	1,554	1,554	
Panel B. Chinese Employment			
Higher Resettlement	0.357	0.352	−0.005
	(0.166)	(0.203)	(0.064)
# County-Years	1,400	1,476	
Panel C. Non-Chinese Employment			
Higher Resettlement	0.235	0.239	0.004
	(0.105)	(0.117)	(0.073)
# County-Years	1,400	1,476	

Notes: This table shows the relationship between sectoral employment in 1980–1991 and county resettlement density. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Panel A shows the effect of resettlement on total employment in the secondary sector (Column 1), the tertiary sector (Column 2), and the difference between the two (Column 3). Panels B and C show the effects on Chinese employment and non-Chinese employment, respectively. The secondary sector is comprised of manufacturing; utility; and construction. The tertiary sector is comprised of wholesale and retail trade; transport and communication; and finance, business, and other services. All regressions are estimated using the Poisson pseudo-maximum-likelihood (PPML) estimator and include state-year fixed effects and the main controls interacted with year: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the county-year. Data from the Census of Population in 1980 and 1991. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.7. Household Asset Ownership, by County Resettlement Density

	Chinese Households (1)	Non-Chinese Households (2)	Difference (1) − (2) (3)
Panel A. Owned the House			
Higher Resettlement	0.045	0.020	0.025
	(0.016)	(0.017)	(0.019)
Mean of Outcome	0.58	0.65	
Panel B. Have Vehicle			
Higher Resettlement	0.059	0.023	0.036
	(0.027)	(0.012)	(0.021)
Mean of Outcome	0.37	0.14	
Panel C. Have Fridge			
Higher Resettlement	0.043	0.037	0.006
	(0.029)	(0.021)	(0.025)
Mean of Outcome	0.52	0.24	
Panel D. Have TV			
Higher Resettlement	0.039	0.006	0.033
	(0.012)	(0.018)	(0.017)
Mean of Outcome	0.66	0.51	
Panel E. Have Phone			
Higher Resettlement	0.038	0.014	0.024
	(0.021)	(0.008)	(0.015)
Mean of Outcome	0.22	0.05	
# Households	10,593	23,269	

Notes: This table shows the relationship between household asset ownership and county resettlement density. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Each panel shows the effect of resettlement density on a different indicator of asset ownership: the occupied house (Panel A); any motor car or van (Panel B); any refrigerator (Panel C); any black or color TV (Panel D); any phone (Panel E). Column 1 reports the estimates for Chinese households, column 2 reports the estimates for non-Chinese households, and column 3 reports the difference between the estimates in columns 1 and 2. All regressions are estimated by OLS and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the household. Data from the 2% individual-level Census of Population microdata in 1980 and Second Malaysian Family Life Survey (1988–1989). Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.8. Differences Between Resettled Chinese and Other Residents

	Employed Agriculture, First Job (1)	Completed Primary Education (2)	Completed Secondary Education (3)	Acres of Land Owned (4)
Panel A. Resettled Chinese vs. Other Chinese				
Resettled Chinese	0.149	−0.125	−0.151	−105.390
	(0.069)	(0.063)	(0.023)	(78.932)
# Observations	896	989	989	395
Panel B. Resettled Chinese vs. Non-Chinese				
Resettled Chinese	−0.155	−0.049	−0.080	−46.747
	(0.062)	(0.053)	(0.057)	(16.517)
# Observations	3,153	3,627	3,627	1,299
Mean of Resettled Chinese	0.36	0.47	0.17	3.03

Notes: This table reports differences in characteristics between resettled Chinese and other residents of similar age living in the same state in 1988, using data from the Second Malaysian Family Life Survey. Resettled Chinese are identified as individuals not born in a New Village but who had "migrated" to one before 1960. Panel A compares them with other Chinese in the same state; Panel B compares them with non-Chinese in the same state. Each column reports the estimated difference in a separate outcome: first job in agriculture (Column 1); completion of primary education (Column 2); completion of secondary education (Column 3); and acres of land owned (Column 4). All regressions include state-by-gender fixed effects and control for age and age squared. The unit of observation is the individual for columns 1–3 and the household for column 4. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.9. Non-Migrant Household Income in 1980, by County Resettlement Density

	Chinese Households (1)	Non-Chinese Households (2)	Difference (1) − (2) (3)
Panel A. Log Earnings			
Higher Resettlement	0.084	−0.007	0.091
	(0.046)	(0.033)	(0.039)
Mean of Outcome	8.52	8.04	
# Households	4,192	9,116	
Panel B. Log Earnings, Agriculture			
Higher Resettlement	0.053	−0.031	0.084
	(0.040)	(0.040)	(0.050)
Mean of Outcome	8.30	7.89	
# Households	1,009	5,159	
Panel C. Log Earnings, Non-Agriculture			
Higher Resettlement	0.105	0.003	0.102
	(0.048)	(0.035)	(0.039)
Mean of Outcome	8.59	8.24	
# Households	3,183	3,957	

Notes: This table shows the relationship between county resettlement density and non-migrant household income. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Panel A, Columns 1 and 2 show the effect of resettlement density on log household earnings for Chinese households (Column 1) and non-Chinese households (Column 2), respectively. Column 3 reports the difference between the estimates in columns 1 and 2. Panel B restricts the sample to households whose head is employed in agriculture, comprised of agriculture and mining. Panel C restricts the sample to households whose head is employed in non-agriculture. All regressions are estimated by OLS and include the main controls: state fixed effects, expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the household. Data from the 2% individual-level Census of Population microdata in 1980. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.10. School Access in 1959 and 2022, by County Resettlement Density

	Chinese Schools		Non-Chinese Schools
	1959	2022	2022
	(1)	(2)	(3)
Panel A. Distance to Schools (km)			
Higher Resettlement	0.395	0.698	−0.087
	(0.338)	(0.314)	(0.096)
Mean of Outcome	6.95	6.72	2.60
# Counties	777	777	777
Panel B. Density of Schools			
Higher Resettlement	0.377	−0.092	1.114
	(0.853)	(0.725)	(1.878)
Mean of Outcome	3.06	3.11	15.82
# Counties	777	777	777
Panel C. Teacher-to-Student Ratio			
Higher Resettlement	–	−0.033	−0.002
	–	(0.018)	(0.003)
Mean of Outcome	–	0.14	0.11
# Counties	–	407	754

Notes: This table shows the relationship between county resettlement density and measures of school access in 1959 and 2022. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Each panel reports the effect of resettlement density on a different measure of school access: distance (in km) to the nearest schools (Panel A); number of schools per 100 square kilometers (Panel B); and average teacher-to-student ratio (Panel C). Columns 1 and 2 report results for Chinese schools in 1959 and 2022, respectively; Column 3 reports results for non-Chinese schools in 2022. All regressions are estimated using OLS and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the county. Chinese school data for 1959 are from the Directory of Singapore and Malaya; school data for 2022 are from the Ministry of Education. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.11. Heterogeneous Effects of County Resettlement Density on 1980 Household Income, by Education of Household Head

	Log Household Earnings		
	Chinese Households (1)	Non-Chinese Households (2)	Difference (1) − (2) (3)
Panel A. No Formal Education			
Higher Resettlement	0.057	0.008	0.050
	(0.041)	(0.034)	(0.047)
Mean of Outcome	8.42	7.93	
# Households	2,232	5,156	
Panel B. Primary Education (1–6 Years)			
Higher Resettlement	0.077	0.035	0.042
	(0.045)	(0.028)	(0.036)
Mean of Outcome	8.49	8.13	
# Households	4,551	10,366	
Panel C. Secondary or Higher Education (7+ Years)			
Higher Resettlement	0.157	0.073	0.085
	(0.039)	(0.039)	(0.039)
Mean of Outcome	8.77	8.43	
# Households	2,814	4,963	

Notes: This table shows the relationship between county resettlement density and household income in 1980 by different educational levels of the household head. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Each panel corresponds to a different subsample: Panel A includes households with no formal education; Panel B includes those with 1–6 years of schooling; and Panel C include those with 7 or more years of schooling. Columns 1 and 2 report the estimated effect for Chinese and non-Chinese households, respectively. Column 3 reports the differences between the estimates in columns 1 and 2. All regressions are estimated by OLS and include state fixed effects and the main controls—expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the household. The sample is restricted to households whose head is age 18 or above. Data from the 2% individual-level Census of Population microdata in 1980. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.12. Road Access in 1961 and 1983, by County Resettlement Density

	1961 (1)	1983 (2)	Difference (2) − (1) (3)
Panel A. Distance to Main Roads			
Higher Resettlement	0.202	−0.196	−0.398
	(0.450)	(0.275)	(0.383)
Mean of Outcome	5.99	3.53	−2.46
Panel B. Distance to Local Roads			
Higher Resettlement	−0.386	−0.401	−0.015
	(0.658)	(0.673)	(0.532)
Mean of Outcome	10.87	7.80	−3.07
Panel C. Density of Main Roads			
Higher Resettlement	−0.001	0.016	0.017
	(0.004)	(0.014)	(0.013)
Mean of Outcome	0.03	0.12	0.10
Panel D. Density of Local Roads			
Higher Resettlement	0.007	0.012	0.005
	(0.003)	(0.008)	(0.006)
Mean of Outcome	0.02	0.05	0.03
# Counties	777	777	

Notes: This table shows the relationship between county resettlement density and measures of road access in 1961 and 1983. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Each panel reports the effect of resettlement density on a different road variable: distance (km) to the nearest main roads (Panel A); distance to other roads (Panel B); main road density (Panel C); and other road density (Panel D). Column 1 reports estimates for roads measured in 1961; column 2 reports estimates in 1983; and column 3 reports their difference. All regressions are estimated using OLS and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road in 1942; road density in 1942; distance to the nearest rail station in 1945; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the county. Road data from the G8031 maps. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.13. Balance of Demographic Composition of the Resettled Population

		Main Occupation	
	Family Size	Agriculture	Rubber
	(1)	(2)	(3)
Higher Resettlement	0.110	0.047	-0.043
	(0.158)	(0.051)	(0.037)
Mean of Outcome	5.33	0.42	0.67
# Counties	159	213	213

Notes: This table examines the relationship between county resettlement density and demographic compositions of resettled population. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. Column 1 reports results for average family size among resettled villagers. Columns 2 and 3 report the share of villages whose main occupation was in non-rubber-related agriculture (Column 2) or rubber tapping/cutting (Column 3). All regressions are estimated by OLS and include state fixed effects and the main controls: expected resettlement density; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the county. The sample includes all counties containing at least one New Village. Data from the Malaysian Christian Council Survey. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.14. Robustness to Specifications of Counterfactual Resettlement Density

	Population, 1980		Employment, 1980–1991		Log Earnings, 1980	
	Log Total Population	Share of Chinese	Agriculture	Non-Agriculture	Chinese Households	Non-Chinese Households
	(1)	(2)	(3)	(4)	(5)	(6)
1. Baseline	0.109	0.050	0.111	0.291	0.110	0.039
	(0.062)	(0.011)	(0.037)	(0.129)	(0.049)	(0.031)
2. Excluding expected resettlement density	0.091	0.047	0.074	0.298	0.095	0.045
	(0.051)	(0.010)	(0.032)	(0.094)	(0.039)	(0.028)
3. Prioritize roads over river up to 10 km	0.093	0.053	0.126	0.298	0.105	0.033
	(0.067)	(0.011)	(0.040)	(0.145)	(0.050)	(0.030)
4. Minimum distance of 1 km between villages	0.093	0.054	0.126	0.298	0.105	0.033
	(0.066)	(0.011)	(0.040)	(0.142)	(0.050)	(0.030)
5. Squatters within 2.5 km of forest	0.107	0.053	0.124	0.289	0.105	0.035
	(0.062)	(0.011)	(0.039)	(0.133)	(0.048)	(0.032)
6. Squatters within 10 km of forest	0.100	0.054	0.129	0.298	0.099	0.029
	(0.065)	(0.012)	(0.040)	(0.143)	(0.050)	(0.030)
7. Lower resettlement cost elasticity ($\psi = 0.5$)	0.095	0.054	0.122	0.292	0.105	0.034
	(0.066)	(0.011)	(0.039)	(0.142)	(0.050)	(0.030)
8. Higher resettlement cost elasticity ($\psi = 0.8$)	0.094	0.054	0.126	0.292	0.103	0.031
	(0.066)	(0.011)	(0.040)	(0.145)	(0.050)	(0.031)
9. Log resettlement density (resettled counties)	0.096	0.042	0.085	0.260	0.131	0.028
	(0.049)	(0.013)	(0.033)	(0.155)	(0.048)	(0.036)

Notes: This table shows robustness to alternative specifications of counterfactual resettlement density in the relationship between county resettlement density and the main outcome variables: log total population in 1980 (Column 1), Chinese population share in 1980 (Column 2), total employment in agriculture in 1980–1991 (Column 3), total employment in non-agriculture in 1980–1991 (Column 4), log earnings for Chinese households in 1980 (Column 5), and log earnings for non-Chinese households in 1980 (Column 6). Row 1 reports the baseline specification, which includes state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. Row 2 excludes expected resettlement density from baseline specification. Row 3–8 add alternative specification of expected resettlement density to the baseline. Row 9 uses log transformations for both actual and expected resettlement density, restricting the sample to resettled counties. The unit of observation is the county in columns 1–4 and the households in columns 5–6. Rows 2–7 additionally control for a variant specification of expected resettlement density. Data from the tabulated population Census in 1980 and 1991, and the 2% sample of microdata in 1980. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.15. Robustness to Additional Controls for Determinants of Resettlement Density

	Population, 1980		Employment, 1980–1991		Log Earnings, 1980	
	Log Total Population (1)	Share of Chinese (2)	Agriculture (3)	Non-Agriculture (4)	Chinese Households (5)	Non-Chinese Households (6)
1. Baseline	0.109	0.050	0.111	0.291	0.110	0.039
	(0.062)	(0.011)	(0.037)	(0.129)	(0.049)	(0.031)
2. Cubic in population and Chinese share	0.103	0.049	0.131	0.275	0.108	0.034
	(0.060)	(0.011)	(0.030)	(0.129)	(0.048)	(0.032)
3. Neighboring roads within 3 km	0.107	0.052	0.113	0.282	0.114	0.044
	(0.062)	(0.011)	(0.037)	(0.132)	(0.051)	(0.034)
4. Neighboring roads within 5 km	0.107	0.050	0.116	0.286	0.113	0.045
	(0.061)	(0.011)	(0.037)	(0.120)	(0.048)	(0.031)
5. Neighboring population within 3 km	0.088	0.050	0.098	0.256	0.120	0.029
	(0.063)	(0.012)	(0.038)	(0.119)	(0.048)	(0.037)
6. Neighboring population within 5 km	0.099	0.049	0.099	0.235	0.103	0.018
	(0.064)	(0.012)	(0.035)	(0.106)	(0.048)	(0.035)
7. All above (rows 2–6)	0.085	0.050	0.119	0.229	0.090	0.022
	(0.058)	(0.012)	(0.031)	(0.118)	(0.038)	(0.031)

Notes: This table shows robustness to adding controls for initial roads and population—key determinants of resettlement density—in the relationship between county resettlement density and the main outcome variables: log total population in 1980 (Column 1), Chinese population share in 1980 (Column 2), total employment in agriculture in 1980–1991 (Column 3), total employment in non-agriculture in 1980–1991 (Column 4), log earnings for Chinese households in 1980 (Column 5), and log earnings for non-Chinese households in 1980 (Column 6). Row 1 reports the baseline specification, which includes state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. Rows 2–7 add additional controls to this baseline. The unit of observation is the county in columns 1–4 and the households in columns 5–6. Data from the tabulated population Census in 1980 and 1991, and the 2% sample of microdata in 1980. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.16. Robustness to Additional Controls for Initial Economic Activities

	Population, 1980		Employment, 1980–1991		Log Earnings, 1980	
	Log Total Population (1)	Share of Chinese (2)	Agriculture (3)	Non-Agriculture (4)	Chinese Households (5)	Non-Chinese Households (6)
1. Baseline	0.110	0.050	0.112	0.292	0.110	0.039
	(0.062)	(0.011)	(0.037)	(0.129)	(0.049)	(0.031)
2. Topographic features	0.115	0.049	0.098	0.300	0.102	0.023
	(0.063)	(0.011)	(0.033)	(0.128)	(0.051)	(0.033)
3. Rice suitability and land use	0.110	0.052	0.113	0.264	0.104	0.044
	(0.062)	(0.012)	(0.037)	(0.134)	(0.053)	(0.033)
4. Coconut suitability and land use	0.114	0.048	0.114	0.288	0.105	0.040
	(0.061)	(0.011)	(0.035)	(0.129)	(0.050)	(0.031)
5. Other land use shares	0.087	0.053	0.123	0.263	0.102	0.029
	(0.056)	(0.012)	(0.035)	(0.143)	(0.046)	(0.031)
6. Distance to prewar industrial sites	0.109	0.050	0.117	0.277	0.099	0.034
	(0.063)	(0.011)	(0.036)	(0.131)	(0.045)	(0.031)
7. Distance to major cities	0.110	0.050	0.111	0.278	0.093	0.042
	(0.063)	(0.011)	(0.037)	(0.129)	(0.048)	(0.032)
8. Road access in 1983	0.103	0.050	0.102	0.301	0.113	0.039
	(0.063)	(0.011)	(0.036)	(0.130)	(0.048)	(0.031)
9. All above (rows 2–8)	0.085	0.054	0.089	0.260	0.087	0.025
	(0.057)	(0.011)	(0.028)	(0.143)	(0.046)	(0.036)

Notes: This table shows robustness to adding controls for location fundamentals and initial economic activities in the relationship between county resettlement density and the main outcome variables: log total population in 1980 (Column 1), Chinese population share in 1980 (Column 2), total employment in agriculture in 1980–1991 (Column 3), total employment in non-agriculture in 1980–1991 (Column 4), log earnings for Chinese households in 1980 (Column 5), and log earnings for non-Chinese households in 1980 (Column 6). Row 1 reports the baseline specification, which includes state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. Rows 2–9 add additional controls to this baseline. The unit of observation is the county in columns 1–4 and the households in columns 5–6. Data from the tabulated population Census in 1980 and 1991, and the 2% sample of microdata in 1980. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.17. Robustness to Sample

	Population, 1980		Employment, 1980–1991		Log Earnings, 1980	
	Log Total Population (1)	Share of Chinese (2)	Agriculture (3)	Non-Agriculture (4)	Chinese Households (5)	Non-Chinese Households (6)
1. Baseline	0.110	0.050	0.112	0.292	0.110	0.039
	(0.062)	(0.011)	(0.037)	(0.129)	(0.049)	(0.031)
2. Exclude top and bottom counties by area	0.102	0.059	0.113	0.255	0.112	0.027
	(0.059)	(0.014)	(0.036)	(0.130)	(0.056)	(0.035)
3. Exclude top and bottom resettled counties	0.103	0.048	0.119	0.286	0.110	0.031
	(0.063)	(0.011)	(0.037)	(0.131)	(0.050)	(0.031)
4. Exclude most densely populated towns	0.109	0.052	0.099	0.261	0.090	0.036
	(0.060)	(0.011)	(0.039)	(0.146)	(0.050)	(0.035)
5. Only counties with sampled Chinese	0.077	0.047	0.102	0.272	0.110	0.053
	(0.061)	(0.013)	(0.038)	(0.135)	(0.049)	(0.033)

Notes: This table shows robustness to alternative county sample restrictions in the relationship between county resettlement density and the main outcome variables: log total population in 1980 (Column 1), Chinese population share in 1980 (Column 2), total employment in agriculture in 1980–1991 (Column 3), total employment in non-agriculture in 1980–1991 (Column 4), log earnings for Chinese households in 1980 (Column 5), and log earnings for non-Chinese households in 1980 (Column 6). Row 1 reports the baseline sample. Row 2 excludes the top and bottom 1% of counties by area. Row 3 excludes the top and bottom 1% by resettlement density. Row 4 excludes counties containing the ten most densely populated towns in 1947. Row 5 restricts the sample to counties with observed Chinese individuals in 2% census microdata. All regressions include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the county in columns 1–4 and the households in columns 5–6. Data from the tabulated population Census in 1980 and 1991, and the 2% sample of microdata in 1980. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.18. Nonlinear Effects of County Resettlement Density on 1980 Household Income

	Chinese Households (1)	Non-Chinese Households (2)	Difference (1) − (2) (3)
Panel A. Log Earnings			
Higher Resettlement	0.099	0.034	0.065
	(0.047)	(0.034)	(0.029)
Higher Resettlement Squared	0.016	0.011	0.005
	(0.026)	(0.026)	(0.018)
Mean of Outcome	8.55	8.15	
# Households	9,634	20,549	
Panel B. Log Earnings, Agriculture			
Higher Resettlement	0.048	−0.003	0.052
	(0.036)	(0.035)	(0.034)
Higher Resettlement Squared	0.037	0.030	0.007
	(0.031)	(0.023)	(0.027)
Mean of Outcome	8.32	7.94	
# Households	2,197	9,359	
Panel C. Log Earnings, Non-Agriculture			
Higher Resettlement	0.119	0.065	0.054
	(0.043)	(0.036)	(0.020)
Higher Resettlement Squared	0.010	−0.020	0.030
	(0.025)	(0.024)	(0.016)
Mean of Outcome	8.63	8.33	
# Households	7,437	11,190	

Notes: This table shows the relationship between county resettlement density and household income in 1980, including a squared term to capture potential nonlinearity. "Higher Resettlement" is the county resettlement density defined in Section IV, standardized to have a standard deviation of 1. "Higher Resettlement Squared" is its squared term. Panel A reports effects on log household earnings for Chinese households (Column 1), non-Chinese households (Column 2), and the difference between the two estimates (Column 3). Panel B restricts the sample to households whose head is employed in the agricultural sector (agriculture and mining). Panel C restricts the sample to households whose head is employed in the non-agricultural sector. All regressions are estimated by OLS and include state fixed effects and the main controls: expected resettlement density; an indicator for any resettlement in the county; distance to the nearest road; road density; distance to the nearest rail station; distance to coastline; Chinese population share in 1947; log population density in 1947; log county area; and the shares of land used for rubber and mining in 1943. The unit of observation is the household. Data from the 2% individual-level Census of Population microdata in 1980. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.19. Over-identification Check: Correlates of Estimated Fundamentals

	Production Fundamental		Amenity Fundamental	
	Agriculture (1)	Non- Agriculture (2)	Chinese (3)	Non-Chinese (4)
Ruggedness	−0.005	0.000		
	(0.002)	(0.002)		
Rice Suitability	0.060	0.039		
	(0.181)	(0.185)		
Caloric Suitability	0.003	−0.001		
	(0.002)	(0.002)		
Mfg. Estab. Density	0.001	0.094		
	(0.019)	(0.036)		
Police Station Density			7.753	1.413
			(1.637)	(1.312)
Post Office Density			8.504	5.683
			(5.420)	(4.744)
Distance to Coast			−0.006	−0.007
			(0.004)	(0.004)
Distance to School			−0.298	−0.128
			(0.064)	(0.062)
School Density			0.684	−0.047
			(0.320)	(0.329)
# Counties	685	685	685	685

Notes: This table shows how estimated production and amenity fundamentals correlate with various productivity and amenity measures listed in each row. Columns 1 and 2 use log production fundamentals as the dependent variable by sector: agriculture (Column 1) and non-agriculture (Column 2), both averaged across ethnic groups. Columns 3 and 4 use log amenity fundamentals as the dependent variable by ethnic group: Chinese (Column 3) and non-Chinese (Column 4). Ruggedness is the terrain ruggedness index (Nunn and Puga, 2012). Rice Suitability is suitability for padi rice cultivation from FAO GAEZ v4. Caloric Suitability is average agricultural suitability in terms of calories (Galor and Özak, 2016). Mfg. Estab. Density is the number of manufacturing establishments per square kilometer in 1970. Police Station Density is the number of police stations per square kilometer in 1945. Post Office Density is the number of post or telegraph office per square kilometer in 1945. Distance to Coast is the average distance to the coastline. Distance to School is the average distance to the nearest school in 2022. School Density is the number of schools per square kilometer in 2022. All regressions control for log county area. The unit of observation is the county. Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.20. Housing Elasticity in 1989

	Log Rents (1989)	
	OLS	IV
	(1)	(2)
Panel A. Year 1980		
Log Population	0.266	0.325
	(0.054)	(0.154)
F-stat (1st Stage)		64.3
Panel B. Year 2000		
Log Population	0.267	0.270
	(0.059)	(0.122)
F-stat (1st Stage)		105.5
# Counties	103	103

Notes: This table shows the relationship between log housing rents in 1989 and log population in years 1980 (Panel A) and 2000 (Panel B). Column 1 reports the OLS estimates. Column 2 reports the IV estimates and the first-stage F statistics. The instrumental variable used is the residual resettlement density, as shown in Figure 3, Panel B. The unit of observation is the household. The sample is restricted to households reporting non-missing rent expenditure. Data from the Malaysian Family Life Survey (1988–1989). Conley standard errors with a 30-kilometer distance cutoff are reported in parentheses.

Table A.21. Robustness to Calibrated Parameter Values: Aggregate Impact of Emergency Resettlement

	Changes in Aggregate Output (% Baseline), by Substitution Elasticity σ			
	$\sigma = 4$	$\sigma = 6$	$\sigma = 8$	$\sigma = 10$
	(1)	(2)	(3)	(4)
Panel A. Trade Cost Distance Elasticity $\xi = 0.18$				
Migration elasticity $\nu = 1.5$	1.058	2.579	3.536	3.963
Migration elasticity $\nu = 2.5$	0.567	1.804	2.517	2.778
Migration elasticity $\nu = 3.5$	0.026	0.976	1.377	1.475
Panel B. Trade Cost Distance Elasticity $\xi = 0.28$				
Migration elasticity $\nu = 1.5$	1.364	3.042	3.612	3.723
Migration elasticity $\nu = 2.5$	0.781	2.049	2.425	2.510
Migration elasticity $\nu = 3.5$	0.182	1.097	1.295	1.364
Panel C. Trade Cost Distance Elasticity $\xi = 0.37$				
Migration elasticity $\nu = 1.5$	1.698	3.076	3.287	3.288
Migration elasticity $\nu = 2.5$	0.944	1.863	1.975	1.983
Migration elasticity $\nu = 3.5$	0.279	0.818	0.798	0.800

Notes: This table reports how the aggregate output impact of the resettlement program varies with alternative values of the three externally calibrated parameters. Columns vary the substitution elasticity across regional goods (σ). Panels correspond to different values of the trade cost elasticity (ξ). Within each panel, rows vary the migration elasticity (ν).

Table A.22. Robustness to Estimated Parameter Values: Aggregate Impact of Emergency Resettlement

	Changes in Aggregate Output (% Baseline), by Agricultural Productivity Spillover γ_A		
	$\gamma_A = -0.18$ (1)	$\gamma_A = -0.12$ (2)	$\gamma_A = -0.06$ (3)
Panel A. Amenity Spillover by Size $\beta = -0.06$			
Amenity spillover by ethnic share $\beta^e = 0.06$	1.882	1.940	2.010
Amenity spillover by ethnic share $\beta^e = 0.13$	1.826	1.884	1.954
Amenity spillover by ethnic share $\beta^e = 0.19$	1.667	1.719	1.782
Panel B. Amenity Spillover by Size $\beta = -0.01$			
Amenity spillover by ethnic share $\beta^e = 0.06$	1.995	2.059	2.139
Amenity spillover by ethnic share $\beta^e = 0.13$	1.945	2.009	2.089
Amenity spillover by ethnic share $\beta^e = 0.19$	1.805	1.862	1.934
Panel C. Amenity Spillover by Size $\beta = 0.04$			
Amenity spillover by ethnic share $\beta^e = 0.06$	2.060	2.127	2.216
Amenity spillover by ethnic share $\beta^e = 0.13$	2.014	2.082	2.172
Amenity spillover by ethnic share $\beta^e = 0.19$	1.868	1.927	2.008

Notes: This table reports how the aggregate output impact of the resettlement program varies with alternative values of parameters estimated with uncertainty. Columns vary the agricultural productivity spillover (γ_A). Panels correspond to different values of the amenity spillover with respect to population size (β). Within each panel, rows vary the amenity spillover with respect to ethnic share (β^e).

Table A.23. Robustness to Calibrated Parameter Values: Aggregate Impact of Reducing Cross-Ethnic Frictions

	Changes in Aggregate Output (% Baseline), by Substitution Elasticity σ			
	$\sigma = 4$ (1)	$\sigma = 6$ (2)	$\sigma = 8$ (3)	$\sigma = 10$ (4)
Panel A. Trade Cost Distance Elasticity $\xi = 0.18$				
Migration elasticity $\nu = 1.5$	4.125	4.092	4.085	4.090
Migration elasticity $\nu = 2.5$	4.314	4.200	4.175	4.171
Migration elasticity $\nu = 3.5$	3.595	3.627	3.526	3.473
Panel B. Trade Cost Distance Elasticity $\xi = 0.28$				
Migration elasticity $\nu = 1.5$	4.109	4.064	4.064	4.072
Migration elasticity $\nu = 2.5$	4.276	4.157	4.141	4.141
Migration elasticity $\nu = 3.5$	3.614	3.611	3.538	3.577
Panel C. Trade Cost Distance Elasticity $\xi = 0.37$				
Migration elasticity $\nu = 1.5$	4.078	4.037	4.043	4.049
Migration elasticity $\nu = 2.5$	4.226	4.147	4.152	4.161
Migration elasticity $\nu = 3.5$	3.677	3.610	3.650	3.673

Notes: This table reports how the aggregate output impact of halving cross-ethnic barriers to productivity spillovers (γ^e) varies with alternative values of the three externally calibrated parameters. Columns vary the substitution elasticity across regional goods (σ). Panels correspond to different values of the trade cost elasticity (ξ). Within each panel, rows vary the migration elasticity (ν).

Table A.24. Robustness to Estimated Parameter Values: Aggregate Impact of Reducing Cross-Ethnic Frictions

	Changes in Aggregate Output (% Baseline), by Agricultural Productivity Spillover γ_A		
	$\gamma_A = -0.18$ (1)	$\gamma_A = -0.12$ (2)	$\gamma_A = -0.06$ (3)
Panel A. Amenity Spillover by Size $\beta = -0.06$			
Amenity spillover by ethnic share $\beta^e = 0.06$	4.413	4.415	4.418
Amenity spillover by ethnic share $\beta^e = 0.13$	4.136	4.143	4.151
Amenity spillover by ethnic share $\beta^e = 0.19$	2.758	2.768	2.780
Panel B. Amenity Spillover by Size $\beta = -0.01$			
Amenity spillover by ethnic share $\beta^e = 0.06$	4.406	4.414	4.424
Amenity spillover by ethnic share $\beta^e = 0.13$	4.127	4.143	4.162
Amenity spillover by ethnic share $\beta^e = 0.19$	2.794	2.819	2.850
Panel C. Amenity Spillover by Size $\beta = 0.04$			
Amenity spillover by ethnic share $\beta^e = 0.06$	4.418	4.432	4.451
Amenity spillover by ethnic share $\beta^e = 0.13$	4.139	4.165	4.198
Amenity spillover by ethnic share $\beta^e = 0.19$	2.844	2.886	2.942

Notes: This table reports how the aggregate output impact of halving cross-ethnic barriers to productivity spillovers (γ^e) varies with alternative values of parameters estimated with uncertainty. Columns vary the agricultural productivity spillover (γ_A). Panels correspond to different values of the amenity spillover with respect to population size (β). Within each panel, rows vary the amenity spillover with respect to ethnic share (β^e).

Table A.25. Robustness to Calibrated Parameter Values: Aggregate Impact of Malay Wage Subsidies in Non-Agriculture

	Changes in Aggregate Output (% Baseline), by Substitution Elasticity σ			
	$\sigma = 4$	$\sigma = 6$	$\sigma = 8$	$\sigma = 10$
	(1)	(2)	(3)	(4)
Panel A. Trade Cost Distance Elasticity $\xi = 0.18$				
Migration elasticity $\nu = 1.5$	0.591	0.510	0.467	0.458
Migration elasticity $\nu = 2.5$	0.747	0.727	0.752	0.788
Migration elasticity $\nu = 3.5$	1.107	1.108	1.252	1.393
Panel B. Trade Cost Distance Elasticity $\xi = 0.28$				
Migration elasticity $\nu = 1.5$	0.573	0.497	0.495	0.513
Migration elasticity $\nu = 2.5$	0.733	0.723	0.746	0.744
Migration elasticity $\nu = 3.5$	1.071	1.112	1.177	1.162
Panel C. Trade Cost Distance Elasticity $\xi = 0.37$				
Migration elasticity $\nu = 1.5$	0.559	0.528	0.565	0.594
Migration elasticity $\nu = 2.5$	0.727	0.751	0.770	0.767
Migration elasticity $\nu = 3.5$	1.051	1.094	1.098	1.083

Notes: This table reports how the aggregate output impact of an 18% wage subsidy for Malays in non-agriculture varies with alternative values of the three externally calibrated parameters. Columns vary the substitution elasticity across regional goods (σ). Panels correspond to different values of the trade cost elasticity (ξ). Within each panel, rows vary the migration elasticity (ν).

Table A.26. Robustness to Estimated Parameter Values: Aggregate Impact of Malay Wage Subsidies in Non-Agriculture

	Changes in Aggregate Output (% Baseline), by Agricultural Productivity Spillover γ_A		
	$\gamma_A = -0.18$ (1)	$\gamma_A = -0.12$ (2)	$\gamma_A = -0.06$ (3)
Panel A. Amenity Spillover by Size $\beta = -0.06$			
Amenity spillover by ethnic share $\beta^e = 0.06$	1.109	0.938	0.768
Amenity spillover by ethnic share $\beta^e = 0.13$	1.035	0.869	0.705
Amenity spillover by ethnic share $\beta^e = 0.19$	0.961	0.801	0.645
Panel B. Amenity Spillover by Size $\beta = -0.01$			
Amenity spillover by ethnic share $\beta^e = 0.06$	1.212	1.050	0.889
Amenity spillover by ethnic share $\beta^e = 0.13$	1.138	0.980	0.824
Amenity spillover by ethnic share $\beta^e = 0.19$	1.085	0.929	0.778
Panel C. Amenity Spillover by Size $\beta = 0.04$			
Amenity spillover by ethnic share $\beta^e = 0.06$	1.381	1.231	1.083
Amenity spillover by ethnic share $\beta^e = 0.13$	1.324	1.177	1.034
Amenity spillover by ethnic share $\beta^e = 0.19$	1.319	1.170	1.028

Notes: This table reports how the aggregate output impact of an 18% wage subsidy for Malays in non-agriculture varies with alternative values of parameters estimated with uncertainty. Columns vary the agricultural productivity spillover (γ_A). Panels correspond to different values of the amenity spillover with respect to population size (β). Within each panel, rows vary the amenity spillover with respect to ethnic share (β^e).